

# Mechanics Applied to Aerospace Engineering

EXERCISES AND EXAM PROBLEMS

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AEROSPACE ENGINEERING DEPARTMENT (UC3M)



## Preface

These exercises and problems are part of the course materials of the second year course *Mechanics Applied to Aerospace Engineering* (MAAE) of the Bachelor in Aerospace Engineering (BAE) at Universidad Carlos III de Madrid (UC3M). **Most of the problems have been presented in previous exams to the students.** Many of the current and former staff of the Aerospace Engineering department are or have been involved at some point in this course and have also contributed to the development of this collection.

Various sources have been used during the preparation of this document, in particular the following books. Some of the figures have been taken or adapted from these sources.

- Introduction to Space Dynamics. W.T. Thomson. Dover. 1986
- Engineering Dynamics. J. Ginsberg. Cambridge Univ. Press. 2008
- Classical Mechanics. T.W.B. Kibble & F.H. Berkshire. Imperial College Press. 2004
- Curso de Mecánica Racional. Vols I & II. Manuel Prieto. ADI Madrid. 1986

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## 1 Particle exercises and problems

1. An out-of-control rocket traces an upward helical trajectory described in cartesian coordinates for  $t > 0$  by

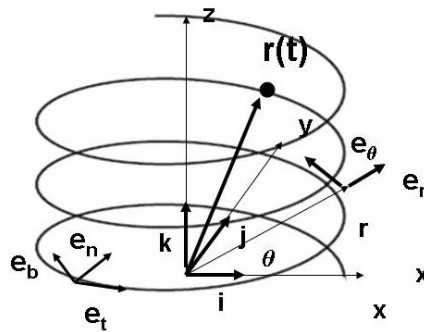
$$x = a \cos(\omega t), \quad (1)$$

$$y = a \sin(\omega t), \quad (2)$$

$$z = v_z t. \quad (3)$$

where  $a$ ,  $v_z$  and  $\omega$  are constants.

- Express the position, velocity and acceleration vectors in the cartesian  $x, y, z$  system using unit vectors  $\mathbf{i}, \mathbf{j}, \mathbf{k}$ .
- Express the position, velocity and acceleration vectors in the cylindrical coordinate system using unit vectors  $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$ .
- Express the velocity and acceleration vectors using  $s, \rho$  and the vector basis  $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ .
- Compute the modulus of the velocity vector
- Express the evolution of the arc-length coordinate as a function of time
- Write down the trajectory as a function of the arc-length coordinate
- Where is the center of curvature for this curve when the particle is located at  $(x, y, z) = (a, 0, 0)$ ?



(Taken from MIT OCW)

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2. A planar spiral is given by the equation  $r = \alpha e^{\beta \theta}$ , where  $\alpha$  and  $\beta$  are constants,  $r$  is the radial coordinate, and  $\theta$  is the angular coordinate. If a particle moves along the spiral according to the equation  $\theta = \delta t^n$  ( $\delta$  is a constant), determine:

- The radial and transverse components of the velocity
- The radial and transverse components of the acceleration
- The tangential and normal components of the velocity
- The tangential and normal components of the acceleration
- The radius of curvature of the spiral

- 
3. A point particle describes the astroid:

$$x = a \cos^3 \varphi$$

$$y = a \sin^3 \varphi$$

with a motion whose tangential acceleration is constant,  $k$ . If the point is initially at rest at  $\varphi = 0$ , compute the evolution of the position as a function of time while the particle is in one of the smooth segments of the astroid. (Hint:  $ds = \sqrt{dx^2 + dy^2}$ ).

4. An aircraft moves at constant altitude  $y_0$  with velocity  $v_0$  in the  $x$  direction. At time  $t = 0$  it is located at  $x = 0$ , and it begins to accelerate in the  $x$  direction at a constant rate  $a_x$ . Describe the position, velocity and acceleration in polar coordinates,  $r$ ,  $\theta$ , and answer the following questions:

- (a) Show that:  $\dot{r} = v(t) \cdot x(t)/r(t)$
- (b) Show that:  $\dot{\theta} = -v(t) \cdot y_0 / (r(t))^2$
- (c) Show that:  $\ddot{r} = (v^2(t) \cdot y_0^2 + a_x \cdot x(t) \cdot r^2(t)) / (r(t))^3$
- (d) Show that:  $\ddot{\theta} = y_0 \cdot (2 v^2(t) \cdot x(t) - a_x \cdot r^2(t)) / (r(t))^4$

From these expressions form the acceleration vector  $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$ . Repeat the problem using cartesian coordinates.

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5. An airplane  $A$  is flying horizontally with a constant speed  $v_A$ , and is towing a glider  $B$ . The glider is gaining altitude. The tow cable has a length  $r$ , and the angle  $\theta$  formed between the tow cable and the horizontal is increasing at a constant rate of  $\dot{\theta}$ . We want to determine the magnitude of the velocity,  $v_0^B$ , and the acceleration,  $a_0^B$ , of the glider as a function of  $\theta$  and its derivatives.
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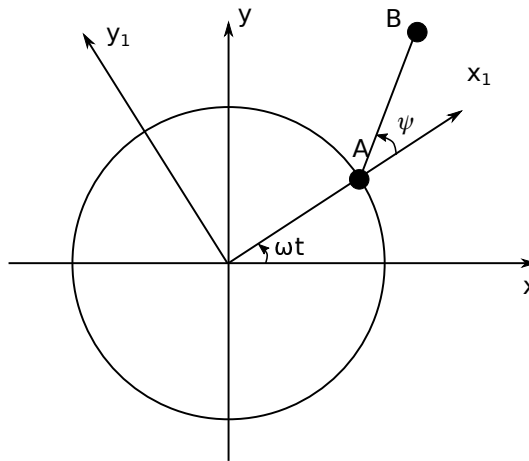
6. A point particle  $P$  travels along a curve given by

$$y = \sin x$$

with a velocity along  $x$  given by  $\dot{x} = 1$ .

- (a) Find the intrinsic, unitary vector basis associated to the curve.
- (b) Express  $x$  as a function of an arc length parameter,  $s$ . You can leave your solution as a function of an integral.
- (c) Find the tangent and normal acceleration components of  $P$  as a function of  $x$ .
- (d) Write the radius of curvature of the trajectory as a function of  $x$ .

7. A skater  $B$  is being towed by a vehicle  $A$  on a perfectly smooth horizontal plane, where we define an inertial reference frame  $Oxy$ . The vehicle is moving along a circumference of radius  $a$  and center at  $O$  with constant angular velocity  $\omega$ . The skater is attached to the vehicle by a bar without mass and constant length  $l = a$ . Initially, the vehicle is located at  $(a, 0)$  and the skater is at rest at  $(2a, 0)$ . Defining a rotating reference frame  $Ox_1y_1$  with origin at  $O$  and axis  $Ox_1$  along  $OA$ , and calling  $\psi$  the angle between the axis  $Ox_1$  and the bar  $AB$ , determine:
- The relative acceleration of the skater with respect to  $Ox_1y_1$  as a function of  $\psi$  and its time derivatives.
  - The centripetal and Coriolis acceleration terms also as a function of  $\psi$  and its time derivatives.
  - Reduce the evolution of  $\psi$  to integrals of the form  $\int[...]\mathrm{d}\psi = \int[...]\mathrm{d}t$ . (consider that  $\dot{\psi}^2 = \int 2\dot{\psi}\ddot{\psi}\mathrm{d}t$ ).
  - Determine the tension on the bar.
  - If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.



- 
8. Two trains,  $A$  and  $B$ , are moving along parallel tracks. The separation between the tracks is  $a$ . The absolute velocity of each train is  $v_A$  and  $v_B$ , respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that the train  $B$  is now moving along a circular track of radius  $R$ . The minimum distance between the circular track and the straight track of  $A$  is  $a$ .

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9. In “2001: A Space Odyssey” an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.
- If the cylinder has radius  $R$ , determine the angular velocity  $\omega$  required to obtain a given level artificial gravity  $g$ . Consider  $R = 10$  m and  $g = 9.81$  m/s<sup>2</sup> for a particular case.
  - An astronaut runs inside the surface of the cylinder with a constant speed  $u$  with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut  $h$  with respect to the radius of the cylinder  $R$ ,  $h \ll R$ ).
  - What is the condition for her to feel no artificial gravity at all?

10. A particle of mass  $m$  moves under a force  $\mathbf{F} = -cx^3\mathbf{i}$ , where  $c$  is a positive constant. Find the potential energy function. If the particle starts from rest at  $x = -a$ , what is its velocity when it reaches  $x = 0$ ? In the subsequent motion, where does it stop again?

11. The potential energy function of a particle of mass  $m$  is  $V(x) = -c(x^2 - a^2)^2/2$ , where  $c$  and  $a$  are positive constants. Sketch this function, and describe the possible types of motion in the three cases (a)  $E > 0$ , (b)  $E < -ca^4/2$ , and (c)  $-ca^4/2 < E < 0$ .

12. A particle moving under a conservative force oscillates between  $x_1$  and  $x_2$ . Show that the period of oscillation is:

$$\tau = 2 \int_{x_1}^{x_2} \sqrt{\frac{m}{2[V(x_2) - V(x)]}} dx,$$

where  $V(x)$  is the potential energy function. In particular, if  $V = (1/2)m\omega_0^2(x^2 - bx^4)$ , show that the period for oscillations of amplitude  $a$  is

$$\tau = \frac{2}{\omega_0} \int_{-a}^a \frac{dx}{\sqrt{a^2 - x^2} \sqrt{1 - b(a^2 + x^2)}}$$

13. A particle of mass  $m$  moves along the  $Ox$  axis under the potential  $V(x) = mgx[(x/a)^2 - 3]$ . Compute the evolution of the particle as a function of time, when it is released from  $x_0 = a$  with an initial velocity  $v_0 = \sqrt{8ga}$  directed along the positive  $Ox$  axis. (tip:  $\int dx / ((x+1)\sqrt{2-x}) = -(2/\sqrt{3})\text{arctanh}(\sqrt{2-x}/\sqrt{3})$ )

14. A particle of mass  $m$  is moving along the  $Ox$  axis under the force  $\mathbf{F}(x) = mKxe^{x/a}\mathbf{i}$ , where  $K$  and  $a$  are known positive constants. The initial position of the particle is  $x = a$  and the initial velocity is  $\mathbf{v} = -v_0\mathbf{i}$ . Describe the possible types of motion of the particle as a function of  $v_0$ .

15. The Stokes' law states that the magnitude of the drag force acting on a sphere of radius  $r$  and velocity  $v$  moving on a fluid at rest of dynamic viscosity  $\mu$  is:  $F(v) = 6\pi\mu rv$ . Assuming the density of a vertically-falling sphere is twice the density of the fluid ( $\rho_0$ ), and considering drag, the weight and the buoyancy of the sphere, compute:

- (a) The terminal velocity of the sphere.  
(b) The evolution of the position as a function of time, if the sphere starts from rest.

Remember that the kinematic viscosity is defined as  $\nu = \mu/\rho_0$ .

16. A football has frontal surface area  $S$ , mass  $m$ , and a drag coefficient  $C_D$ . It falls vertically through the air of density  $\rho$ . It is assumed that the forces acting on the ball are only gravity and drag. The drag can be modeled as:  $D = (1/2)\rho v^2 SC_D$ , with  $v$  the speed of the ball. Compute its terminal speed and its position as a function of time. You may find the following integrals useful:

$$\int \frac{dx}{x_0^2 - x^2} = \frac{1}{x_0} \text{atanh} \frac{x}{x_0}; \quad \int \tanh x dx = \ln \cosh x.$$

17. A force attracts a heavy particle  $M$  of mass  $m$  to a point  $O$ . The attractive force is proportional to the mass of the particle and to the distance  $OM$ . The proportionality coefficient is  $k^2$ . The coordinate system  $Oxyz$  has the origin at the centre of attraction  $O$  and the axis  $Oz$  is vertically oriented. Initially, the coordinates of  $M$  are  $(x_0, y_0, z_0) = (\sqrt{3}g/k^2, 0, 0)$  and its initial velocity is  $(\dot{x}_0, \dot{y}_0, \dot{z}_0) = (0, 2g/k, 0)$ .

- (a) Describe the motion of  $M$ , specifying clearly the shape of the path.
  - (b) What is the velocity of the particle as function of  $t$ ?
- 

18. A heavy particle of mass  $m$  is moving without friction on a cone which is described as:

$$z^2 = x^2 + y^2$$

The frame of reference  $Oxyz$  is inertial and  $Oz$  is the ascending vertical. The initial coordinates of the particle are  $(a, 0, a)$  and its velocity is  $\sqrt{ag}\mathbf{j}$ . Reduce the problem to integral equations and determine the maximum and minimum value of  $z$  along the motion.

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19. A heavy particle of mass  $m$  is moving without friction on the paraboloid which is described as:

$$az = r^2$$

where  $a$  is a positive constant and  $r$  is the distance to the  $Oz$  vertical axis. If  $r = a$  initially, and the particle is released with a velocity  $v_0$  tangent to the local parallel, reduce the problem to integrals of the form  $\int[\dots]dr = \int[\dots]dt$ . Determine the maximum and minimum of  $r$  along the motion.

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20. A heavy particle is moving along a smooth circumference of radius  $R$  and center  $O$ . One of the diameters of the circumference is along the local vertical, and the circumference is rotating with constant angular velocity  $\omega$  about it. The position of the particle with respect to the circumference is described by the angle  $\theta$  between the particle and the lowest point of the circumference.

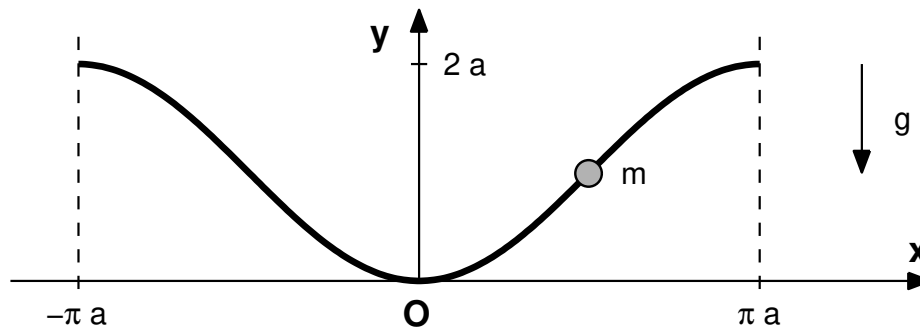
- (a) Discuss the conservation of mechanical energy in the moving reference frame and compute its value.
  - (b) Compute the reaction of the circumference in the direction normal to its plane, as a function of  $\theta$ .
- 

21. On a spinning spherical planet of radius  $R$  and gravitational acceleration  $g_p$ , the bodies weight twice as much on the poles as on the equator. Determine the duration of the day on that planet.
- 

22. A heavy point particle  $P$  of mass  $m$  is constrained to move without friction along the curve

$$y = a \left( 1 - \cos \left( \frac{x}{a} \right) \right), \quad (4)$$

in the range  $-\pi a \leq x \leq \pi a$ , where  $a$  is a positive constant. Initially, the particle is located at  $x = 0$  with velocity  $v_0$  along the positive  $x$  axis.



- (a) Reduce the problem to integrals.
- (b) If  $v_0 = \sqrt{ga}$  determine the minimum and maximum value of  $x$  along the motion.
- (c) What should be the minimum value of  $v_0$  in order to reach  $x = \pi a$ ?

Let's assume now that the curve containing the point particle rotates around the  $OY$  axis with constant angular velocity  $\omega = \Omega \mathbf{j}$ .

- (d) Define a convenient moving reference frame with origin in  $O$ ,  $OX_1Y_1Z_1$ . Determine the inertia force that acts on the particle in this moving reference frame as a function of  $m, \Omega, x_1$  and  $\dot{x}_1$ , where  $x_1$  represents the position coordinate of  $P$  along  $OX_1$ .
- (e) Reduce the problem to quadratures and determine the value of the reaction force.
- (f) Find out a transcendental equation whose solutions are the equilibrium positions of the point particle in  $OX_1Y_1Z_1$ .
- (g) If  $\Omega$  were a function of time, would it be possible to use the energy equation in the same way as before? Justify your answer.

- 
23. A simple pendulum consists of a mass  $m$  hanging from a string of length  $L$  and fixed at a pivot point  $O$ . The simple pendulum is equivalent to the problem of a heavy particle constrained to move along a smooth circumference. A unilateral constraint is considered, such that the particle move along the interior of the circumference and it might fall under certain conditions. Analyze the motion of the particle when it is pushed from the lowest point of the circumference with velocity  $v_0$ . At what point does the particle leave the circumference? Is this point ever reached?

If the circumference rotates around the vertical axis with a constant angular velocity equal to  $\omega$ , write down the new equations of motion. Compute the locations along the circumference in which the particle is in equilibrium.

24. A heavy particle  $P$  is constrained to move along a circumference and it is attached to a spring as shown in the figure. The mass of the particle is  $m$ , and the radius of the circumference is  $a$ . The other end of the spring is attached to a point of the  $y$  axis with coordinates  $(0, \beta a)$  and the constant of the spring is equal to  $k$ .

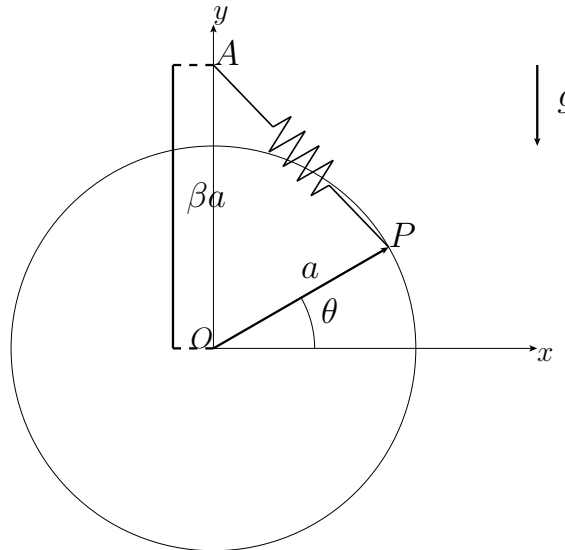
- (a) Compute the potential of the forces acting on the particle as a function of the angle  $\theta$  (shown in the figure).
- (b) Identify the equilibrium positions.

The particle departs from the point  $(a, 0)$  of the circumference with an initial velocity  $\mathbf{v}_0 = v_0 \mathbf{j}$ .

- (c) Reduce the motion to integrals (i.e., quadratures).
- (d) If  $\beta = mg/(ka)$ , obtain the position of the particle as a function of time.

Consider now the situation in which an additional force acts on the particle. The force is proportional to, and in the same direction as, the instantaneous velocity. The constant of proportionality is  $2m\mu$ . The initial conditions are the same as before.

- (e) Obtain the position of the particle as a function of time.
- (f) Obtain the reaction of the circumference.



25. A spacecraft  $S$  is orbiting a celestial body  $F$ . The trajectory of the spacecraft with respect to an inertial reference frame  $Fxy$  is given by

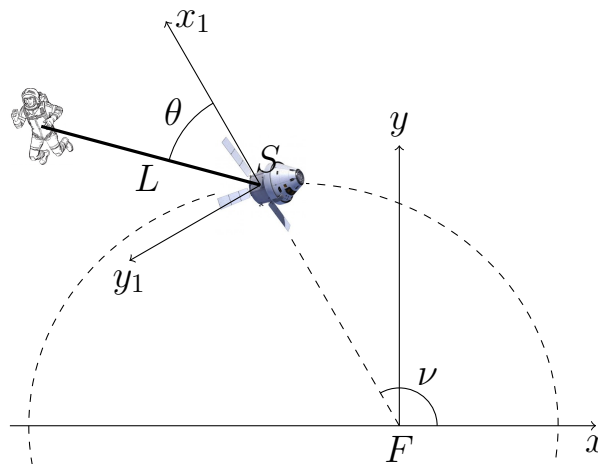
$$r = \frac{p}{1 + e \cos \nu},$$

where  $p$  and  $e$  are known constants and  $\nu(t)$  is a known function of time. The space explorer Ijon Tichy needs to perform some extra-vehicular activities during his Seventh Voyage<sup>1</sup>. He attaches himself with a massless cord of length  $L$  to the spacecraft and goes for a space walk. The cord remains in tension at all times. The astronaut is very light compared to the spacecraft, so his motion does not perturb the motion of the spacecraft. Furthermore, we assume that we can consider him as a point particle of mass  $m$ . We define a moving reference frame  $Sx_1y_1$  as depicted in the figure.

- Determine the acceleration of the spacecraft with respect to  $Fxy$  as a function of time.
- Determine the tangent vector to the trajectory of  $S$  as a function of time.
- Determine the relative velocity and relative acceleration of Ijon Tichy with respect to  $Sx_1y_1$  as a function of  $\theta$  and its time derivatives.
- Determine the fictitious forces acting on Ijon Tichy if we analyze the problem with respect to the frame  $Sx_1y_1$ .

The length  $L$  is very small compared to the distance between  $F$  and  $S$ . This allows us to take an approximation:<sup>2</sup> in the following, assume that the gravitational acceleration on the astronaut and on the spacecraft are exactly the same vector (magnitude and direction) as a function of time.

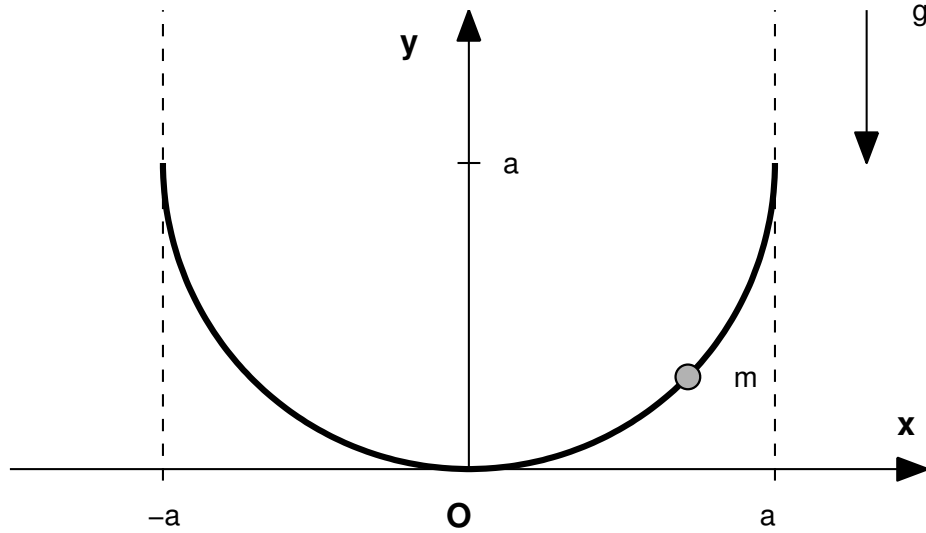
- Write down one ordinary differential equation that once solved (with given initial conditions) would yield the angle  $\theta$  as a function of time (you do not need to solve it)
- Assuming that you solved the previous equation, and therefore knew  $\theta(t)$ , determine the value of the tension on the cord as a function of time.



<sup>1</sup> *The Star Diaries* by Stanislaw Lem.

<sup>2</sup> In actual problems the small differences in gravity at the position of the astronaut and the spacecraft can give rise to small but important effects, and this approximation is usually not taken for that reason.

26. A heavy point particle  $P$  of mass  $m$  is constrained to move without friction along the half-circumference wire of radius  $a$  shown in the figure. The wire containing the point particle rotates around the  $OY$  axis with constant angular velocity  $\omega = \Omega \mathbf{j}$ . Initially, the particle is located at  $x = 0$  with velocity  $v_0$  along the positive  $x$  axis.

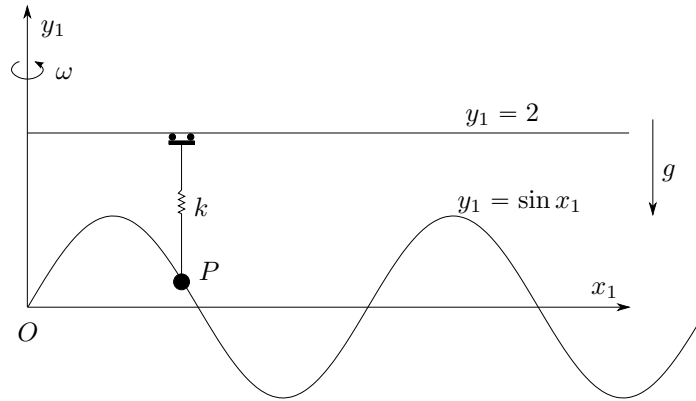


- (a) Determine the number of degrees of freedom of the problem.
- (b) Define a convenient moving reference frame with origin in  $O$ ,  $OX_1Y_1Z_1$ . Express the inertia force that acts on the particle in this moving reference frame.
- (c) Express also the components of the weight of the particle with respect to  $OX_1Y_1Z_1$ .
- (d) Determine if there are any equilibrium points.
- (e) Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.
- (f) Is it possible to use the energy equation? If  $\Omega$  were a function of time, would it be possible to use the energy equation? Justify your answer.

27. A heavy bead  $P$  of mass  $m$  slides without friction along a thin wire with shape given by the following equation:

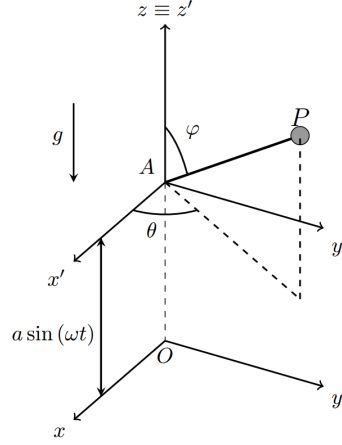
$$y_1(x_1) = \sin x_1.$$

Moreover, the bead is joined to a straight rail located at  $y_1 = 2$  by a spring of stiffness  $k$ . The connection of the spring with the rail can move freely along the rail, so that the spring is always vertical. The whole setup rotates with respect to the  $y_1$  axis with a constant angular velocity  $\omega$ . A sketch of the setup and the definition of the right-handed, rotating  $Ox_1y_1z_1$  reference frame is given below.



- Write down the general expression for the components of the relative velocity and acceleration of  $P$  in the reference frame  $Ox_1y_1z_1$  as functions of the axial position of the bead,  $x_1$ , and its time derivatives  $\dot{x}_1$  and  $\ddot{x}_1$ .
- Identify all forces acting on the bead and then write down the three components of Newton's second law in the  $Ox_1y_1z_1$  reference frame. From these equations, obtain an explicit expression of the components of the reaction force on the wire as a function of  $x_1$ ,  $\dot{x}_1$  and the parameters of the problem only, and a second order differential equation for  $x_1$  that does not depend on the reaction force.
- Is it possible to use the equation of conservation of mechanical energy of the relative motion of  $P$ ? If so, write it down in terms of  $x_1$ ,  $\dot{x}_1$  and the parameters of the problem. What is the relation of this equation with the differential equation for  $x_1$  in the previous point?
- For  $\omega = 0$  and  $x_1 \in [0, 2\pi]$ , discuss the existence of equilibrium positions as a function of the parameter  $\lambda = mg/k$ . Are these equilibrium positions stable?
- Consider now that  $\omega \neq 0$ , and  $k = 0$ . We drop the bead  $P$  at  $x_1 = \pi$  with no initial velocity. Sketch the shape of the potential and discuss qualitatively the motion of  $P$  if the parameter  $\beta = \omega^2/g$  is small or large. Write down the system of equations that would allow to solve for the critical value of  $\beta$  that divides between the two behaviors.

28. A point particle  $P$  of mass  $m$  is attached with a massless cord of length  $L$  to the point  $A$ , as shown in the figure. The point  $A$  is oscillating vertically with respect to the fixed reference frame  $Oxyz$  according to the law  $z = a \sin \omega t$ , where  $a$  and  $\omega$  are known constants. The position of the particle  $P$  with respect to the moving reference frame  $Ax'y'z'$  is completely specified by the angles  $\varphi$  and  $\theta$  shown in the figure.



At  $t = 0$  the relative position and relative velocity of the particle are defined by the values

$$\begin{aligned} \theta_0 &= 0, \quad \varphi_0 = 30^\circ, \\ \dot{\theta}_0 &= \sqrt{\frac{2g}{\sqrt{3}L}}, \quad \dot{\varphi}_0 = 0, \end{aligned}$$

where  $g$  is the gravitational acceleration.

We define the following unit vectors that define a spherical coordinate system

$$\begin{aligned} \mathbf{e}_r &= \sin \varphi \cos \theta \mathbf{i} + \sin \varphi \sin \theta \mathbf{j} + \cos \varphi \mathbf{k}, \\ \mathbf{e}_\varphi &= \cos \varphi \cos \theta \mathbf{i} + \cos \varphi \sin \theta \mathbf{j} - \sin \varphi \mathbf{k}, \\ \mathbf{e}_\theta &= -\sin \theta \mathbf{i} + \cos \theta \mathbf{j}. \end{aligned}$$

The partial derivatives of these vectors with respect to  $\varphi$  and  $\theta$  are

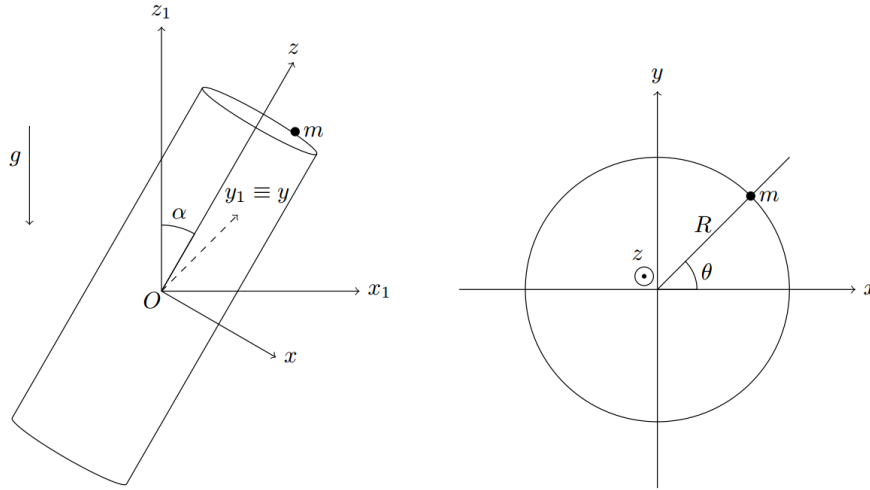
$$\begin{aligned} \frac{\partial \mathbf{e}_r}{\partial \varphi} &= \mathbf{e}_\varphi, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \varphi} = -\mathbf{e}_r, \quad \frac{\partial \mathbf{e}_\theta}{\partial \varphi} = 0, \\ \frac{\partial \mathbf{e}_r}{\partial \theta} &= \sin \varphi \mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \theta} = \cos \varphi \mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\theta}{\partial \theta} = -\sin \varphi \mathbf{e}_r - \cos \varphi \mathbf{e}_\varphi. \end{aligned}$$

- (a) Show that the relative acceleration of  $P$  with respect to  $Ax'y'z'$  can be expressed as

$$\mathbf{a}'_P = -L \left( \ddot{\varphi}^2 + \dot{\theta}^2 \sin^2 \varphi \right) \mathbf{e}_r + L \left( \ddot{\varphi} - \dot{\theta}^2 \sin \varphi \cos \varphi \right) \mathbf{e}_\varphi + L \left( \ddot{\theta} \sin \varphi + 2\dot{\theta} \dot{\varphi} \cos \varphi \right) \mathbf{e}_\theta.$$

- (b) Write down a system of three ordinary differential equations that once solved would yield the angles  $\theta$  and  $\varphi$ , and the tension on the cord as a function of time (you do not need to solve it)
- (c) At  $t = 0$ , determine the relative acceleration of  $P$  with respect to  $Ax'y'z'$ .
- (d) At  $t = 0$ , determine the value of the tension on the cord.

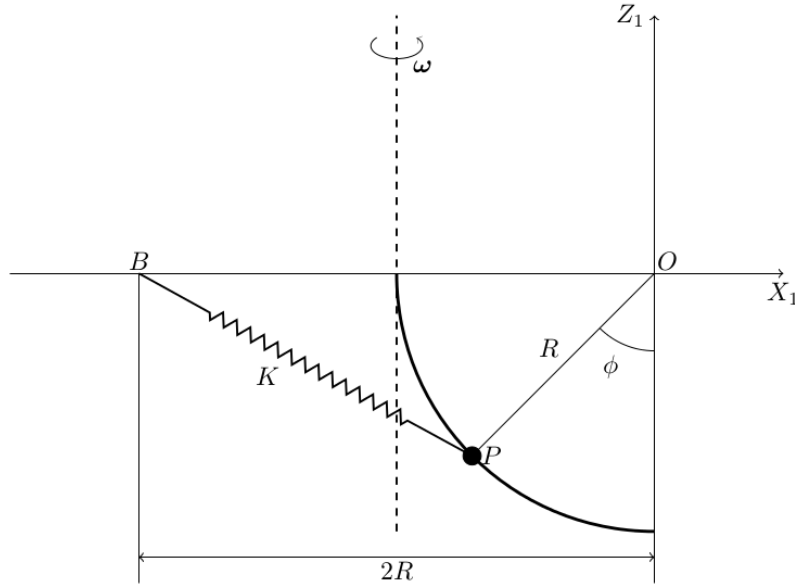
29. A heavy particle of mass  $m$  is constrained to move without friction along the surface of an infinitely-long, circular cylinder of radius  $R$ . The cylinder axis ( $Oz$ ) forms an angle  $\alpha$  with the vertical direction ( $Oz_1$ ). A sketch is shown on the left panel of the figure below. The position of the particle is completely specified by providing the coordinate  $z$  along the axis  $Oz$  and the angle  $\theta$  as shown on the right panel. Initially, the particle is located at  $z = 0$ ,  $\theta = \pi/2$  and the time derivatives are  $\dot{z} = v_0$  and  $\dot{\theta} = \omega_0$ .



- Determine the value of  $z$  as a function of time,  $z(t)$ .
- Reduce the determination of the angle  $\theta(t)$  to integrals of the form  $\int[\dots]d\theta = \int[\dots]dt$ .
- Determine the reaction from the cylinder as a function of  $\theta$ .
- For the particular case,  $\omega_0^2 = g \sin \alpha / R$  describe qualitatively the motion of the particle.
- For the particular case,  $\omega_0^2 = 4g \sin \alpha / R$  describe qualitatively the motion of the particle.

30. A point particle  $P$  of mass  $m$  is constrained to move on a smooth quarter of circumference as shown in the figure below. We define a moving reference frame  $OX_1Y_1Z_1$  that rotates with respect to an inertial reference frame  $OXYZ$ . The rotation of the moving reference frame is performed along a vertical axis defined as  $x_1 = -R$  and  $y_1 = 0$ , the angular velocity is  $\omega = \omega \mathbf{k} = \omega \mathbf{k}_1$ . The particle is attached to a linear spring with the other end attached to the point  $B$  of the reference frame  $OX_1Y_1Z_1$  with coordinates  $(-2R, 0, 0)$ . The constant of the spring is  $K = \alpha m \omega^2$ . The angle between  $\overline{OP}$  and the negative side of the  $z_1$  axis,  $\phi$ , is used to define the position of the particle. Neglect the action of the gravity on the particle.

Initially the particle is located at  $\phi = 0$  and it is set to motion with a relative velocity  $-\beta \omega R \mathbf{i}_1$ . Both  $\alpha$  and  $\beta$  are positive nondimensional parameters.



- Compute the relative velocity and the relative acceleration of the particle in the path variable reference frame  $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$  as functions of the coordinate  $\phi$  and their derivatives.
- Write down the three components of Newton's second law expressed in the path variables reference frame  $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ .
- Express the energy conservation equation in the relative reference frame. Discuss if this equation is equivalent to one of the above ones.

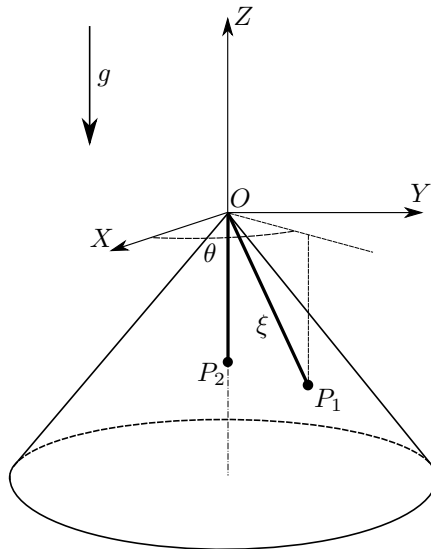
Let us now consider a different problem. Depending on the values of the parameters, we may find the equilibrium of the forces at different positions.

- Obtain the value of  $\alpha$  to have an equilibrium position when the particle is positioned at  $\phi$  equal to  $\frac{\pi}{6}$ .

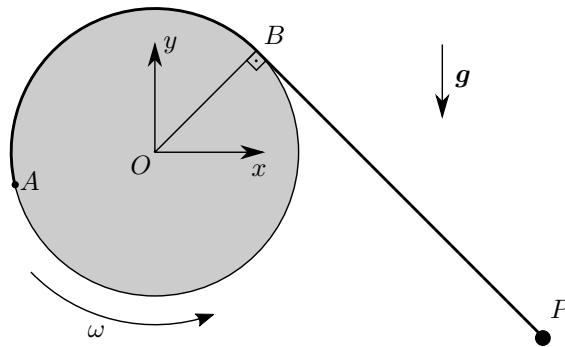
For this value of  $\alpha$ , the particle is again located at  $\phi = 0$  and is set to motion with the relative velocity defined before.

- Describe the possible types of motion for the different values of  $\beta$ . In particular, obtain the critical value of  $\beta$  that makes the particle reach the position  $\phi = \frac{\pi}{2}$  and the reaction force of the curve to the particle at that instant.
- For a value of  $\beta$  higher than the critical one, the particle leaves the arc of circumference and its motion is not anymore constrained. Under these conditions compute the maximum height reached by the particle,  $z_{max}$ , as a function of  $\beta$ . Note that the particle is still under the action of the spring.

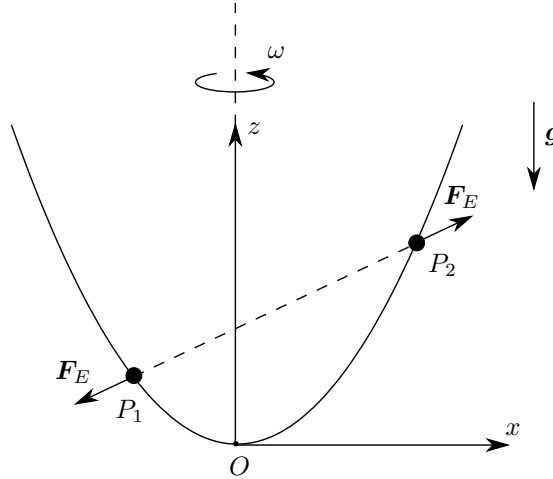
31. Two heavy point particles  $P_1$  and  $P_2$  of mass  $m$  each are connected by an inextensible, massless string of total length  $\ell$ . Particle  $P_1$  is constrained to move without friction over the surface of a cone of semiangle  $\alpha$  (the particle cannot leave the surface of the cone). The string passes through an aperture at the cone vertex  $O$ , and it can slide through it without friction. Particle  $P_2$  hangs along the cone axis, and is constrained to move along it. We define an inertial reference frame  $S_0$  as shown in the figure. We call  $\xi$  the length of the fraction of string between  $O$  and  $P_1$ , and  $\theta$  the angle formed between the  $Ox$  axis and the projection onto the  $Oxy$  plane of the  $OP_1$  vector. In the following, assume that the string is always tense and that the particles do not reach point  $O$  during motion.
- Write down the general expression of the position, velocity and acceleration vectors of  $P_1$  and  $P_2$  as a function of  $\theta$ ,  $\xi$  and the parameters of the problem
  - Write down the expression of all forces that act on  $P_1$  and  $P_2$
  - Write down the differential equations that, if solved, would yield the time evolution of  $\theta$ ,  $\xi$ , the tension on the string and any other reaction force in the problem
  - What is the mathematical condition that the function  $\xi(t)$  must satisfy for the string to remain tense all the time?
  - How fast should  $P_1$  rotate in the  $\theta$  direction to have a circular trajectory over a cone parallel? What is the value of the string tension and any other reaction force in the problem in that case? Give your answer as a function of  $\xi$  and the parameters of the problem
  - Taking these equations as the starting point (or based on the physical considerations of the problem) write down two first-integrals of the problem that do not depend on the string tension or any other reaction force in the problem



32. A heavy particle  $P$  of mass  $m$  is connected by a string of total length  $l = 4\pi a$  to a cylindrical spool of radius  $a$  and center at  $O$  as shown in the figure. The spool rotates at a known constant angular rate  $\omega$  counterclockwise. The string is massless, inelastic, thin, and remains always in tension during the motion of the particle. We define the inertial reference frame  $Oxy$  as shown in the figure. The other end of the string is attached to the spool at a point  $A$  of its perimeter, and we shall call  $B$  the instantaneous point of tangency between the straight segment of the string (the segment  $BP$ ) and the spool. We will denote the angle between the local horizontal (axis  $Ox$ ) and the radius  $OB$  as  $\phi$ . Initially,  $\mathbf{OA} = -a\mathbf{i}$ , and  $B$  never reaches  $A$ .
- Discuss the number of degrees of freedom of the motion of the particle.
  - Write down the position, velocity and acceleration vectors of  $P$  with respect to  $Oxy$  as a function of the constants of the problem, time, the angle  $\phi$ , and its derivatives.
  - Write down the mathematical expression of the forces acting on  $P$ .
  - Is the mechanical energy of  $P$  conserved? If so, write down the equation of conservation of mechanical energy; otherwise, explain why it is not conserved.
  - Write down a closed set of equations that, if solved, would yield  $\phi$  (and the other unknowns of the problem, if any) as a function of time. Note that you do not need to solve the equations.
  - Consider now that  $\omega = 0$ . Can you write down a first integral of motion that only involves  $\phi$  and  $\dot{\phi}$  (but not  $\ddot{\phi}$ ) in this case? You may want to rethink your previous answers for this particular case.



33. Two heavy point particles,  $P_1$  and  $P_2$ , of mass  $m$  each, slide without friction along a rigid wire parabola given by equation  $z = x^2/(2L)$ , where  $L$  is a known length. The wire rotates about its axis with a fixed, known angular velocity  $\omega$ . The two particles repel each other electrically with a force of magnitude  $F_E = k/d^2$ , where  $k$  is a known constant and  $d$  is the distance between the particles. We define the rotating reference frame  $Oxyz$  attached to the parabola as shown in the figure below.

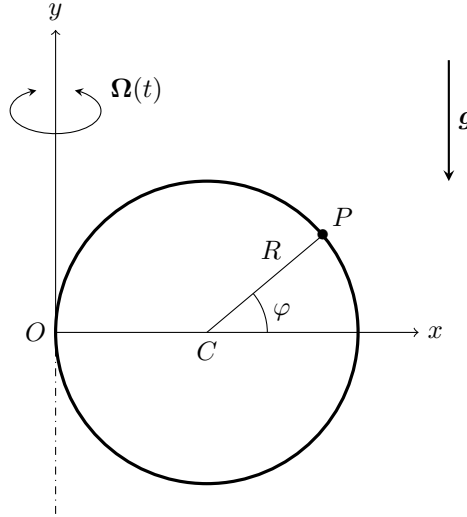


- Write down the position, velocity, and acceleration vectors of each particle relative to  $Oxyz$  in a generic instant of time, as a function of  $x_1$ ,  $x_2$  (the position of  $P_1$  and  $P_2$  along the  $Ox$  axis) and their derivatives.
- Write down Newton's second law for each particle in the  $Oxyz$  reference frame.
- If the rotation of the wire is maintained by a motor, how much torque must it produce in the  $Oz$  direction to keep  $\omega$  constant as a function of the variables of the problem and their derivatives?
- Find out the equilibrium configurations of the system in the reference frame  $Oxyz$  as a function of the parameters  $\gamma = g/(\omega^2 L)$  and  $\kappa = k/(m\omega^2 L^3)$ .

Consider now that  $P_1$  and  $P_2$  are constrained to move symmetrically on each branch of the parabola, and that  $\gamma = \kappa = 2$ . Initially,  $x_1(0) = L$  and  $v_1(0) = L\omega$  (with  $\dot{x}_1(0) > 0$ ).

- Write down the equation that gives the minimum and maximum values of  $x_1$  occurring in this case. Hint: this equation is a third-order polynomial in  $x_1$ .

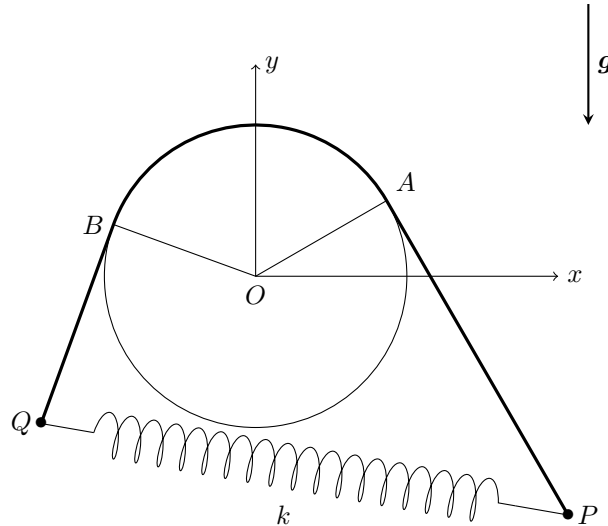
34. A heavy point particle  $P$  of mass  $m$  slides without friction on a circular wire of center  $C$  and radius  $R$ . The wire is located on a vertical plane  $Oxy$  that rotates about the vertical axis  $Oy$  with a prescribed angular frequency  $\boldsymbol{\Omega}(t) = A\omega \sin(\omega t)\mathbf{j}$ , where  $A$  and  $\omega$  are known positive constants, as shown in the figure. The angle that the vector  $\mathbf{CP}$  makes with the  $Ox$  axis is denoted  $\varphi$ .



Answer the following questions in a reasoned way:

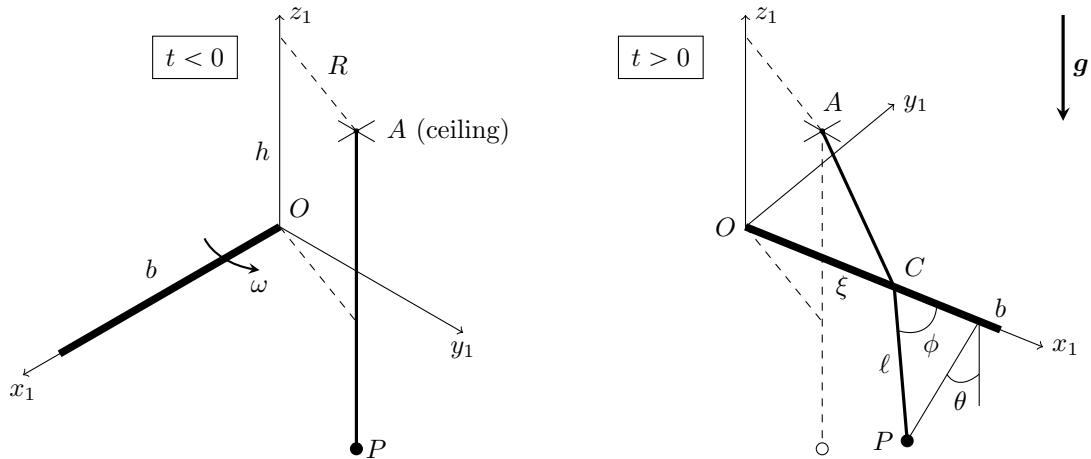
- Write down the position, velocity and acceleration vectors of  $P$  relative to the reference frame  $Oxyz$ , as a function of  $\varphi$  and its derivatives.
- Identify the forces that act on  $P$ , and write down their mathematical expression for a generic instant of time  $t$ .
- Is the mechanical energy of  $P$  conserved as seen from the  $Oxy$  reference frame? Why?
- Write down a single second order differential equation to calculate  $\varphi$  as a function of time. Note that you do not need to solve this equation.
- Compute the reaction of the wire on the particle  $P$  as a function of  $\varphi$ ,  $\dot{\varphi}$  and  $t$ .
- Assume now that  $A^2 \ll g/(R\omega^2)$  and that the motion of  $P$  is such that  $\varphi = \delta - \pi/2$  with  $\delta \ll 1$ . Around what value of  $\delta$  does the particle oscillate? Are there any values of  $\omega$  for which the approximation  $\delta \ll 1$  will not hold?

35. Two heavy particles  $P$  and  $Q$  of mass  $m$  each are connected by an inextensible massless string of constant length  $L$ . Part of this string rests on top of a cylinder of radius  $R$  and center at  $O$ . The string can slide over the cylinder without friction, and remains tense at all times. Furthermore,  $P$  and  $Q$  are also connected by a spring of zero natural length and stiffness  $k$ . We define the inertial reference frame  $S_0$  as shown in the figure. We denote as  $A$  and  $B$  the points of tangency of the string with the cylinder.



- How many degrees of freedom does this system have? Define your own generalized coordinates to parametrize the configuration of the system. You will use them in the rest of the problem.
- Write down the general expression of the position, velocity and acceleration vectors of  $P$  and  $Q$  relative to the  $S_0$  reference frame as a function of your chosen coordinates and their time derivatives.
- Write down the expression of all forces acting on  $P$  and  $Q$ .
- Discuss whether the mechanical energy of  $P$  is conserved; if so, write down the expression of the corresponding potential energy.
- Consider now that we restrict the motion of the system to symmetric configurations about the  $Oy$  axis only. What is the number of degrees of freedom now? Find the mathematical relations that exist between your previously-defined coordinates in this case.
- Find the approximate value of your coordinates when the system is in equilibrium and the configuration is symmetric about the  $Oy$  axis. To do so, assume that the deviation from the vertical of the  $AP$  and  $BQ$  segments is small, and linearize the equations of motion. Find the angular frequency of the oscillations around this equilibrium configuration.

36. A heavy point particle  $P$  with mass  $m$  hanging from a fixed point  $A$  on the ceiling with a massless string of constant length  $L$ . The particle is initially at rest. A bar  $b$  rotates about its center  $O$  in a horizontal plane at a distance  $h$  from the ceiling, with a known, constant angular velocity  $\omega$ . The horizontal distance between  $O$  and  $A$  is  $R$ . At time  $t = 0$ , the bar collides with the string and drags it along its motion. The string slides over the bar without friction. We call  $C$  the instantaneous point of contact of the string with the bar. We define a reference frame  $S_1$  that rotates with the bar as shown in the figure. We call  $\xi$  the distance between  $O$  and  $C$ ;  $\ell$  the length of the string segment between  $C$  and  $P$ ;  $\theta$  the angle between the vertical plane  $Ox_1z_1$  and the plane defined by the bar and  $P$ ; and  $\phi$  the angle formed between the string segment  $CP$  and the bar. You are asked to study only the part of the motion in which the string remains tense and in contact with the bar.

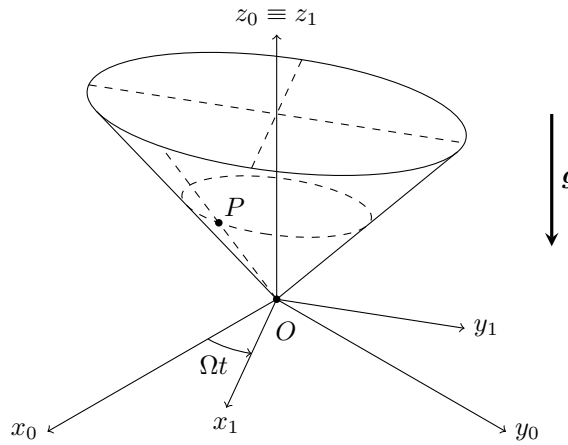


- Write down the position and velocity of point  $A$  in  $S_1$  as a function of time.
- Obtain a general expression for the position, velocity, and acceleration of  $P$  in  $S_1$  as a function of  $\xi$ ,  $\ell$ ,  $\theta$ ,  $\phi$ , their derivatives, and time.
- Express  $\mathbf{AC}$ ,  $\ell$ , and  $\phi$  as a function of  $\xi$  and time.
- Identify and express all the forces acting on  $P$ .
- Write down a set of second order differential equations for the evolution of the unknowns of the problem.
- Compute the initial conditions for the previous equations at  $t = 0$ .
- Is the mechanical energy of the particle conserved in time? Justify your answer.
- If/when the string separates from the bar, express the geometric condition as a relation between  $\theta$  and time.

37. A heavy point particle  $P$  with mass  $m$  moves without friction on a cone of elliptic cross-section as shown in the figure. We define an auxiliary reference frame  $S_1 : \{O; B_1\}$  attached to the cone. The cone is defined in term of two parameters  $\zeta, \beta$  as

$$\text{Elliptic cone: } \begin{cases} x_1 = \zeta \cos \beta \\ y_1 = 2\zeta \sin \beta \\ z_1 = \zeta, \end{cases} \quad (5)$$

The cone spins about its  $Oz_1$  axis at a constant angular rate  $\Omega$  with respect to the inertial reference frame  $S_0 : \{O; B_0\}$ . At time  $t = 0$ , the reference frame  $S_1$  coincides with  $S_0$ , the particle  $P$  is located at  $\mathbf{r}_{1,0}^P = 2\mathbf{j}_1 + \mathbf{k}_1$ , and has a velocity  $\mathbf{v}_{1,0}^P = \mathbf{i}_1$ .

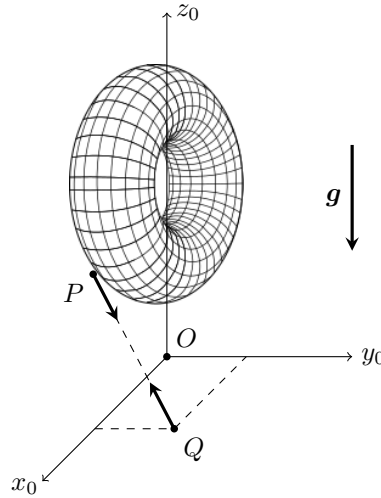


- How many degrees of freedom does the particle motion have?
- Obtain the general expression for the position, velocity, and acceleration of  $P$  **in reference frame**  $S_1$  as a function of  $\zeta, \beta$  and their derivatives. Remember to define any unit vectors that you use in terms of the vectors of  $B_0$ .
- Identify and express mathematically all the forces acting on  $P$  as seen from  $S_1$ .
- Obtain two second-order differential equations for the evolution of  $\zeta, \beta$  in time, which do not depend on any other unknowns in the problem.
- Find, justify, and write down one conservation law in this problem, including the value of any constants there might be.
- Find the equilibrium positions for  $P$  in  $S_1$  for  $\zeta > 0$  as a function of the constants of the problem.

38. A heavy point particle  $P$  with mass  $m^P = 1$  moves without friction on a vertically-standing torus as shown in the figure. The particle cannot separate from the surface of this torus. The torus is defined in the Cartesian coordinates of the inertial reference frame  $S_0$  by the following expressions:

$$\begin{cases} x_0 &= (2 + \cos \phi) \sin \theta \\ y_0 &= \sin \phi \\ z_0 &= 4 + (2 + \cos \phi) \cos \theta \end{cases}$$

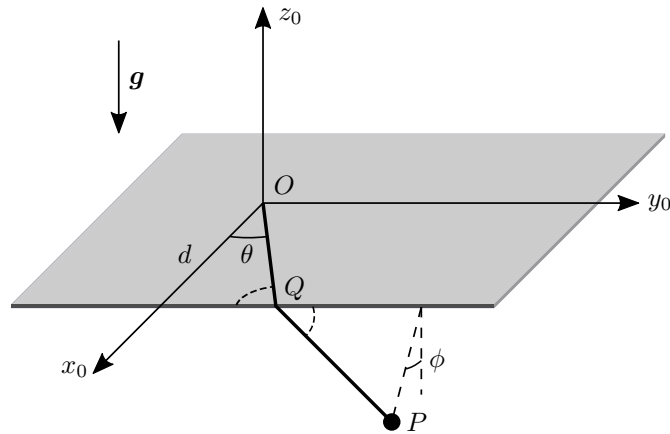
A second particle  $Q$  with mass  $m^Q = 2$  is constrained to move on the  $Ox_0y_0$  plane without friction. The two particles  $P$  and  $Q$  attract each other with a force that is proportional to the distance between them. The proportionality constant of this force is  $k = 2$ . Initially, the particle  $P$  is located at  $\mathbf{r}_0^P(t_0) = 2\mathbf{i}_0 + \mathbf{j}_0 + 4\mathbf{k}_0$  with  $\mathbf{v}_0^P(t_0) = -\mathbf{k}_0$  and the particle  $Q$  is located at  $\mathbf{r}_0^Q(t_0) = 3\mathbf{i}_0 + \mathbf{j}_0$  with  $\mathbf{v}_0^Q(t_0) = -\mathbf{i}_0 + \mathbf{j}_0$ .



- Pick a minimal, valid set of coordinates to describe the configuration of the system. Indicate what is the geometrical meaning of the chosen coordinates.
- Obtain a general expression for the position, velocity, and acceleration vectors of  $P$  and  $Q$  in  $S_0$  using the chosen coordinates and their derivatives.
- Obtain two independent tangent vectors and one normal vector to the torus as a function of the chosen coordinates.
- Identify and express mathematically all the forces acting on  $P$  and  $Q$ .
- Find down a set of second order differential equations to determine the evolution of the chosen coordinates in time, free of any other unknowns.
- Obtain a scalar conservation law.
- Attending to this conservation law alone, can  $P$  ever reach the highest point of the torus?

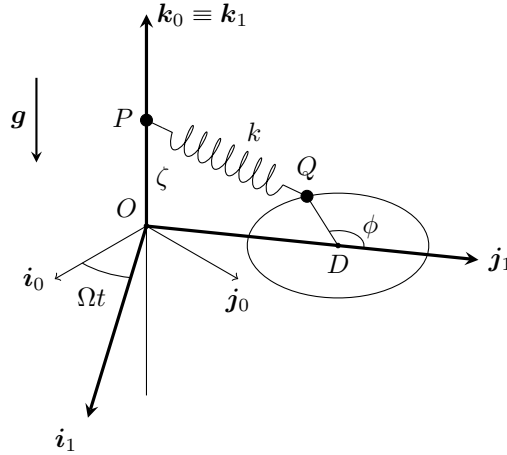
39. A heavy point particle  $P$  with mass  $m$  is connected with point  $O$  on the upper surface of a thin square plate by a massless, inextensible string of length  $\ell$ . The distance between  $O$  and the edge of the plate is  $d$ . The string hangs over the edge of the plate and can slide without friction. You can assume that the string remains tense and in contact with the plate at all times. We denote the geometric point where the string is in contact with the plate edge as  $Q$ . We define the inertial reference frame  $S_0 : \{O, B_0\}$ , with  $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$  as shown in the figure. We define the angle  $\theta$  between the  $Ox_0$  axis and the string segment  $OQ$ , and the angle  $\phi$  between a vertical plane containing the plate edge and the line perpendicular to the plate edge that passes through  $P$ . Initially at time  $t = 0$ ,  $\theta(0) = \phi(0) = \pi/3$  rad, and the modulus of the velocity of  $P$  is  $v_0^P(0) = \sqrt{\ell g}$ , where  $g$  is the magnitude of the acceleration due to gravity.

**Hint:** Note that under the conditions of the problem, the two angles marked with dashed lines are equal.



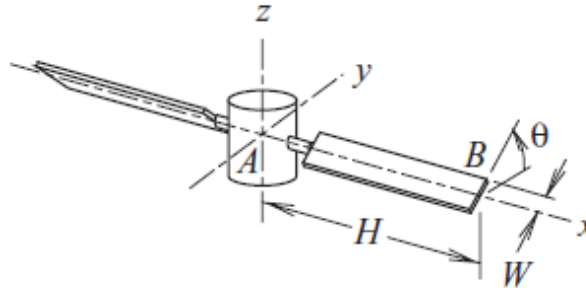
- How many degrees of freedom does the system have?
- Obtain the general expression of a unit vector in the direction from  $Q$  to  $P$  as a function of  $\theta$  and  $\phi$ .
- Obtain the general expression for the position, velocity, and acceleration vectors of  $Q$  and  $P$  in  $S_0$  as a function of  $\theta$ ,  $\phi$ , their derivatives, and the parameters of the problem.
- Identify and express mathematically all the forces acting on  $P$ .
- Find a set of second order differential equations to determine the evolution of  $\theta$  and  $\phi$  in time, free of any other unknowns.
- Does the tension force exert work on  $P$ ? Provide proof of your answer.
- Derive the general expression of the mechanical energy of  $P$  in  $S_0$  and compute its value initially. Is this a conserved quantity of motion? Justify your answer.

40. Two heavy particles  $P$  and  $Q$ , each of mass  $m$ , are connected with a spring of stiffness  $k$  as shown on the figure. An inertial reference frame  $S_0$  is defined with origin  $O$  and vector basis  $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ . Particle  $P$  slides without friction along the vertical axis, while particle  $Q$  slides without friction on a ring of radius  $R$  contained in the horizontal plane defined by  $O$ ,  $\mathbf{i}_0$  and  $\mathbf{j}_0$ . The center  $D$  of this ring is located a distance  $\ell$  from  $O$  and rotates around the vertical axis of  $S_0$  with a known constant angular rate  $\Omega$ . A rotating reference frame  $S_1$  is defined so that the  $\mathbf{j}_1$  vector follows the position of  $D$ . We define the generalized coordinates  $\zeta$  (vertical position of  $P$ ) and  $\phi$  (angle of  $Q$  on the circle from  $\mathbf{j}_1$ ) as shown.



- How many degrees of freedom does the system have? Define the vectors  $\mathbf{i}_1$ ,  $\mathbf{j}_1$  and  $\mathbf{k}_1$  in terms of the vectors of  $B_0$ .
- Write down the general expression of the position, velocity, and acceleration of  $P$  and  $Q$  with respect to  $S_1$  in terms of  $\zeta$ ,  $\phi$  and their derivatives.
- Identify and write down the general expression of all the forces acting on  $P$  and  $Q$ .
- Using  $S_1$  for your analysis, write down the scalar equations of motion of  $P$  and  $Q$ .
- Starting from these equations, write down a system of two differential equations for  $\zeta$  and  $\phi$ , that do not depend on any other unknowns of the problem.
- Find one conservation law of the problem (i.e., a first integral of motion) in reference frame  $S_1$ .

41. The angle  $\theta$  describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis  $x$  is an arbitrary function of time. The satellite spins about the  $z$  axis at the constant angular rate  $\omega$ . Derive expressions for the absolute velocity and acceleration of point  $B$  in the figure relative to an inertial reference frame centered in  $A$ .



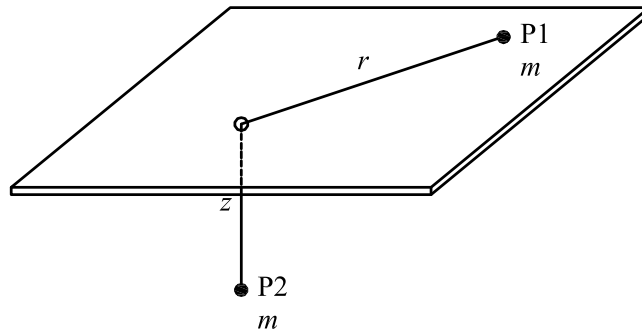
(Taken from Ginsberg, "Engineering Dynamics", Cambridge University Press, 2008)

42. (*Advanced problem:*) A point particle  $P$  of mass  $m$  is attracted by the origin of coordinates  $O$  of the inertial reference frame  $S_0$  by a force with magnitude  $F = m\alpha/r^\beta$ , where  $r$  is the distance between  $O$  and  $P$  and  $\alpha, \beta$  are known constants.
- Show that the angular momentum vector and the mechanical energy of  $P$  are conserved quantities of motion, and that the motion of  $P$  occurs in a plane.
  - Using polar coordinates  $(r, \theta)$  in the plane of motion, find a second-order differential equation to solve for  $u = 1/r$  as a function of  $\theta$  (i.e.,  $u = u(\theta)$ ).
  - Find a first integral of this equation and discuss the limits of motion of  $u$  as a function of  $\alpha/h^2$ ,  $\beta$ , and the value of  $u$  and  $u' = du/d\theta$  at  $t = 0$ , where  $h$  is the magnitude of the angular momentum vector about the origin per unit mass. For  $\beta = 1, 2, 3$ , and  $4$ , determine under which conditions:
    - The particle escapes to infinity ( $u = 0$ ).
    - The particle reaches  $r = 0$  ( $u \rightarrow \infty$ ).
    - The particle oscillates between two values of  $u$ .

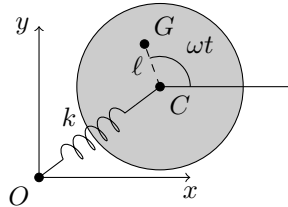
Hint: try to solve first the problem with  $\beta = 2$ , and then try the other values.

43. We consider a smooth horizontal plane with a small hole in it. One of the ends of a massless cord of length  $2a$  is passed through the hole and then, two point particles,  $P_1$  and  $P_2$ , each with the same mass  $m$ , are attached to the ends of the cord. Therefore,  $P_1$  is constrained to move on the plane and  $P_2$  moves along a vertical line below the hole. Initially, half of the cord lies under the plane and half of the cord is on the plane. At that instant,  $P_1$  is pushed with a speed  $v_0$  perpendicular to the cord, counterclockwise as seen from above. Assume that the cord is always in tension.
- (a) What is the value of  $v_0$  if  $P_1$  describes a circumference?
  - (b) In that case, what is the value of the tension in the cord?
  - (c) If  $v_0 = \sqrt{8ag/3}$ , reduce the problem to integrals (i.e., quadratures).
  - (d) Show that for this value of  $v_0$ ,  $P_1$  oscillates between  $r = a$  and  $r = 2a$ .
  - (e) Apply Newton's law to  $P_2$  and show that the tension of the cord is  $T = mg(\frac{1}{2} + \frac{4a^3}{3r^3})$
  - (f) Write down the new equations of motion if the plane on which  $P_1$  moves is rough, with a friction coefficient  $\mu$ .

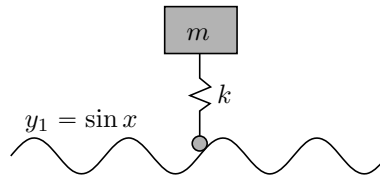
(Hint: the roots of the polynomial  $-3x^3 + 7x^2 - 4$  are  $x_1 = 1$ ,  $x_2 = 2$  and  $x_3 = -2/3$ )



44. The little Prince is on his spherical asteroid, which has radius  $R$  and mass  $M$ , and does not rotate. While he sits at the equator, he throws a baobab seed with an initial velocity of magnitude  $v_0$  and angle  $\gamma = 60^\circ$  with respect to the local horizontal. Knowing that it reaches an apocenter of  $r = 3R$  before falling back, determine  $v_0$  as a function of the parameters of the problem and the universal gravitational constant  $G$ .
45. To understand the vibrations that occur in a turbojet when the mass distribution of the rotor is slightly unbalanced, we can consider the turbojet as a rigid disk of mass  $m$  that rotates at a fixed angular velocity  $\omega$ , with the center of mass  $G$  being separated from the geometric center  $C$  by a fixed distance. An elastic restoring force (with constant  $k$ ) tries to keep  $C$  at its design position  $O$ . We call  $x_0^G, y_0^G$  the Cartesian coordinates of  $G$ , and  $x_0^C, y_0^C$  the Cartesian coordinates of  $C$ . Considering the weight of the disk negligible, write down the equation of the dynamics of  $G$ . Assuming that the homogeneous solution is damped over time and only the particular solution remains, write down the expression of  $x_0^G(t)$  and  $x_0^C(t)$ . Can we affirm that  $|OC| > |OG|$  if  $\omega > \sqrt{k/m}$ ?

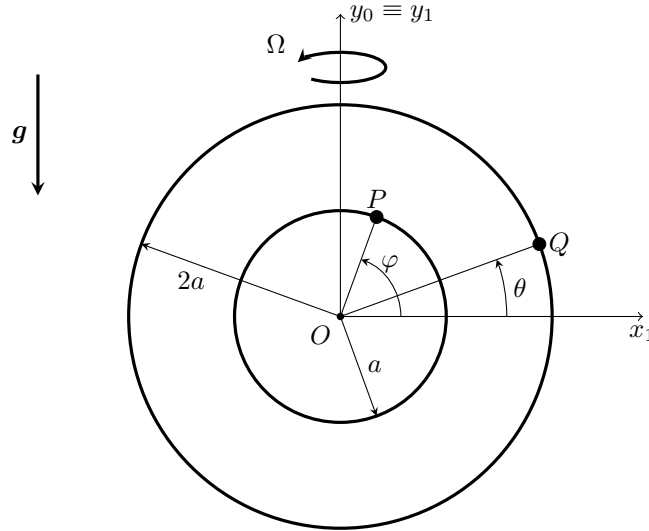


46. An airplane is taxiing over a bumpy runway of vertical profile  $y_1 = \sin(x)$  with a horizontal velocity  $c$ . The vertical motion of the airplane can be modeled as that of a heavy particle  $P$  of mass  $m$  supported on a landing gear of nominal length  $L$  when unloaded. Consider that the landing gear has stiffness constant  $k$  and negligible damping.



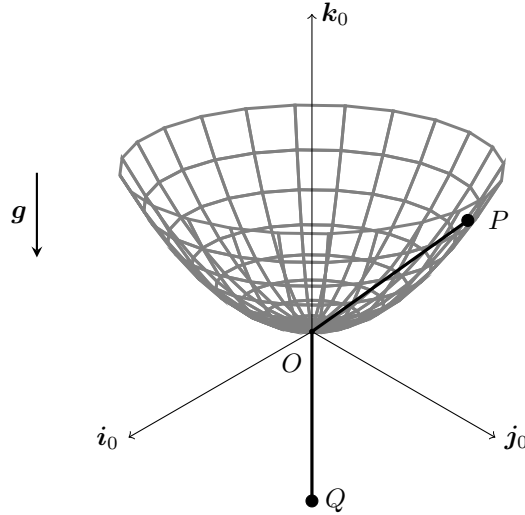
- At what taxiing speed is the system in resonance with the bumpy road?
- Find the expression of the amplitude of the airplane oscillations when the taxiing velocity  $c$  is very high. Hint: the expression depends on  $c$ .
- Consider now that the landing gear has damping constant  $\lambda$ . How does the results above change? Consider the underdamped, critical, and overdamped cases. What value of  $\lambda$  would you choose for this application and why? If the mass of the aircraft changes (e.g. more or less payload), does this affect your conclusion?

47. We consider a system of two heavy particles of equal mass  $m$ . The first particle  $P$  is constrained to move without friction along a circular wire of radius  $a$ . The second particle  $Q$  is constrained to move without friction along a circular wire of radius  $2a$ . Both wires are contained in a plane  $Ox_1y_1$  that rotates at a constant, known angular rate  $\Omega$  about the axis  $Oy_1 \equiv Oy_0$  with respect to a fixed reference frame  $S_0$  with axes  $Ox_0y_0z_0$  (not shown in the figure). We also define a moving reference frame  $S_1$  with axes  $Ox_1y_1z_1$ . The two particles repel each other with a force that is proportional to their distance, with constant of proportionality,  $k$ .



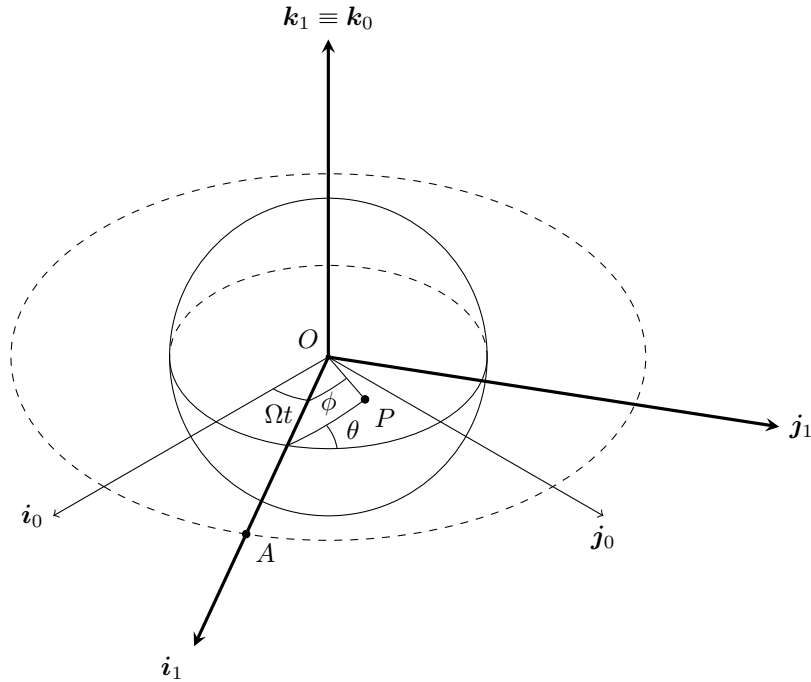
- How many configuration degrees of freedom has this system?
- How many independent components of the reaction forces on the particles need to be considered? Write down the reaction force on each particle.
- Write down the relative accelerations of  $P$  and  $Q$  with respect to  $S_1$  as a function of the coordinates of the problem and their derivatives.
- Express the applied forces on the particles as a function of the coordinates of the problem.
- Express the fictitious forces of inertia on the particles as seen from  $S_1$ , as a function of the coordinates of the problem and their derivatives.
- Write down a system of ordinary differential equations to determine the evolution of the coordinates of the problem as a function of time, free of any other unknowns. You do not need to solve this system (since no initial conditions have been provided) but we assume that we know how to solve it, so that from this point we know the coordinates as a function of time.
- Determine the values of the reaction forces from the wires as a function of time.
- Is the mechanical energy of this system conserved in  $S_1$ ? If so, write down the corresponding equation.
- Determine the values of  $\alpha = \frac{ka}{mg}$  and  $\beta = \frac{a\Omega^2}{g}$  so that  $\varphi = 30^\circ$ ,  $\theta = -60^\circ$  is an equilibrium position of the system in the relative motion.

48. Two heavy particles  $P$  and  $Q$ , each of mass  $m$ , are connected with an inextensible, massless string of length  $\ell$  as shown in the figure. Particle  $P$  is constrained to stay on a paraboloid, given implicitly by equation  $z = cR^2$ , where  $c$  is a known constant and we have introduced the usual cylindrical coordinates  $\{R, \theta, z\}$ . The string passes through a small hole at the origin  $O$ , and particle  $Q$  is restricted to hang along the vertical axis. The reference frame  $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$  is inertial. There is no friction in this problem. Assume that the string is long enough so that  $Q$  never reaches  $O$  and that it is always tense.



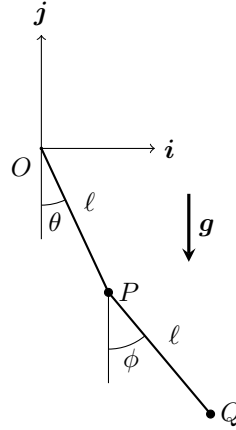
- How many effective degrees of freedom does the system have? justify your answer.
- Give the expressions for the position, velocity, and acceleration of  $P$  and  $Q$  with respect to  $S_0$  as a function of  $R$ ,  $\theta$  and their derivatives. If you use any vector basis other than  $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$  to write your solution, define it first.
- Find a normal vector and two linearly independent tangent vectors to the paraboloid as a function of  $R$ ,  $\theta$ . They do not need to be unitary.
- Identify and write down the general expression of all the forces acting on  $P$  and  $Q$ , introducing new unknowns as needed.
- Find the condition that the solution  $R(t)$  must satisfy so that the string is indeed always in tension.
- Using  $S_0$  for your analysis, write down two second-order differential equations for  $R$  and  $\theta$ , that do not depend on any other unknowns.
- Find two scalar conservation laws (i.e., first integrals).
- Assuming that initially  $R = 1/c$ ,  $\theta = 0$ ,  $\dot{R} = 0$ ,  $\dot{\theta} = d$ , where  $d \neq 0$  is a given value, write down an equation in the form  $f(R) = 0$  whose roots determine the minimum and maximum values of  $R$  in the subsequent motion. You do not need to solve this equation.

49. A point particle  $P$  of mass  $m$  slides without friction on a sphere of radius  $a$  centered at the origin of the inertial reference frame  $S_0$  as shown in the figure. The particle is attracted by a point  $A$  in the equatorial plane, located a distance  $a$  from the surface of the sphere as shown. The force of attraction is linearly proportional the distance between this point and  $P$ , with known constant of proportionality  $k > 0$ . The sphere and the point  $A$  rotate rigidly about the vertical with a known angular velocity  $\Omega \mathbf{k}_0$ . We define the auxiliary reference frame  $S_1$  that rotates with the sphere and point  $A$ , with  $\mathbf{k}_1 = \mathbf{k}_0$  and  $\mathbf{i}_1$  pointing in the direction of  $A$ . We introduce the angles  $\phi$  and  $\theta$  as shown to locate  $P$  on the sphere. Note that there is no gravity in this problem.



- Express the position, velocity, and acceleration of  $P$  with respect to  $S_1$  in terms of  $\phi$ ,  $\theta$  and their derivatives. Remember to define any unit vectors that you want to use in terms of  $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$  first.
- Write down the (vector) equation of motion of  $P$  in  $S_1$ , and derive from there a set of two scalar second-order differential equations for  $\phi$ ,  $\theta$  that do not depend on any other unknowns. To do so, first identify and write down all the forces needed in your analysis.
- Is the mechanical energy of  $P$  conserved in  $S_1$ ? Explain. If so, write down the corresponding conservation law.
- Identify all equilibrium configurations of the system, and discuss them as a function of the ratio  $m\Omega^2/k$ .
- Assuming the functions of time  $\phi(t)$  and  $\theta(t)$  are known, write down the expression of the reaction of the sphere on  $P$ .

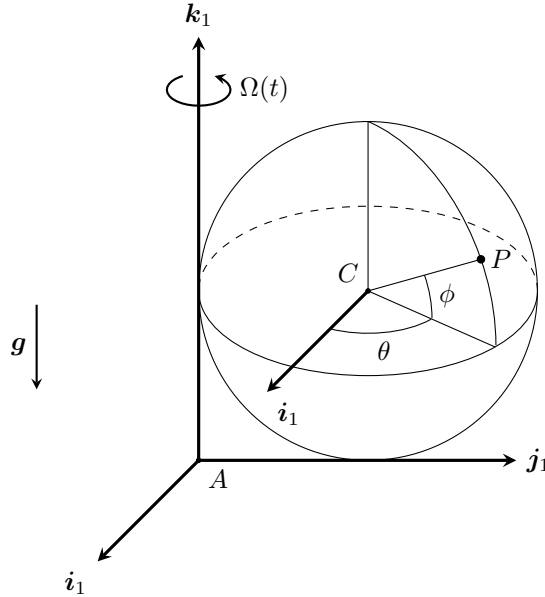
50. A double pendulum consists of two heavy point masses  $P$  and  $Q$ , each of mass  $m$ , connected to the origin  $O$  and to each other by two rigid, massless bars of length  $\ell$  as shown in the figure. We define the inertial frame  $S_0$  with the origin  $O$  and vector basis  $B_0$ . We define the two angles  $\theta$  and  $\phi$  shown in the figure, to describe the configuration of the system.



- Write down the general expressions of the position, velocity, and acceleration vectors of  $P$  and  $Q$ , using  $\theta$ ,  $\phi$  and their derivatives.
- Write down all forces acting on  $P$  and  $Q$ , introducing new unknowns as needed.
- Write down the equations of motion for  $\theta$  and  $\phi$ , free of other unknowns.
- Find and write down one conservation law of this problem.
- If the initial conditions are  $\theta = \phi = \pi/4$ ,  $\dot{\theta} = \dot{\phi} = 0$ , will  $Q$  ever cross the  $Ox$  axis? Justify your answer.
- Consider now that we impose the additional constraint  $\phi = 2\theta$ . Compute the natural frequency of oscillation of the system under the assumption of small angles.

**Hint:** The mechanical energy of an ideal harmonic oscillator of natural frequency  $\omega_0$  and mass  $m$  can be written as  $m(\dot{x}^2 + \omega_0^2 x^2)/2$ .

51. A heavy point particle P of mass  $m$  is constrained to move inside the surface of a hollow sphere of radius  $R$  and center C located at  $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$ . The surface of the sphere is smooth. The reference frame  $S_1$  shown in the figure is a non-inertial reference frame, because it rotates with a known angular velocity  $\boldsymbol{\omega}_{10} = \Omega(t)\mathbf{k}_1$  with respect to inertial space. We introduce the two coordinates  $\theta$  and  $\phi$  as shown in the figure, in order to characterize the configuration of P.



- Give the expressions for the position, velocity, and acceleration vectors of P with respect to  $S_1$ , as functions of  $\theta$ ,  $\phi$  and their derivatives. Remember that you may use any unit vectors in your solution, provided that you define them first.
- Identify and write down all forces (actual and fictitious) on P as seen from  $S_1$ .
- Write down two differential equations for  $\theta$  and  $\phi$ , free of any other unknowns.
- Write down an expression for the reaction unknown. This expression may depend on  $\theta, \dot{\theta}, \phi, \dot{\phi}$  and the known parameters of the problem. What is the mathematical condition for the particle to separate from the surface?
- In the case that  $\Omega = \text{const}$ , determine and discuss the value of  $\theta, \phi$  and the reaction unknown for the equilibrium configurations of P.

## 2 Rigid body exercises and problems

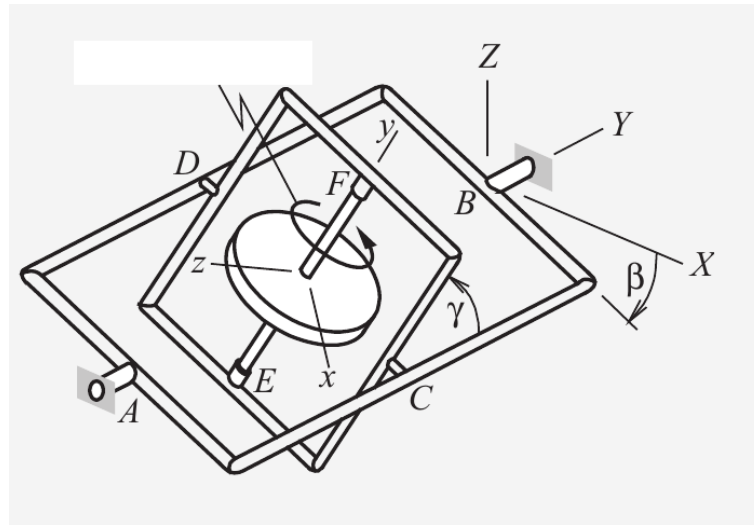
1. A rigid body moves with respect to a reference frame  $Oxyz$ . At a given instant, the velocity of the point  $A(a, 0, 0)$  is  $v_0 \mathbf{j}$ , the velocity of a point  $B(0, a, 0)$  of the rigid body is  $v_0(\mathbf{j} + \mathbf{k})$  and the velocity of a third point of the rigid body,  $C(0, 0, a)$ , is  $v_0(\mathbf{i} + \mathbf{j} + \mathbf{k})$ . Where is the instantaneous axis of rotation? Does the body have a slip velocity at this instant? Compute also the angular velocity of the rigid body at this instant. Find the minimum velocity of all points of the rigid body.

2. A rigid body (body reference frame  $Px_B y_B z_B$ ) is moving with respect a reference frame  $Oxyz$  in a way such that:

- the point  $P$  of the solid is located along  $Oy$
- the axis  $Px_B$  passes through  $A(a, 0, 0)$  (coordinates with respect to  $Oxyz$ )
- the angle  $\theta$  between  $Py_B$  and the plane  $Oxy$  is equal to the angle between  $Ox$  and  $Px_B$

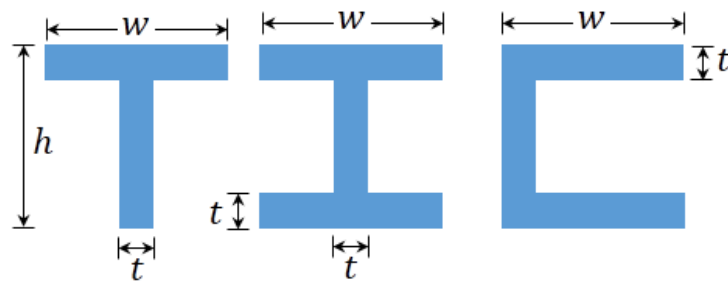
Determine the angular velocity of the body ( $\omega$ ) and the velocity of  $P$  ( $\mathbf{v}_0^P$ ) as a function of  $\theta$  and  $\dot{\theta}$ .

3. A free gyroscope consists of a flywheel that rotates relative to the inner gimbal at the constant angular speed of  $\omega_\theta$ , and the rotation of the inner gimbal relative to the outer gimbal is  $\gamma = A \sin(\omega_\gamma t)$ . The rotation of the outer gimbal is  $\beta = B \sin(\omega_\beta t)$ . Determine the angular velocity and angular acceleration of the flywheel. Express the results in terms of components relative to the body-fixed reference frame  $S_b$  ( $xyz$ ) and with respect to the external reference frame  $S_0$  ( $XYZ$ ), where the  $x$  axis was directed from bearing  $D$  to bearing  $C$  at  $t = 0$ .



4. Consider the following planar shapes (T,I,C shapes). Assuming the figures have uniform surface density  $\sigma$ , calculate:
  - (a) the area of each figure and the position of the center of mass  $G$ . Once you have found it, define axes  $x, y$  with origin on  $G$ .
  - (b) the moment of inertia relative to the  $x$  axis,  $I_{Gx}$ .
  - (c) the moment of inertia relative to the  $y$  axis,  $I_{Gy}$ .

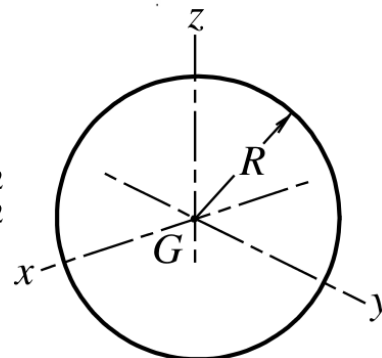
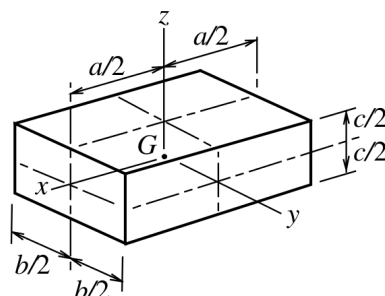
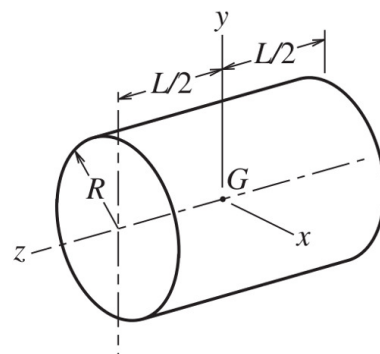
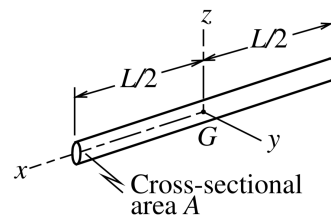
What is the value of the product of inertia in  $P_{xz,yz}$ ? Can you infer its value from the geometrical properties of the shapes?

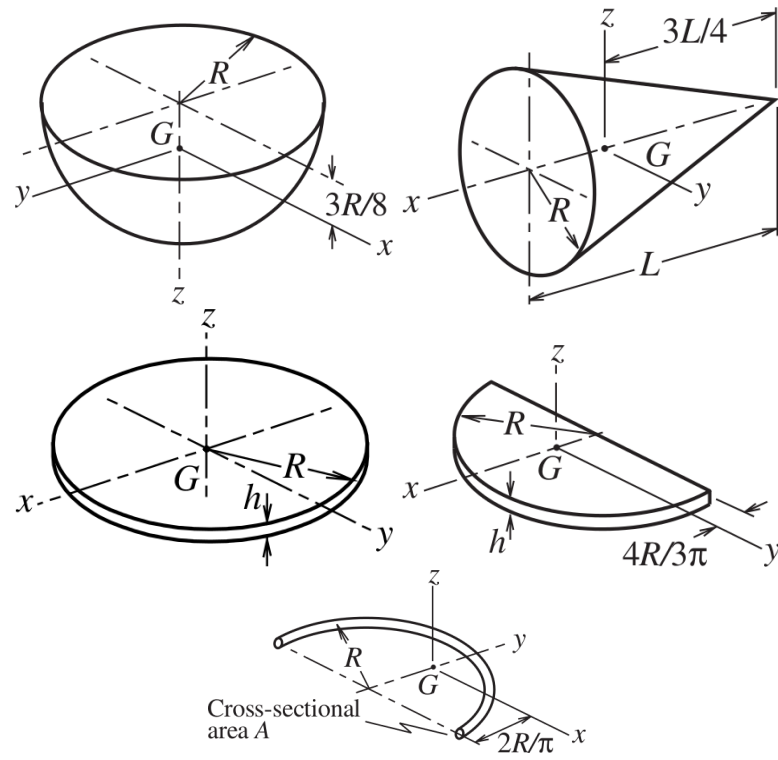



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5. Compute the full three dimensional inertia tensor of the following homogeneous rigid bodies with respect to their center of mass:

- (a) Slender bar
- (b) Cylinder
- (c) Rectangular parallelepiped
- (d) Sphere
- (e) Hemisphere
- (f) Right angular cone
- (g) Thin disk ( $h \ll R$ )
- (h) Semicircular plate ( $h \ll R$ )
- (i) Semicircular wire arc
- (j) A flat sheet of length  $L$  and width  $w$ , with respect a vertex  $A$  and with respect to the center of mass  $G$  (use the Steiner theorem)





6. Los Angeles 2019. Deckard is a special policeman (a blade runner) specialized in “retiring” Nexus 6 replicants. He goes around in a spinner, his flying car. We want to analyze the flying motion of the spinner with respect to an inertial reference frame  $S_0 : \{O; B_0\}$  with  $B_0 : \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$  as shown in the figure. We are going to consider the spinner as a rigid body. A body-fixed reference frame  $S_B : \{G; B_B\}$  with  $B_B : \{\mathbf{i}_B, \mathbf{j}_B, \mathbf{k}_B\}$  is also defined. In the instant considered,

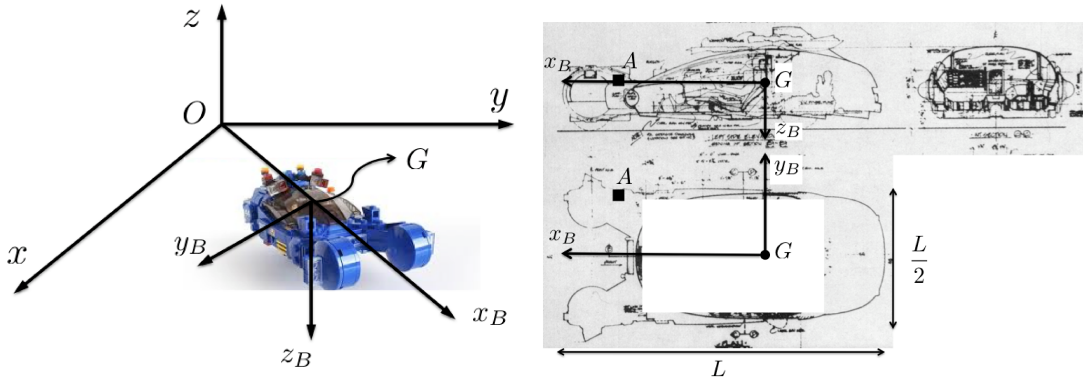
- the center of mass  $G$  and the axes  $Gx_B$  and  $Gy_B$  are contained in the plane  $Oxy$ ,
- the distance between  $G$  and  $O$  is  $3L/2$ , where  $L$  is the length of the spinner as indicated in the figure,
- the angle between the axis  $Ox$  and  $Gx_B$  is  $45^\circ$
- the velocity of  $G$  is  $\mathbf{v}_0^G = 3\sqrt{2}\Omega L(-\mathbf{i} + \mathbf{j})/4$ ,
- the velocity of point  $A$  of the body is  $\mathbf{v}_0^A = \Omega L(\mathbf{i}_B - 8\mathbf{j}_B + \sqrt{2}\mathbf{k}_B)/4$ , where the position vector of  $A$  is given by  $\mathbf{r}_B^A = (2\mathbf{i}_B + \mathbf{j}_B)L/4$ ,
- we also know that  $\mathbf{v}_0^G \cdot \boldsymbol{\omega}_{B0} = 0$ , where  $\boldsymbol{\omega}_{B0}$  is the angular velocity of the spinner.

The mass of the spinner is  $M$ . The inertia tensor about  $G$  is

$$\bar{\bar{I}}_G = ML^2 \begin{bmatrix} B & 0 & E \\ 0 & C & F \\ E & F & D \end{bmatrix}_b,$$

where  $B, C, D, E$  and  $F$  are known constants. Compute

- The instantaneous axis of rotation with respect to  $S_0$ .
- The kinetic energy of the spinner with respect to  $S_0$ .
- The angular momentum of the spinner about  $G$  with respect to  $S_0$ .
- The angular momentum of the spinner about  $A$  with respect to  $S_0$ .



7. We want to analyze the motion of the ring with respect to an inertial reference frame  $S_0 : \{O; B_0\}$  (with unit vectors  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$ ) as shown in the figure. We are going to consider the ring as a rigid body, with a body-fixed reference frame  $S_B : \{G; B_B\}$  (with unit vectors  $\mathbf{i}_B$ ,  $\mathbf{j}_B$  and  $\mathbf{k}_B$ ) attached to it. In the instant considered,

- the ring is rolling without slipping.
- the centre of mass  $G$  and the axes  $Gx_B$  and  $Gz_B$  are contained in the plane  $Oxz$
- the angle between the axis  $Ox$  and  $Gx_B$  is  $45^\circ$
- the contact point of the ring with the plane  $Oxy$  is  $C$ . The coordinates of  $C$  in the body-fixed reference frame are  $R(-1, 0, -19)/20$ .
- the velocity of  $G$  is  $\mathbf{v}_G = \Omega R \left( -\mathbf{i} + \frac{\sqrt{5}}{2} \mathbf{j} - \frac{9}{10} \mathbf{k} \right)$
- the angular velocity fulfills the condition  $\boldsymbol{\omega} \cdot \mathbf{i}_B = \boldsymbol{\omega} \cdot \mathbf{k}_B$

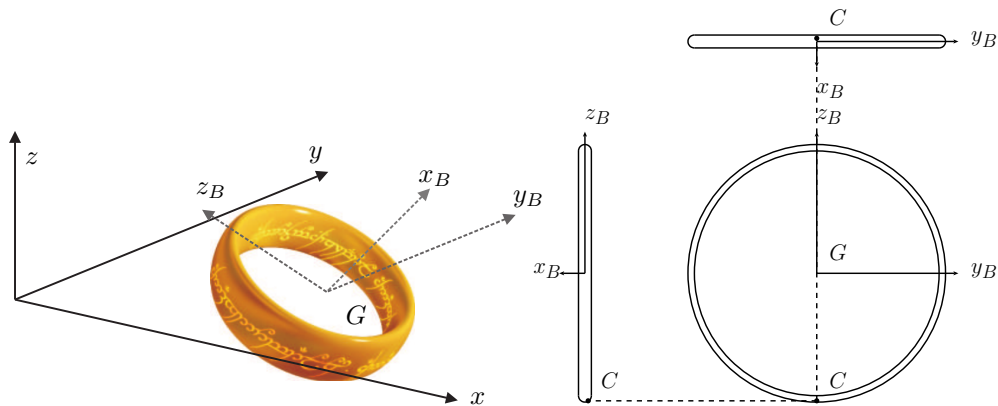
The mass of the ring is  $M$  and the inertia tensor with respect to the reference frame  $S_B$  is approximately given by

$$\bar{\bar{I}}_G = MR^2 \begin{bmatrix} B & 0 & 0 \\ 0 & D & 0 \\ 0 & 0 & D \end{bmatrix}$$

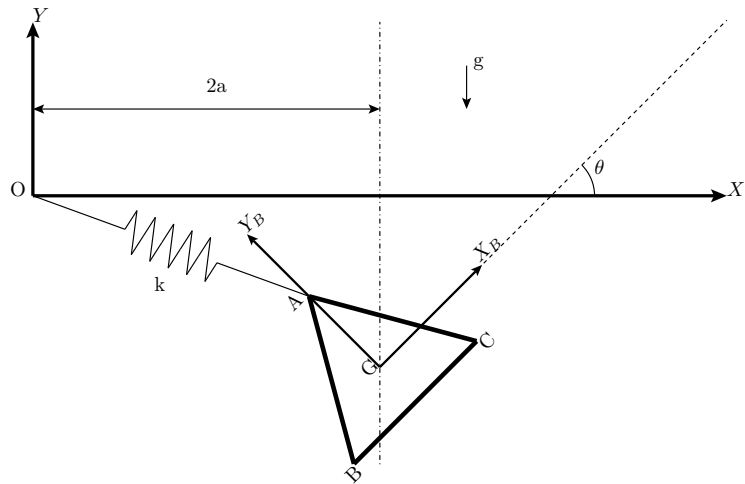
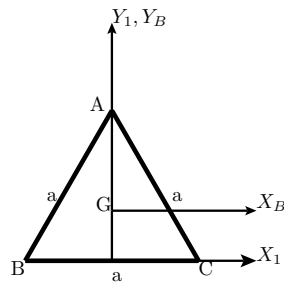
where  $B$ , and  $D$  are considered as constants.

- Compute the instantaneous axis of rotation
- Compute the kinetic energy of the ring and the angular momentum of the ring with respect to  $G$  as seen from  $S_0$
- Assume Mordor is located at the origin and attracts the ring with a force. The force applied at each differential mass is proportional to the differential mass and the distance, with proportionality constant being  $\beta$ . Compute the force and the moment with respect to  $G$  due to the presence of Mordor.
- If we know that  $B = 2D$  and the moment of inertia with respect to  $G$  is  $I_G = MR^2$ , solve for  $B$  and  $D$ .

...One Ring to rule them all

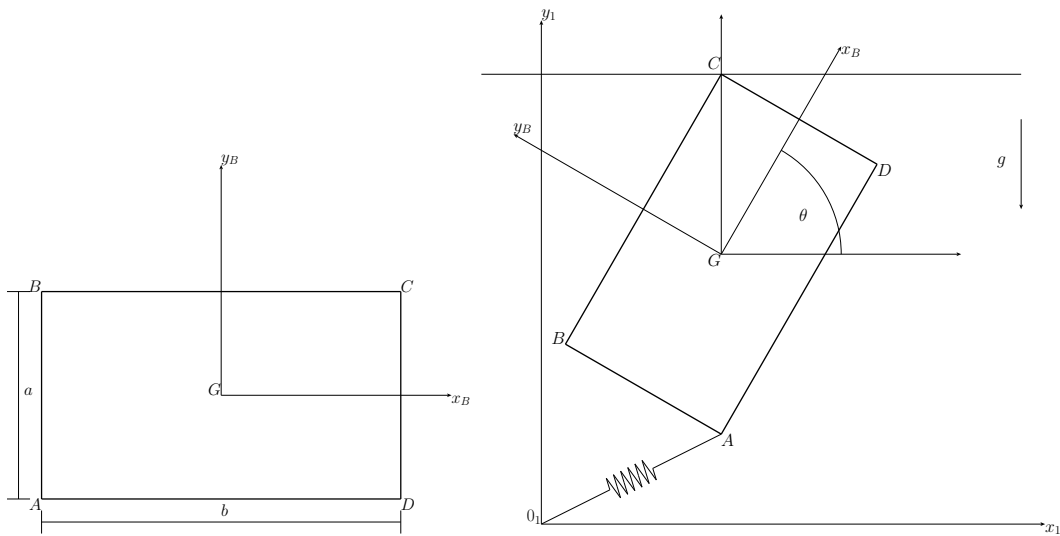


8. The planar equilateral triangle has side length  $a$  and uniform mass  $m$ . It moves in the vertical plane  $Oxy$  of the inertial frame  $S_0 : \{O, B_0\}$  with  $B_0 : \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ . The vertex  $A$  of the triangle is connect to  $O$  by a spring with constant  $k$  and zero nominal length. The center of mass  $G$  of the triangle is constrained to move along the vertical line  $x = 2a$  without friction.
- Determine the position of  $G$  with respect to the coordinate system  $x_1y_1$ .
  - Determine the moment of inertia with respect to point  $G$ .
  - Write down the equations of motion for the triangle.
  - At time  $t = 0$ , the side  $BC$  of the triangle coincides with the axis  $Ox$ , the vertex  $A$  is on the line  $x = 2a$  with  $y > 0$ , and the velocity of  $A$  is  $\mathbf{v}_0^A = u_0\mathbf{i} + v_0\mathbf{j}$ .
    - Obtain the required initial conditions for the equations of the motion.
    - Determine also the instantaneous axis of rotation at  $t = 0$ .
  - Now, we consider a different situation. The triangle is in the equilibrium position and at rest, with the orientation  $\theta = \pi/4$ . Determine the value of the spring constant  $k$ , the position of  $G$ , and the reaction force.
  - In the equilibrium position computed before, we now fix the position of  $G$  and apply a small perturbation to the orientation of the triangle, such that  $\theta = \pi/4 + \phi$  and  $\phi \ll 1$ . Determine the oscillation frequency of the triangle with respect to the equilibrium position.

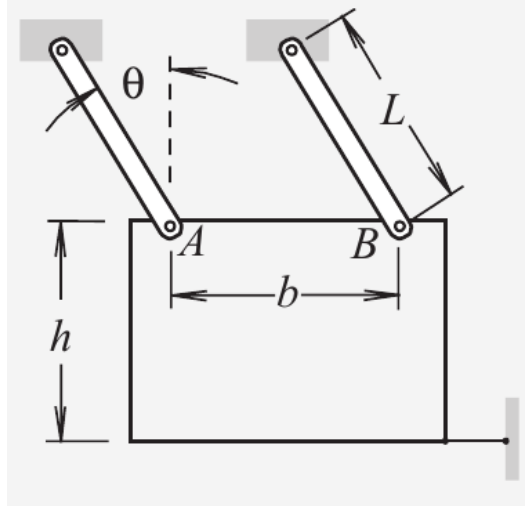


9. We consider the planar motion in a vertical plane  $OXY$  of a heavy homogeneous rectangle of mass  $m$ , length  $a$  and width  $b = \sqrt{3}a$ . The motion of the rectangle is influenced by the spring of constant  $k$  which connects the origin of the inertial reference frame  $O_1X_1Y_1$  to the vertex  $A$  of the rectangle, as shown in the figure. In addition, the vertex  $C$  of the rectangle is constrained to move along the horizontal line  $y = 5a/4$ .
- Determine the moment of inertia with respect to the center of mass of the rectangle  $G$ .
  - If at time  $t = 0$ , the side  $BC$  of the rectangle is located on the axis  $O_1Y_1$ , the vertex  $A$  has a positive  $x_1$  component and the velocity of the vertex  $A$  is  $\mathbf{v}_A = u_0\mathbf{i}_1 + v_0\mathbf{j}_1$  (where  $\mathbf{i}_1, \mathbf{j}_1$  are the unit vectors along the axes  $O_1X_1$  and  $O_1Y_1$ ).
    - Write down the equations of motion for the rectangle.
    - Specify the initial conditions that need to be imposed on the previous equations in order to completely specify the problem. Determine also the instantaneous axis of rotation at  $t = 0$ .
  - If at time  $t = 0$ , the rectangle is at rest, the value of the angle is  $\theta = \pi/3$  and we know that the body is in equilibrium. Determine the position of the centre of mass and the reaction force.
  - In the equilibrium position computed before, we now do the following actions:
    - fix the position of  $G$
    - release the vertex  $C$  from the constraint
    - apply a small perturbation to the orientation of the rectangle. Such that  $\theta = \pi/3 + \phi$  and  $\phi \ll 1$ .

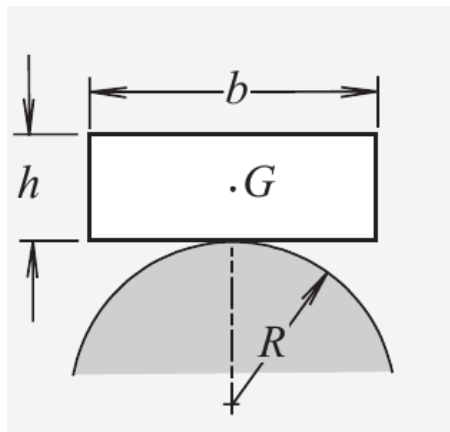
Determine the frequency of oscillation of the rectangle with respect to the equilibrium position.



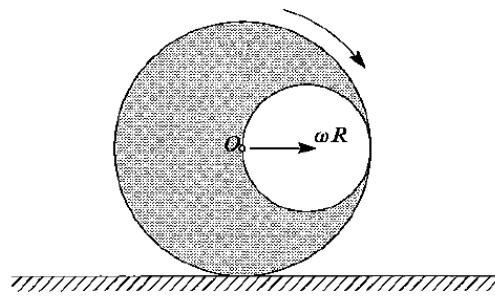
10. A heavy fire door is a rectangular plate with mass  $m$ . The door is suspended from the ceiling by two rigid links with negligible mass. The door is also tied to the wall by a cable. In case of an emergency, this cable is severed and the door swings under the restraint of the rigid links. Derive the first-order differential equation of motion that governs the angle of inclination  $\theta$  of the rigid links. Also derive the expressions for the forces exerted by the links on the plate.



11. A disc with radius  $R$  is rolling (without slipping) on the plane  $Oxy$  of a reference frame  $S_0 : \{O, B_0\}$  with  $B_0 : \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ . The center of the disc describes a circumference with radius  $R$ , in the counterclockwise direction seen from  $Oz$  and with constant velocity  $\omega R$ . This circumference is contained in the plane  $z = R$ , with the center being at  $(0, 0, R)$ . The disc always remains perpendicular to plane  $Oxy$ . The perpendicular axis of the disc always intersects  $Oz$ . Determine:
- (a) Angular velocity and acceleration of the disc with respect to  $S_0$ .
  - (b) Acceleration of the point of the disc that is in contact with  $Oxy$ .
  - (c) Instantaneous axis of rotation.
12. The homogeneous box, whose mass is  $m$ , is placed horizontally on the semicylindrical surface, such that contact is below the centroid  $G$ . Assuming that the box does not slip, derive the differential equation of motion governing the angle  $\varphi$  by which the box rotates away from horizontal.

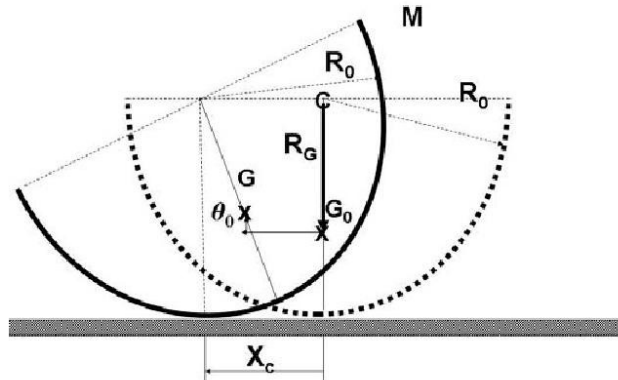


13. A heavy homogeneous sheet is in the form of a disc that has a circular hole with diameter equal to the disc radius  $R$ , as shown in the figure. The mass of the sheet is  $m$ . The sheet is in contact with a horizontal rough straight line with friction coefficient  $\mu$ . Initially, the vector from the disk geometrical center  $O$  to the center of the hole is in the horizontal direction, as shown in the figure. The initial angular velocity of the sheet is  $\omega$  and the initial velocity of  $O$  is  $\omega R$ . In the subsequent motion, the sheet can roll with or without slip or even can lose contact with the line (assuming unilateral constraint).
- Determine the position of the center of mass.
  - Determine the moment of inertia about  $O$ .
  - Determine the moment of inertia about the center of mass.
  - Determine the initial velocity of the disk point that is in contact with the line.
  - Reduce the motion to quadratures (i.e., integrals) for  $\mu = 0$ .
  - Reduce the motion to quadratures for  $\mu \rightarrow \infty$ .

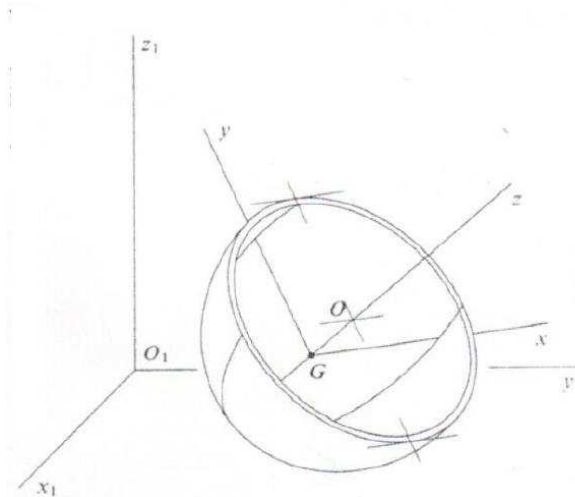


(From Prieto's book)

14. A thin 2D semi-circular arc of mass  $M$  and radius  $R_0$  rests on a flat plane. It is displaced through an angle  $\theta_0$  and held at rest. It maintains rolling contact with the plane. (i.e. it rolls without slipping.). The location of the center of mass for the arc is  $R_G = 2R_0/\pi$ .
- What is the moment of inertia of the semicircular arc about its center,  $I_C$  ?
  - Using the parallel axis theorem, find the moment of inertia about the center of mass,  $I_G$ .
  - What is the displacement of the point of contact,  $X_C$ ?
- As it rolls, the center of mass displaces in both the horizontal and vertical directions.
- What is the vertical displacement of the center of mass,  $y_{CM}$  ?
  - What is the horizontal displacement of the center of mass  $x_{CM}$  ?
  - What is the potential energy in the displaced  $\theta = \theta_0$  position with the arc at rest?
- Now the arc is released and returns to and past  $\theta = 0$  in an oscillatory motion.
- What is the total kinetic energy in both translation and rotation as the arc passes through  $\theta = 0$ .



15. An empty homogeneous semi-sphere of mass  $m$ , radius  $a$  and negligible thickness is moving on a smooth horizontal plane. An auxiliary reference frame  $Gxyz$  can be defined in the following way: the  $Gz$  axis is the axis of revolution of the semi-sphere and is oriented towards the edge of the semi-sphere; the  $Gx$  is always parallel to the plane  $O_1x_1y_1$  and perpendicular to  $Gz$ ; and finally  $Gy$  is orthogonal to  $Gz$  and  $Gx$ , constituting a right-handed reference system. The position of the sphere is determined by five parameters: the coordinates  $\xi, \eta$  of the projection of  $G$  on the plane  $O_1x_1y_1$ ; the angle  $\psi$  between  $Gx$  and  $O_1x_1$ ; the angle  $\theta$  between  $Gz$  and  $O_1z_1$ ; and the angle  $\varphi$  of the rotation of the semi-sphere about the  $Gz$  axis.
- (a) Compute the inertia tensor of the semi-sphere with respect to its center  $O$  and a set of axis parallel to  $Gxyz$ . It can be deduced from symmetry considerations, knowing that the moment of inertia of a spherical shell of radius  $R$  and mass  $M$  with respect to an axis that passes through its center is  $I = 2MR^2/3$ .
  - (b) Determine the position of the center of mass of the semi-sphere.
  - (c) Determine the inertia tensor at  $G$
  - (d) Determine the angular velocity of the semi-sphere expressed as a function of  $\psi, \theta, \varphi$  and its time derivatives.
  - (e) Determine the kinetic energy of the semi-sphere as a function of the parameters that determine its position and their time derivatives.
  - (f) Write down and integrate the linear momentum equation (a) with respect to the axis  $O_1x_1$  and  $O_1y_1$
  - (g) Write down and compute the first integral (b) of the angular momentum equation with respect to the  $Gz$  axis.
  - (h) Write down and compute the first integral (c) of the angular momentum equation with respect to the vertical along  $G$ .
  - (i) Write down and get a first integral (d) from the energy equation.



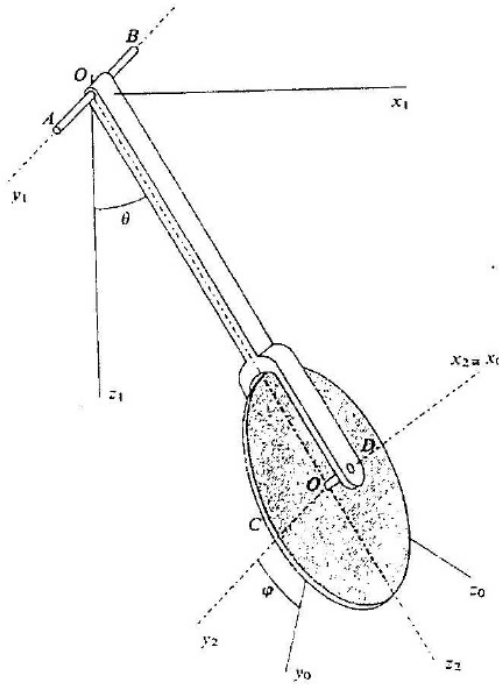
16. A homogeneous sphere of mass  $m$  and radius  $R$ , rolls without slipping on a fixed plane. The angle between this plane and the horizontal plane is equal to  $\alpha$ . The sphere is subject to the action of gravity and an attraction which is directed towards a point of the plane  $O$ , and it is proportional to the mass of the sphere and the distance to  $O$ . Analyze the motion of the sphere and compute the reactions on the plane.

17. The system of the figure comprises a fork and a disc. The fork can turn around the horizontal axis  $AB$ . The disc turns around the axis  $CD$  mounted on the fork perpendicularly to  $AB$ . Thus, the disc always contains the horizontal  $AB$ . The mass of the fork is  $M$ ,  $a$  is the distance from the center of mass of the fork  $G$  to the axis  $AB$ ,  $m$  is the mass of the disc,  $b$  the minimum distance between the axis  $AB$  and  $CD$ ,  $R$  the radius of the disc. The tensor of inertia of the fork in  $O_1$  has the following form:

$$\bar{\mathcal{I}}_{O_1}^F = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}_{22}$$

Assuming that the thickness of the disc is negligible and that gravity  $\mathbf{g} = g\mathbf{k}_1$ ,

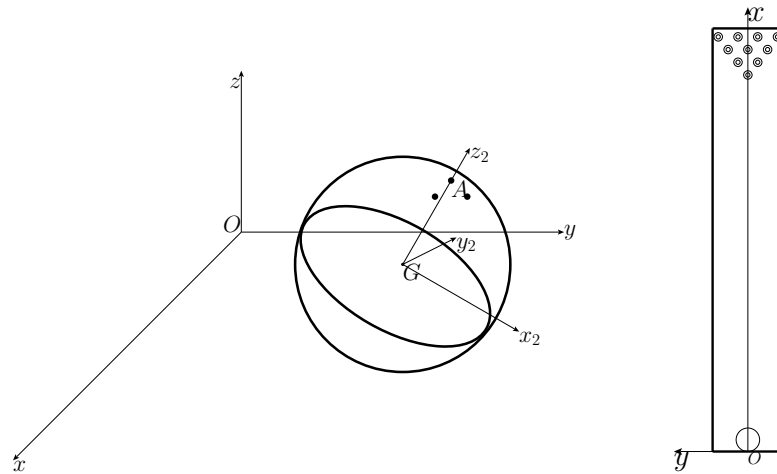
- Compute the kinetic energy of the system fork-disc with respect to frame  $S_1 : \{O_1; B_1\}$ .
- Compute the angular momentum of the disc about  $O$  with respect to  $S_1$ . Express its components using the vector basis  $B_2 : \{\mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$ .
- Compute the angular momentum of the fork-disc about  $O_1$  with respect to  $S_1$ . Express its components using the vector basis  $B_1 : \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ .
- Formulate the equations of motion of the system and reduce them to quadratures (i.e., integrals).
- Determine the resultant force and torque that the axis  $AB$  applies on the fork-disc system at  $O_1$ .
- Determine the resultant force and torque that the axis  $CD$  applies on the disc at  $O$ .



18. A bowling player, Jesús Quintana, is obsessed with the bowling league semifinals. In order to do the best possible throw, he decides to analyze the motion of the bowling ball along the lane. The bowling ball is modeled as a heavy sphere of mass  $M$  and radius  $a$ . The inertial frame  $S_0 : \{O, B_0\}$  with  $B_0 : \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$  is defined as  $Oxyz$  in the figure. The lane contains in the plane  $Oxy$ , with width  $W$ , length  $l$ , and friction coefficient  $\mu$ . The projection of the center of mass  $G$  on  $Oxy$  plane is given by the coordinates  $x_G$  and  $y_G$ . The orientation of the ball is given by three Euler angles  $\psi$ ,  $\theta$ , and  $\phi$ , with the order of rotation being 3 – 2 – 3. After the second simple rotation, the  $Gz_2$  axis of  $S_2 : \{G, B_2\}$  is in the direction from  $G$  to  $A$  ( $A$  is a point on the surface of the ball at which one of the holes is drilled, as shown in the figure).
- For the condition of rolling with slip: determine the degree of freedom and the number of constraints; identify the necessary variables of the problem.
  - For the condition of rolling without slip: determine the degree of freedom and the number of constraints; identify the necessary variables of the problem.
  - Express the angular momentum of the ball about  $G$ , with respect to  $S_0$ , as function of the variables of the problem.
  - Express force and torque as function of the variables of the problem.
  - For rolling with slip, obtain the set of ordinary differential equations and algebraic equations to solve the motion.
  - For rolling without slip, reduce the problem to quadratures (i.e., integrals). What is the shape of the trajectory of  $G$ ?
  - If Jesús Quintana throws the ball with the following initial conditions:

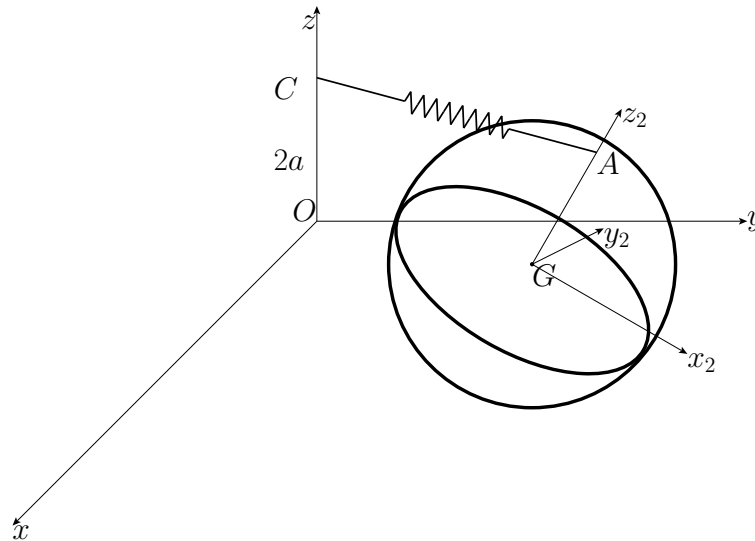
$$\begin{aligned}
 x_G(t=0) &= y_G(t=0) = 0, \\
 \dot{x}_G(t=0) &= \dot{x}_{G0}, \\
 \dot{y}_G(t=0) &= 0, \\
 \psi(t=0) &= \theta(t=0) = \phi(t=0) = 0, \\
 \dot{\psi}(t=0) &= \dot{\phi}(t=0) = 0, \\
 \dot{\theta}(t=0) &= \frac{\dot{x}_{G0}}{4a},
 \end{aligned}$$

solve the whole motion of the ball along the lane and determine: 1) when and where the ball starts to slip; 2) for  $l = \dot{x}_{G0}^2 / (10\mu g)$ , what are the velocity and angular velocity of the ball when it hits the pocket.



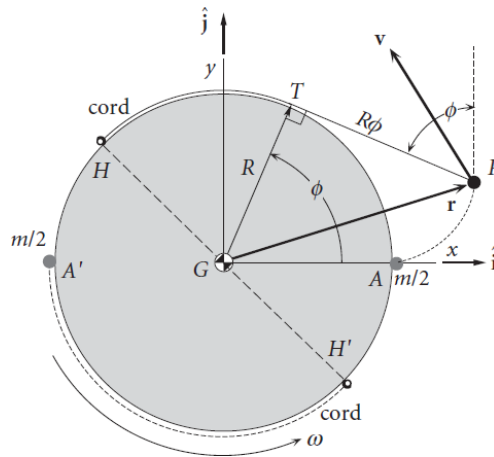
19. The motion of a heavy sphere of mass  $M$  and radius  $a$  is to be analyzed. The sphere rests on a horizontal plane  $Oxy$ . Moreover, a point  $A$  on its surface is attached to a point  $C$  on the vertical axis  $Oz$  by means of a spring with zero natural length and constant  $k$ . The distance  $|OC|$  is equal to  $2a$ . An intermediate reference frame is defined in the following way: axis  $Gz_2$  is the direction of the line connecting the center of mass of the sphere,  $G$ , and  $A$ ; axis  $Gy_2$  perpendicular to both  $Gz_2$  and  $Oz$  and, therefore, parallel to the horizontal plane  $Oxy$ ; and  $Gx_2$  forming a right-handed reference frame. The position of the sphere is given by  $\xi, \eta$ , coordinates of the projection of  $G$  on the horizontal plane  $Oxy$ , angle  $\psi$  between  $Gy_2$  and  $Oy$ , angle  $\theta$  between  $Gz_2$  and  $Oz$ , and angle  $\varphi$  corresponding to the rotation of the sphere about the  $Gz_2$  axis. Note that the difference between the  $Gx_2y_2z_2$  reference frame and a truly body-fixed reference frame is a rotation  $\varphi$  about the  $Gz_2$  axis.

- (a) Considering the horizontal plane a smooth surface:
- express the angular momentum of the sphere with respect to  $G$  as a function of the variables of the problem,
  - express force and torque of the spring as a function of the variables of the problem,
  - obtain the set of five ordinary differential equations to solve the motion,
  - compute also the reaction as a function of the above-mentioned coordinates.
- (b) If the horizontal plane is rough, and assuming that the sphere rolls without slipping, indicate the changes on the set of required equations to solve the motion and the reactions.



20. The Pioneer III lunar probe is an axisymmetric space vehicle, as shown in the figure. The mass of the probe is  $m$ , radius is  $R$ , and moment of inertia with respect to the axis of symmetry is  $I$ . It is launched with an initial constant spin rate  $\omega_0$ . By using a yo-yo with two weights of mass  $m/2$  each, it is desired to reduce the spin rate to  $\omega_f$ . Each mass is connected to the spacecraft by a cord, as shown in the figure for one of the mass. The cords are attached to the spacecraft at  $H$  and  $H'$ . Both masses are released simultaneously by explosive bolts, so that the cords unwrap in the manner shown in the figure. In doing so, the points of tangency,  $T$  and  $T'$ , move around the circumference towards  $H$  and  $H'$  (respectively). When  $T$  and  $T'$  reach the hinges  $H$  and  $H'$ , the cords are automatically released from the spacecraft. The gravitational interaction with all celestial bodies (Sun, Earth, etc.) may be considered negligible.

- Analyze the motion after the release of the yo-yo masses and compute the evolution of the probe's angular velocity as function of time.
- In order to obtain the desired final angular velocity, what is the required length of the cord that joins the yo-yo masses and the probe?
- Compute the evolution of the tension in the cord as function of time.



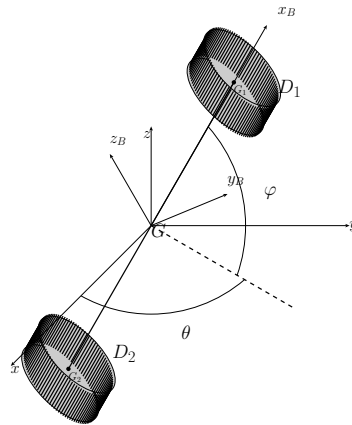
21. A reaction wheel is a type of flywheel used for attitude control of spacecraft. Consider a spacecraft  $b$ , modeled as a homogeneous rigid rod of mass  $m^b = 6M$ , and length  $L$ , and negligible diameter. Two reaction wheels  $w_1$  and  $w_2$  with radius  $R$  and mass  $m^{w_1} = m^{w_2} = M$  are located at the ends of the spacecraft. We define an inertial reference frame  $S_0 = \{O; B_0\}$  (not shown in the figure), and a parallel reference frame  $S_G = \{G; B_0\}$  that translates but does not rotate with the spacecraft. Observe that  $G$  is the center of mass of  $b$  and also the center of mass of the whole assembly. Furthermore, we define a spacecraft-fixed reference frame  $S_b = \{G; B_b\}$ . This reference frame is defined with the axis  $Gx_b$  along the spacecraft in the direction from wheel 2 to wheel 1. The direction of  $Gx_b$  is defined with the angles  $\theta$  and  $\varphi$  as in spherical coordinates, as shown in the figure. The reaction wheels can rotate with respect to the spacecraft  $b$  about the  $Gx_b$  axis, with unknown angular rates  $\dot{\phi}_1$  and  $\dot{\phi}_2$  that may vary in time, respectively. The (degenerate) tensor of inertia of the spacecraft body  $b$  alone (without the wheels) about its center of mass  $G$ , and the tensor of inertia of one wheel  $w_i$  about its center of mass  $G_i$ , expressed in basis  $B_b$ , are:

$$\bar{I}_G^b = \frac{1}{12}m^b L^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_b, \quad \bar{I}_{G_i}^{w_i} = \frac{1}{4}m^{w_i} R^2 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_b, \quad i = 1, 2$$

We assume that the system is in deep space, so all the external forces and torques are negligible. Lastly, consider the following initial conditions ( $\Omega$  is a known constant):

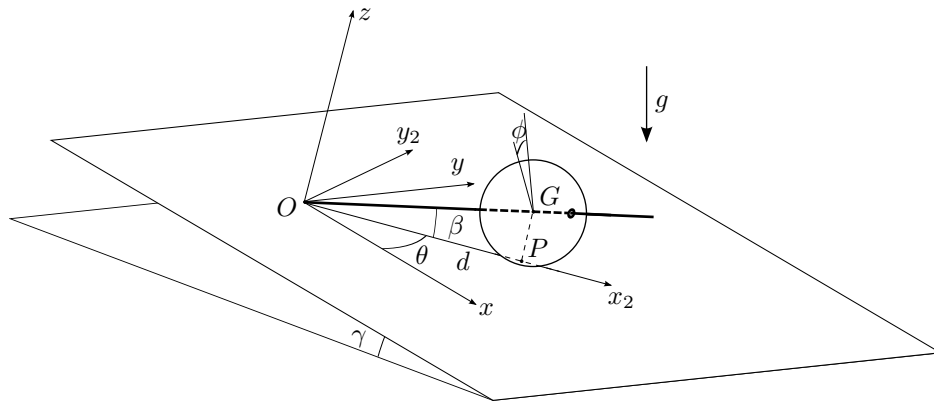
$$\begin{aligned} \mathbf{r}_0^G(t=0) &= \mathbf{0} & \mathbf{v}_0^G(t=0) &= \mathbf{0} \\ \theta(t=0) &= 0 & \varphi(t=0) &= \pi/4 \\ \phi_1(t=0) &= 0 & \phi_2(t=0) &= 0 \\ \boldsymbol{\omega}_{b0}(t=0) &= \frac{\Omega}{\sqrt{2}} \left[ \frac{1}{\sqrt{2}} \mathbf{i}_B + \mathbf{j}_B + \frac{1}{\sqrt{2}} \mathbf{k}_B \right] & \dot{\phi}_1 = \dot{\phi}_2 &= \frac{\Omega}{2} \end{aligned}$$

- Express the angular velocities of the spacecraft and reaction wheels  $\boldsymbol{\omega}_{b0}$ ,  $\boldsymbol{\omega}_{10}$ , and  $\boldsymbol{\omega}_{20}$  as a function of  $\theta$ ,  $\varphi$ ,  $\dot{\theta}$ ,  $\dot{\varphi}$ ,  $\dot{\phi}_1$ ,  $\dot{\phi}_2$ .
- Express the absolute velocities of  $G_1$  and  $G_2$  as a function of the absolute velocity of  $G$ ,  $\theta$ ,  $\varphi$ ,  $\dot{\theta}$  and  $\dot{\varphi}$ .
- Compute the angular momentum of each of the reaction wheels about their respective center of mass ( $G_1$  and  $G_2$ ).
- Compute the angular momentum of the whole system about the center of mass of the whole system.
- Compute the kinetic energy,  $T$ , of the whole assembly.
- Pose the set of ordinary differential equations for the position of  $G$  and for  $\theta$ ,  $\varphi$ ,  $\phi_1$ ,  $\phi_2$ .
- Write down the reaction forces and torques at the joints between the rod and the reaction wheels.



22. A heavy, solid, homogeneous sphere of mass  $M$  and radius  $R$  moves on top of an inclined plane that forms an angle  $\gamma$  with the horizontal. The sphere is always in contact with the plane. A stiff rod of negligible mass traverses the sphere through a small diametrical perforation. The sphere can slide along and rotate about the rod freely, without any friction. The interaction between the rod and the sphere can be reduced to a resultant force and a resultant torque with respect to the center of mass of the sphere,  $G$ . These are unknown reactions until the problem is solved. One end of the rod is fixed at point  $O$  of the inclined plane. The rod is free to turn in any direction but it does not rotate along its longitudinal axis.

To study the motion of the sphere, we define the inertial reference frame  $S\{Oxyz\}$  with origin at  $O$ , the  $x$  axis pointing downward along the maximum slope line of the plane, and the  $z$  axis pointing upward perpendicular to the plane. We also define an auxiliary rotating reference frame  $S_2\{Ox_2y_2z_2\}$ , where  $x_2$  always contains the point  $P$  of contact of the sphere with the plane, and  $z_2 \equiv z$ . In order to describe the state of the system, we will use the angle  $\theta$  between  $x$  and  $x_2$ , the distance  $d = |OP|$ , and the rotation angle  $\phi$  of the sphere about the rod. For convenience, we also define the angle  $\beta$  between the plane and the rod. Note that  $\tan \beta = R/d$ .



You are asked to:

- Choose the minimum number of variables (linear or angular coordinates) in order to fully determine the position and orientation of the sphere.
- Write the general expression for the angular velocity and angular acceleration of the sphere ( $\omega$  and  $\alpha$ ), and the velocity and acceleration of  $G$  ( $v_G$  and  $a_G$ ), as a function of  $\theta$ ,  $d$ ,  $\phi$  and their derivatives.

Consider first that the friction of the sphere with the inclined plane is infinite, so that it cannot slip with respect to the plane.

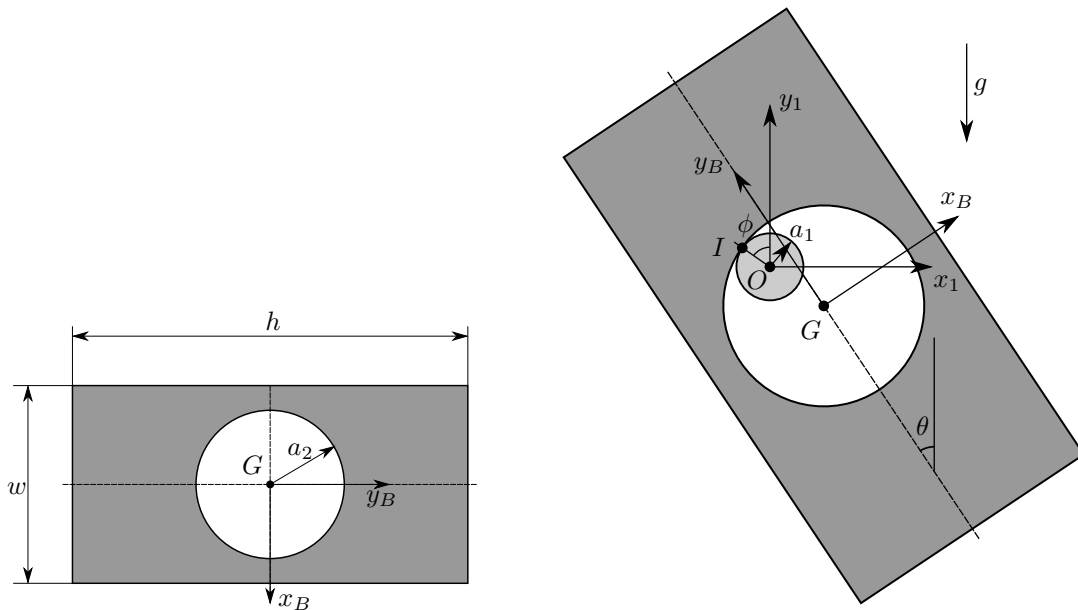
- Relate the velocity of the center of mass  $G$  and the velocity of  $P$ , and use this to express the time derivatives  $\dot{d}$  and  $\dot{\phi}$  in terms of  $\dot{\theta}$  and the angle  $\beta$  formed by the rod with the plane.
- Identify all the components of reaction forces and reaction torques acting on the rod alone. Using the balance of forces and torques on the rod, express all reaction forces and torques as a function of only two force components.
- Identify all the components of forces and torques acting on the sphere, and reduce them to a sum of forces and a sum of torques on  $G$  in terms of the unit vectors of  $S_2$ .
- Obtain a closed set of six differential equations that would provide the evolution of  $\theta$  and the 5 unknowns mentioned in the previous question. Note that you do not need to solve any of these equations.
- If the system is in equilibrium when  $\theta = 0$ , then compute the reaction force between the rod and the sphere. Compute also the normal reaction and the friction force between the sphere and the inclined plane.

- (h) Linearize the equation obtained in the previous question and study the small oscillations about this equilibrium position when the system is released with zero initial velocity and a small nonzero value of  $\theta \ll 1$ . What is the frequency of these oscillations as a function of the parameters of the problem?

Consider now that the inclined plane is frictionless instead:

- (i) What changes in this case? Adapt the differential equations to the new case.  
 (j) If the center of the sphere is initially located at  $\mathbf{r}_{G0} = 2R\mathbf{i} + 3R\mathbf{j} + R\mathbf{k}$  and has an initial velocity  $\mathbf{v}_{G0} = -\sqrt{Rg}(\mathbf{i} + \mathbf{j})$ , derive the initial conditions that are necessary to integrate the differential equations.

23. A heavy rectangular piece of wood of mass  $m$  with a circular hole of radius  $a_2$  is positioned on a fixed cylindrical knob of radius  $a_1$  as shown below. We want to study the motion of the piece as a rigid body with respect to the inertial reference frame  $Ox_1y_1z_1$  of the figure, assuming that the internal surface of the hole is always in contact with the knob at one point  $I$  and that the piece rolls without slip over the knob. We call  $\theta$  the angle between the longitudinal symmetry axis of the piece and the vertical, and we call  $\phi$  the angle between the vector  $OI$  and the vertical. Note that the two angles,  $\theta$  and  $\phi$ , do not need to coincide in general. The dimensions of the problem satisfy  $h = 2w$ ,  $a_2 = w/\sqrt{2\pi}$ ,  $a_1 = a_2/3$ .



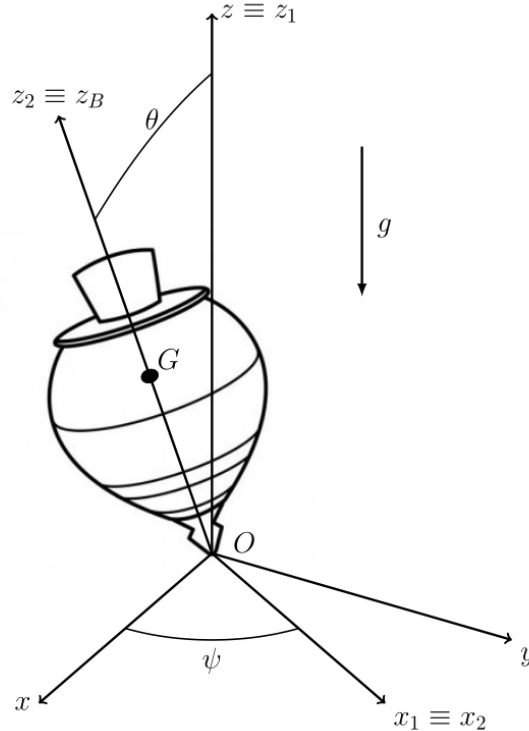
- (a) Compute  $I_z$ , the moment of inertia of the piece along  $z_1$  at  $G$ , as a function of  $m$  and  $w$ .  
 (b) Write down the expressions of the velocity and acceleration of point  $G$  as a function of  $\phi$ ,  $\theta$  and their time derivatives.  
 (c) Find the relation between  $\dot{\phi}$  and  $\dot{\theta}$  imposing the no-slip condition at  $I$ .  
 (d) Write down the equations of motion that yield  $\phi$  and the reaction forces as functions of time. Note that you do not need to solve these equations.  
 (e) From these equations, find the second-order differential equation for the evolution of  $\phi$  that does not depend on the reaction forces.  
 (f) If initially the wood piece is placed such that  $G$  and  $O$  are along the same vertical with  $\theta_0 = 0$ , the speed of  $G$  is  $|\mathbf{v}_{G0}| = 2wA/3$  and the piece is rotating counterclockwise, find the initial conditions that should be imposed to integrate the equation from the previous question for  $\phi$ .

24. In this problem, we are going to study the motion of a heavy, axisymmetric pegtop of mass  $m$ . We will assume that during the precession, nutation and spin of the pegtop, the vertex  $O$  of the pegtop remains fixed at the origin of the inertial reference frame  $Oxyz$ . The distance  $\ell$  between  $O$  and the center of mass of the pegtop  $G$  is known. Also the moment of inertia tensor with respect to the principal axes of inertia  $Gx_B y_B z_B$  is known, and given by

$$\bar{\bar{I}}_G = m\ell^2 \begin{bmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & B \end{bmatrix}.$$

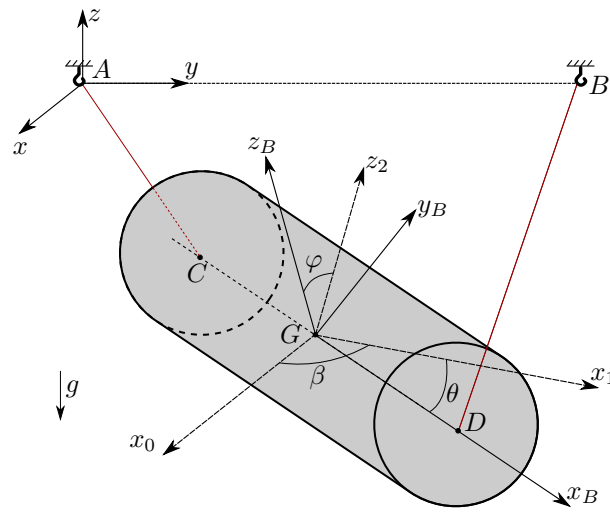
The orientation of the body fixed reference frame  $Gx_B y_B z_B$  with respect to the inertial reference frame  $Oxyz$  is given by three sequential rotations. First, a rotation  $\psi$  with respect to the axis  $Oz$ , (leading to an intermediate reference frame  $Ox_1 y_1 z_1$ ). Second, a rotation  $\theta$  with respect to the axis  $Ox_1$ , (leading to an intermediate reference frame  $Ox_2 y_2 z_2$ ). And finally a rotation  $\varphi$  with respect to the axis  $Gz_2$ , leading to  $Gx_B y_B z_B$ . The initial condition is  $\psi = 0$ ,  $\theta = \pi/6$ ,  $\varphi = 0$ ,  $\dot{\psi} = 0$ ,  $\dot{\theta} = \Omega_1$ ,  $\dot{\varphi} = \Omega_2$ , where  $\Omega_1$  and  $\Omega_2$  are known constants.

- Compute the inertia tensor,  $\bar{\bar{I}}_O$  with respect to a reference frame, parallel to  $Gx_B y_B z_B$  with origin in  $O$ , i.e.  $Ox_B y_B z_B$ .
- Compute the angular momentum of the peg-top with respect to  $O$  as a function of the variables of the problem at an arbitrary instant of time.
- Compute the kinetic energy of the peg-top as a function of the variables of the problem.
- Is the mechanical energy of the system conserved? Why?
- Is there any conserved component of the angular momentum with respect to  $O$ ? Why?
- Obtain three **first order** ordinary differential equations that would allow to obtain the values of  $\psi$ ,  $\theta$  and  $\varphi$  as a function of time.
- Assuming that  $\psi(t)$ ,  $\theta(t)$  and  $\varphi(t)$  are known, explain how you would obtain the value of the reaction at  $O$ . You do not need to obtain the final result, but the procedure needs to be sufficiently clear.



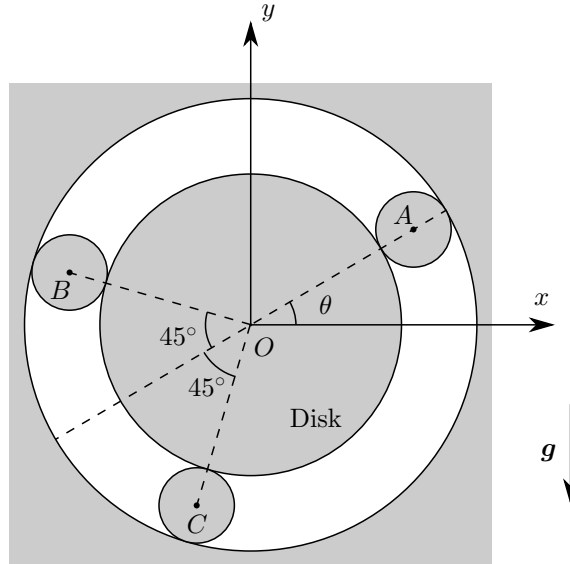
25. A heavy, homogeneous cylinder of mass  $m$ , length  $2l$ , and radius  $l/3$  is hanging from the ceiling from points  $A$  and  $B$ , supported by two massless strings of length  $l$  each. The distance between  $A$  and  $B$  is  $3l$ . The strings are always straight and connected to the cylinder at points  $C$  and  $D$  as shown in the figure. We call  $G$  the center of mass of the cylinder. In order to describe the motion of the cylinder, we define the inertial reference frame  $S_0 : \{A, B_0\}$  with  $B_0 : \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ , as  $Axyz$  in the figure. The body-fixed reference frame  $S_B : \{G, B_B\}$  with  $B_B : \{\mathbf{i}_B, \mathbf{j}_B, \mathbf{k}_B\}$  is defined as  $Gx_By_Bz_B$  in the figure. We denote the position of the center of mass of the cylinder  $G$  as  $\mathbf{r}_0^G = x_G\mathbf{i} + y_G\mathbf{j} + z_G\mathbf{k}$ . We define  $\beta$ ,  $\theta$ , and  $\phi$ , which are 3-2-1 Euler angles of the cylinder with respect to  $S_0$ , positive in the direction shown in the figure ( $Gx_0$  is parallel to  $Ax$ ). Observe that, for clarity, the figure only displays part of the auxiliary axes of these rotations. Lastly, the tensor of inertia of the cylinder about  $G$  is given in terms of two known constants ( $\lambda$ ,  $\mu$ ) :

$$\bar{\bar{\mathcal{I}}}_G = ml^2 \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & \mu \end{bmatrix}_{BB}$$



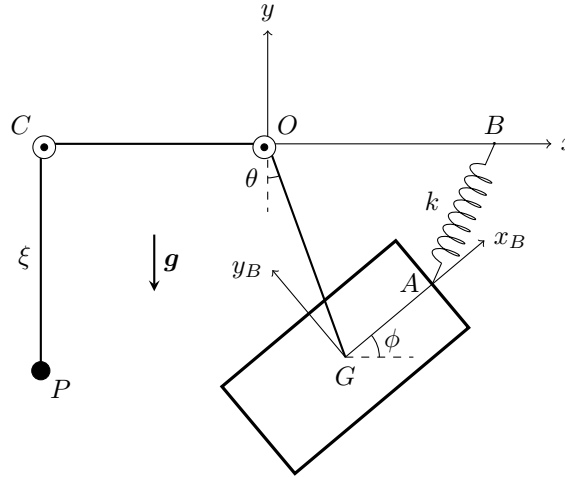
- What is the degree of freedom of this problem? What is the total number of *unknowns* in the problem? How many equations are needed to solve the problem?
- Write down the position vectors of  $C$  and  $D$  as a function of the variables of the problem.
- Calculate the angular velocity and angular acceleration of the body as a function of the variables of the problem and their derivatives.
- Write down the equation of the constraints imposed by the strings.
- Write down the expressions of all forces and moments of forces at  $G$  acting on the cylinder.
- Write down the equations of motion of the cylinder (plus any other necessary equations) that if integrated, would yield the evolution in time of these unknowns. Note that you do not need to integrate them.
- Is the mechanical energy of the cylinder conserved in this problem? If so, write down its equation of conservation of energy. Is this an independent equation with respect to the set of equations of the previous question?

26. A heavy disk of radius  $d$  and mass  $2m$  is allowed to rotate inside a circular cavity thanks to three heavy cylindrical bearings with center at  $A$ ,  $B$ , and  $C$ , each with a radius  $d/4$  and mass  $m$ , as shown in the figure. The disk-bearing contacts and bearing-cavity contacts allow rolling but no slipping. We define the inertial reference frame  $Oxy$  fixed to the cavity. We denote  $\theta$  the angular position of  $A$  with respect to  $Ox$ . Initially,  $\theta_0 = 0$  and all points of the system have zero velocity.



- Discuss the number of degrees of freedom of the system and choose a set of independent parameter(s) to specify its state at any instant of time.
- Express the linear velocity of the center of each bearing, and the angular velocity of the disk and the bearings, as a function of your chosen set of parameter(s) and its/their derivatives in a generic instant of time.
- Write down the position of the center of mass of the system as a function of your chosen set of parameter(s) in a generic instant of time.
- Compute the kinetic energy of each body in the system in a generic instant of time.
- Reduce the motion of the system to quadratures (i.e., integrals).
- Find the equilibrium positions of the system and discuss their stability.
- If there are any stable equilibrium positions, compute the angular frequency of the oscillations of the system about it/them, if only very small deviations from equilibrium are allowed.

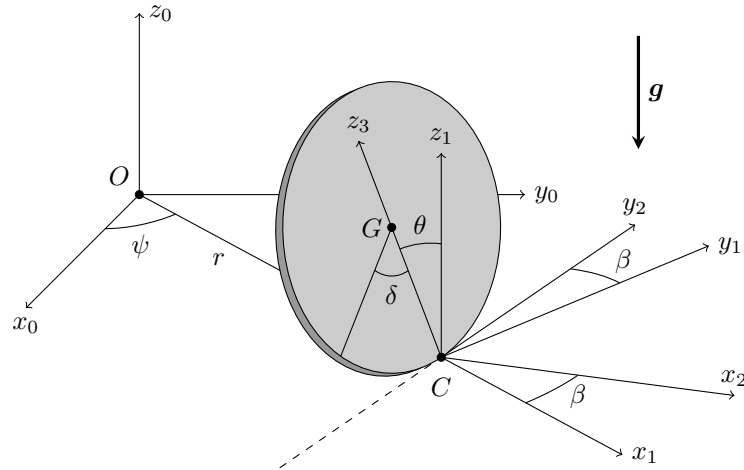
27. A heavy homogeneous rectangular block of mass  $2m$  has height  $L$  and width  $2L$ . The middle point  $A$  of one of the short edges of the block is connected to a fixed point  $B$ , located at a distance  $2L$  to the right of  $O$ , with a spring of stiffness  $k$  and zero length in the absence of forces. The center  $G$  of the block is connected to a point weight  $P$  of mass  $m$  with a string of constant length  $6L$  and negligible mass. The string passes by two frictionless pulleys of negligible radius at  $O$  and  $C$  and is always tense. The distance between the two pulleys is  $2L$ . The weight  $P$  is constrained to move along the vertical that passes by  $C$ . We define the right-handed inertial reference frame  $Oxyz$  and the right-handed body-fixed reference frame  $Gx_By_Bz_B$  attached to the block as shown in the figure. The block moves only in the  $Oxy$  plane. The length  $\xi$  and the angles  $\theta, \phi$  are also defined there.



Answer the following questions in a reasoned way:

- How many degrees of freedom does the system have?
- Express the position, velocity and acceleration of  $G$  and the angular velocity and angular acceleration of the block with respect to  $Oxy$ , as a function of  $\xi, \theta, \phi$  and their derivatives.
- Write down the expression of all forces acting on the weight  $P$  and the block.
- Write down three second order differential equations for the evolution of  $\xi, \theta, \phi$  with time. Note that you do not need to solve them.
- Can you apply the law of conservation of mechanical energy to this problem? if so, write down the corresponding equation.

28. A heavy, homogeneous coin  $b$  of mass  $m$ , radius  $R$  and negligible thickness rolls on a horizontal plane at rest. The coin never takes off from the plane. We call  $C$  the instantaneous contact point between the coin and the plane, and  $G$  the center of mass of the coin. An inertial reference frame  $S_0 = \{O; B_0\}$ , with  $O$  a point of the plane and  $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ , is given. We define  $r$  and  $\psi$ , which are the polar coordinates of  $C$  in the  $Ox_0y_0$  plane, and we call  $S_1 = \{C; B_1\}$  the local reference frame associated to these polar coordinates. Lastly, we define  $\beta$ ,  $\theta$ ,  $\delta$ , which are a set of Euler angles of the coin with respect to  $S_1$ , positive in the direction shown in the figure (the coin is contained in the  $Cy_2z_3$  plane).



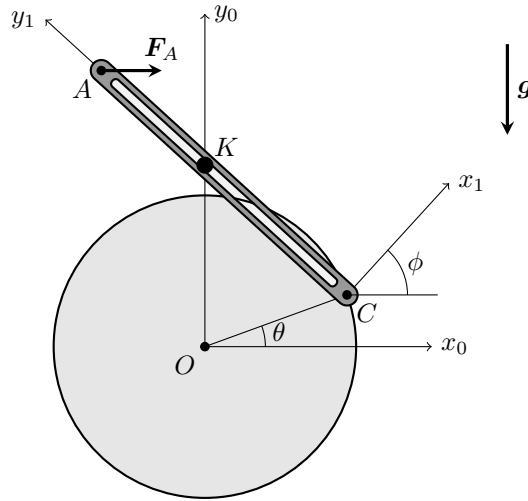
- How many degrees of freedom does the motion of the coin have, when the coin can roll and slip, and when the coin can roll but not slip?
- Assuming that the coin can slip, obtain the general expression of  $\boldsymbol{\omega}_{B_0}$ ,  $\boldsymbol{\alpha}_{B_0}$ ,  $\mathbf{r}_0^G$ ,  $\mathbf{v}_0^G$ , and  $\mathbf{a}_0^G$  as a function of the coordinates of the problem and their time derivatives.

From this point on, consider that the friction coefficient between the coin and the plane is  $\mu = \infty$ , so the coin can roll but not slip:

- Derive the scalar constraint equation(s) between the time derivatives of the coordinates of the problem that result from the no-slipping condition.
- Identify and express all the external forces acting on the coin and the torque that they produce about  $G$ , introducing any necessary additional unknowns.
- Write down the equations of motion of the coin in scalar form. Discuss whether any simple conservation law exists in the problem and justify your answer.
- Finally, consider the particular case in which the coin moves with  $\beta = 0$  and  $\mathbf{r}_0^G = h\mathbf{k}_0$ , where  $h < R$  is constant. Give the value of  $r$ ,  $\theta$ , and the reaction forces in this case. What is the relation between  $h$  and  $\dot{\psi}$ ? Comment on this result.

**NOTE:** The moment of inertia of a disk of mass  $m$  and radius  $R$  about its axis of rotational symmetry is  $mR^2/2$ .

29. A homogeneous disk  $D$  of radius  $R$  and mass  $m$  can rotate freely about its center  $O$ , which is fixed to the wall. We define the inertial reference frame  $S_0$  as shown in the figure. A homogeneous, heavy bar  $B$  of mass  $2m$  and length  $L$  has a longitudinal groove that allows it to slide without friction over a small knob  $K$  fixed to the wall and located at a height  $h$  above  $O$ . One of the end points of the bar is connected to perimeter of the disk at point  $C$  as shown. This connection allows the bar to rotate freely about  $C$  with respect to the disk. The other end point  $A$  of the bar is attracted to the  $Oy$  axis by a force  $\mathbf{F}_A$  that is proportional to its distance to  $Oy$ . The proportionality constant of this force is  $k$ .

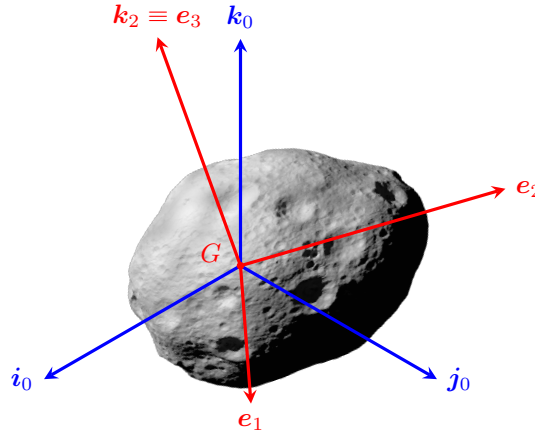


- How many degrees of freedom does this planar problem have? Justify your answer.
- Find the geometric relation between  $\theta$  and  $\phi$ . Compute  $\dot{\phi}$  and  $\ddot{\phi}$ .
- Express the general position, velocity, and acceleration of point  $A$  and the center of mass of the bar  $G$  as a function of  $\theta$ ,  $\phi$  and their time derivatives.
- Write down the general expression of all forces acting on the bar and on the disk, be they applied or otherwise.
- Compute the general expression of the kinetic energy of the bar and the kinetic energy of the disk.
- If  $\theta_0 = 0$ ,  $\dot{\theta}_0 = 0$  initially, obtain one equation on  $\theta$ ,  $\phi$ ,  $\dot{\theta}$ ,  $\dot{\phi}$  to describe the motion of the system that does not depend on any additional unknowns.

30. An irregularly-shaped, rigid asteroid  $b$  with center of mass  $G$  is floating freely in space. We define an inertial reference frame  $S_0 : \{G; B_0\}$ , and a body-fixed reference frame  $S_b : \{G; B_b\}$  as shown in the figure, where  $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$  and  $B_b = \{\mathbf{i}_b, \mathbf{j}_b, \mathbf{k}_b\}$ . One can reach  $B_b$  from  $B_0$  by a sequence of Euler rotations 1-2-3, with angles that we shall call  $\psi$ ,  $\theta$ , and  $\phi$  (in that order). The asteroid has mass  $m$  and a tensor of inertia at  $G$  that expressed in the vector basis  $B_b$  is

$$\bar{I}_G = m\ell^2 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}_b,$$

where  $\ell$  is a characteristic length of the asteroid. Initially,  $(\psi_0, \theta_0, \phi_0) = (\pi/3, 0, \pi/2)$  and  $(\dot{\psi}_0, \dot{\theta}_0, \dot{\phi}_0) = (-2, 0, 1)/\tau$ , where  $\tau$  is a characteristic time.



- Express the vectors  $\mathbf{i}_b, \mathbf{j}_b, \mathbf{k}_b$  in the vector basis  $B_0$  as a function of  $\psi$ ,  $\theta$ , and  $\phi$ .
- Write down the angular velocity vector of the asteroid,  $\boldsymbol{\omega}_{b0}$ , in basis  $B_b$  as a function of  $\psi$ ,  $\theta$ ,  $\phi$  and their derivatives.

In the following, we shall denote the components of  $\boldsymbol{\omega}_{b0}$  in  $B_b$  as  $\omega_{xb}, \omega_{yb}, \omega_{zb}$ , and define a point  $P$  with position vector  $\mathbf{r}_b^P = x_b \mathbf{i}_b + y_b \mathbf{j}_b + z_b \mathbf{k}_b = \tau \ell (\omega_{xb} \mathbf{i}_b + \omega_{yb} \mathbf{j}_b + \omega_{zb} \mathbf{k}_b)$ .

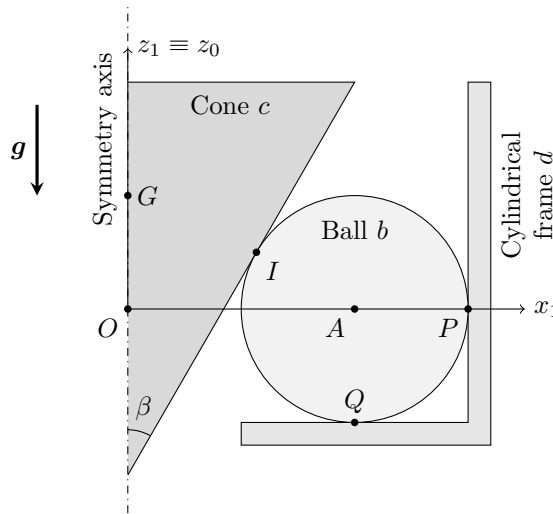
- Compute the kinetic energy  $T_0^b$  of the asteroid. Demonstrate that  $P$  must remain on the surface of a body-fixed ellipsoid at all times. What are the sizes of the semi-axes of this ellipsoid?
- Compute the angular momentum vector  $\mathbf{H}_{G0}^b$  of the asteroid. Demonstrate that the component of  $\boldsymbol{\omega}_{b0}$  in the direction of  $\mathbf{H}_{G0}^b$  is constant. Show that  $P$  must remain on an invariant plane of  $S_0$  at all times, and compute the distance to this plane from  $G$ .

Since the ellipsoid is body-fixed, it rotates with the same angular velocity vector as the asteroid, and we can study the motion of the asteroid with the ellipsoid only.

- Show that the ellipsoid and the plane are tangent at  $P$  at all times.
- Using the results of previous questions, describe qualitatively the motion of the ellipsoid with respect to the plane.

**HINT:** An ellipsoid can be expressed as  $x^2/a + y^2/b + z^2/c = d$  with  $a, b, c, d > 0$ .

31. A rigid, homogeneous, heavy ball  $b$  of mass  $m$  and radius  $R$  is always in contact with a rigid, homogeneous, heavy cone  $c$  of half-angle  $\beta$ , radius at the base  $2R$ , and mass  $2m$ . The ball is also always in contact with a rigid cylindrical frame  $d$  of inner radius  $3R$  that has a base at the bottom and the same axis of symmetry as the cone. We call the instantaneous contact points between the bodies  $I$ ,  $P$  and  $Q$ , as shown in the figure. We call the center of mass of the ball and cone  $A$  and  $G$ , respectively. The cylinder is at rest in the inertial reference frame  $S_0$ . The cone can rotate about its axis, which is constrained to coincide with  $Oz_0$  at all times. We define the auxiliary reference frame  $S_1$  that initially coincides with  $S_0$  and rotates with respect to  $\mathbf{k}_0$ , such that  $Ox_1$  follows the center of the ball in its motion. The angle between  $Ox_0$  and  $Ox_1$  is called  $\theta$ , and the angle of rotation of the cone with respect to  $S_0$  is called  $\phi$ . Both angles are positive in the counterclockwise direction as seen from  $Oz_0$ . The friction coefficient of the ball with the other bodies is  $\mu$ . The ball can pivot freely at each contact point with the cone and the cylinder.



- How many degrees of freedom does the system have in the limit  $\mu = 0$ ? And in the limit  $\mu = \infty$ ?
- Write down the general expression of the position, velocity and acceleration of  $A$  with respect to  $S_0$  as a function of  $\theta$  and its derivatives.

Assuming first that no slipping occurs between the ball and the other objects:

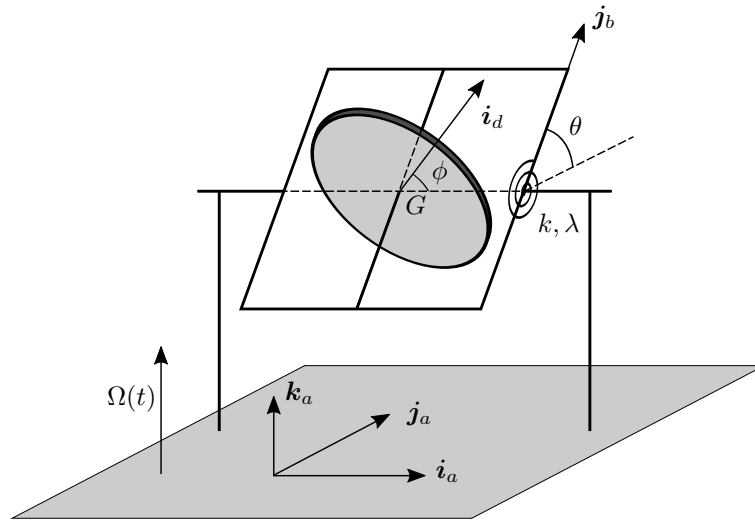
- Locate the instantaneous axis of rotation (IAR) of the ball in  $S_0$ . Write down the general expression of the angular velocity and acceleration of the ball with respect to  $S_0$  as a function of  $\theta$  and its derivatives.
- Find the relation between  $\dot{\theta}$  and  $\dot{\phi}$ .
- Identify and express all forces acting on the ball, and clearly state their point of application.
- Obtain the equations of motion of the system and prove that the problem is undetermined (i.e., more unknowns than equations).

Considering now that the cone is rotating at a constant angular velocity, and assuming that the component in the  $Ox_1z_1$  plane of the friction forces at points  $P$  and  $I$  are zero:

- What is the maximum value of  $\dot{\theta}$  for which the no-slipping condition holds?

**Note:** The moment of inertia of the sphere about any axis that passes through its center is  $2mR^2/5$ . The moment of inertia of the cone about its axis is  $12mR^2/5$ .

32. An aircraft  $a$  is equipped with a rate gyroscope, which consists of a spinning disk  $d$  mounted on a single gimbal  $b$ , which is in turn attached to the aircraft chassis as shown in the figure. Each of these elements can be considered to be a rigid body. The principal moments of inertia of the disk about its center of mass  $G$  are  $2I$  along its axis of spin, and  $I$  along the perpendicular directions. The gimbal has negligible mass and moments of inertia. The figure defines the nutation angle  $\theta$  and the spin angle  $\phi$ . Moreover, it defines the aircraft-fixed  $S_a$ , the gimbal-fixed  $S_b$  (simple rotation of angle  $\theta$  with respect to  $S_a$ ), and the disk-fixed  $S_d$  reference frames (simple rotation of angle  $\phi$  with respect to  $S_b$ ). We shall denote an arbitrary inertial reference frame as  $S_0$  (not shown). There is a torsional spring which exerts a restoring torque on the gimbal proportional to angle  $\theta$ , with constant of proportionality  $k$ . Additionally, there is a torsional damper which exerts a damping torque on the gimbal proportional to  $\dot{\theta}$ , with constant of proportionality  $\lambda$ . The aircraft performs a known yaw maneuver with angular velocity  $\omega_{a0} = \Omega(t)\mathbf{k}_a$ .



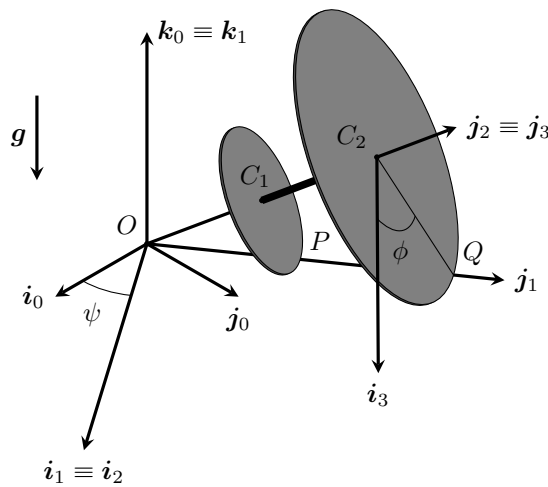
- Express the vectors of the bases of  $S_b$  and  $S_d$  as a function of the vectors of the basis of  $S_a$  and the angles  $\theta$ ,  $\phi$
- Obtain the general expressions for the angular velocity and angular acceleration of  $S_b$  and  $S_d$  with respect to the inertial reference frame,  $S_0$ .
- Identify all the torques acting on the rigid bodies of the problem and provide their general mathematical expression.
- Derive the equations of motion for  $\theta$  and  $\phi$  as a function of the parameters of the problem. No other unknowns shall appear in the equations.
- Prove that the component of the angular momentum of the disk about its spin axis is a constant of motion.

In the following, assume that  $\Omega \ll \dot{\phi}$  and that  $k, \lambda$  are chosen so that  $\theta$  remains small ( $\theta \ll 1$ ) during the maneuver.

- Linearize the equation of motion of  $\theta$  under these assumptions. What is the relation between  $\Omega$  and  $\theta$  when  $\Omega$  is constant?
- Using the linearized equation of motion, find out the value of  $\lambda$  for the system to be critically damped, as a function of the rest of parameters of the problem.

33. Two thin, heavy, homogeneous wheels, of center  $C_1$  and  $C_2$  and radius  $a$  and  $2a$  respectively, are connected to each other with a massless, perpendicular axle of length  $\sqrt{3}a$  forming a single rigid body  $b$  as shown in the figure. The surface density of the wheel material is  $\sigma$ . We label the points of the rigid body that are in contact with the ground at each instant as  $P$  and  $Q$ . The friction coefficient  $\mu$  between the wheels and the ground is sufficiently large to ensure that the wheels always roll without slipping. We define the inertial reference frame  $S_0$  and the auxiliary, rotating reference frame  $S_1$  as shown. We also define other auxiliary vectors in the figure,  $\mathbf{i}_2, \mathbf{j}_2, \mathbf{i}_3, \mathbf{j}_3$ . The axis  $Oy_1$  follows points  $P$  and  $Q$  as they move. The angle  $\psi$  is measured between the axes  $Ox_0$  and  $Ox_1$ . The angle  $\phi$  is the angle that the wheels have rotated about the axle since the initial instant of time. Initially,  $\psi = 0$ ,  $\phi = 0$ , and vector  $\mathbf{i}_3$  points in the direction of  $C_2Q$ .

**Hint:** Observe that the points  $O$ ,  $C_1$ , and  $C_2$  are always aligned. Remember to define all the vector bases that you use as a function of the vectors  $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ , or other, already defined vectors.



- (a) Compute the total mass  $m$ , the distance between point  $C_1$  on the axle and the center of mass  $G$ , and the tensor of inertia about  $G$  of the object.

**Hint:** the tensor of inertia of a thin, homogeneous disk of mass  $M$  and radius  $L$  about its center  $C$ , expressed in a vector basis **with the second unit vector perpendicular to the disk**, is given by:

$$\mathcal{I}_C = \frac{ML^2}{4} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In the rest of the problem, you can name the resulting inertia tensor components with letters  $A$ ,  $B$ , etc. and use them instead of their numerical expressions.

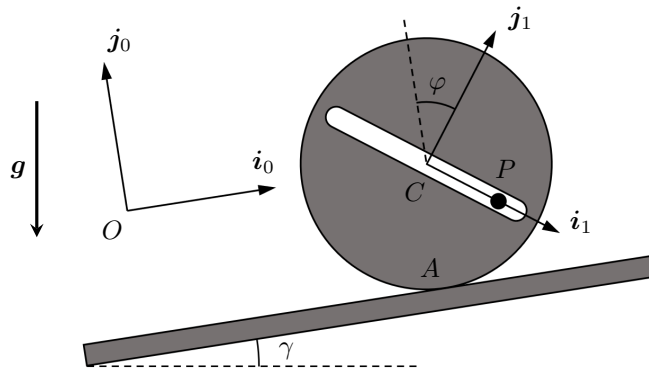
- (b) Write down the position, velocity, and acceleration of  $G$ , and the angular velocity and acceleration of the rigid body  $b$ , with respect to  $S_0$ .
- (c) Find out the kinematic relation between  $\psi$  and  $\phi$ .
- (d) Identify and write down the general expression of all the forces acting on the rigid body, and reduce them to the resultant force and resultant moment of force about  $G$ .
- (e) Write down the equations of motion for the rigid body in the form of 6 scalar differential equations.
- (f) Find out the maximum value of  $\dot{\psi}$  for which both wheels remain in contact with the ground.
- (g) Discuss whether the mechanical energy of the rigid body in  $S_0$  is conserved in this problem. Note that you do not need to write down the equation of mechanical energy.
- (h) Is the system of equations of question (e) closed? Why, or why not?

34. At a given instant of time a rigid body  $b$  is in pure rotation. A point  $A$  and a point  $B$  of the rigid body have position and velocity vectors  $\mathbf{r}_0^A = \mathbf{0}$ ,  $\mathbf{r}_0^B = \mathbf{i}$ ,  $\mathbf{v}_0^A = \mathbf{i} + 2\mathbf{k}$  and  $\mathbf{v}_0^B = c\mathbf{i} + \mathbf{j}$ , where  $c$  is an unknown value. Determine  $c$ , the angular velocity  $\boldsymbol{\omega}_{b0}$  of the body, and the position vector of a point in its instantaneous axis of rotation.
- 

35. Of a spaceship floating in space we know that its center of mass  $G$ , located at  $\mathbf{r}_0^G = \mathbf{0}$ , has a velocity  $\mathbf{v}_0^G = 8\mathbf{i} + 4\mathbf{j}$ . A point  $A$  on its solar array has  $\mathbf{r}_0^A = -2\mathbf{i} + 4\mathbf{j}$  and  $\mathbf{v}_0^A = 4\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$ . Point  $C$  on the rear side has  $\mathbf{r}_0^C = -8\mathbf{i} + 2\mathbf{k}$  and  $\mathbf{v}_0^C = 10\mathbf{i} - 6\mathbf{j} + 8\mathbf{k}$ .

- (a) Is the data compatible with the rigid body constraint?
- (b) What is the minimal velocity that any point of the spacecraft has at that instant?

36. To prevent the effect of martian dust on internal mechanisms, a new concept of a hermetically-sealed rolling robot for the exploration of Mars is proposed, which consists of a homogeneous, heavy, rigid cylinder  $b$  of mass  $M$ , radius  $R$ , and center of mass at its geometric center  $C$ . The moment of inertia of the cylinder about an axis through  $C$  perpendicular to the plane of motion is  $I_z$ . The cylinder has an internal diametrical slot in it of total length  $2\ell < 2R$  as shown in the figure. Inside this slot there is a heavy point particle  $P$  of mass  $m$ , whose position  $x_1^P$  in the slot is varied in time according to a prescribed law  $x_1^P = x_1^P(t)$ . The motion of  $P$  is imposed by an internal motor (not shown) of negligible mass that is mounted on the cylinder  $b$ . We determine the configuration of the cylinder with the position  $x_0^C$  of its center  $C$  and its rotation angle  $\varphi$ . The angle of the ground is denoted  $\gamma$ , the instantaneous contact point of the robot with the ground is denoted  $A$ , and the friction coefficient is denoted  $\mu$ . The reference frame  $S_0$  is inertial and the reference frame  $S_1$  is fixed to the cylinder.



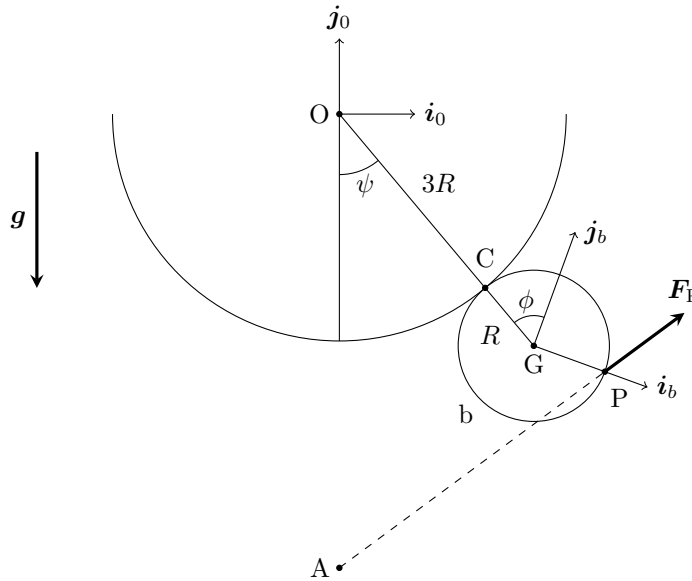
- Find the general expression for the position, velocity, and acceleration vectors of  $P$  with respect to  $S_0$  as a function of  $x_0^C$ ,  $\varphi$ ,  $x_1^P$ , and their derivatives.
- Give the expression of the force  $\mathbf{F}$  on the cylinder  $b$  caused by the particle  $P$  and its motion.  
**NOTE:** observe that as the motor pushes on  $P$ , an equal and opposite force appears on the cylinder along the same line of action.
- Reduce the system of all forces and torques on the cylinder to a single resultant force and a single resultant torque about its geometric center  $C$ .
- Consider now that we want keep the robot at rest, i.e., in equilibrium, on the slope. What is the maximum value of  $\gamma$  that allows this condition if we have  $\varphi = \gamma$  and  $x_1^P$  can be set within the range  $[-\ell, \ell]$ ?

In the following, assume that  $\mu$  is sufficiently large to prevent slipping, that  $\gamma = 0$ , and that the control law  $x_1^P(t) = \ell \sin(\Omega t)$  for the position of  $P$  is prescribed, where  $\Omega$  is a known constant and  $\ell = R/2$ .

- Write down a closed system of equations for all unknowns in the problem, and identify which initial conditions would be needed to integrate it. Explain in what order you would solve this system.
- Consider finally that at a particular instant of time  $\dot{\varphi} = 0$  and  $\ddot{\varphi} = 0$ . What is the maximum value of  $\Omega$  that we can have and still ensure that the robot never takes off the ground?

37. A homogeneous, heavy disk  $b$  of mass  $m$ , radius  $R$ , and center  $G$  rolls without slip on a stationary cylinder of radius  $3R$  and center  $O$ . We call  $C$  the instantaneous point of contact between the two. We assume that the friction coefficient  $\mu$  between the disk and the cylinder is infinite. To study the motion of the disk, we define an inertial reference frame  $S_0$  and a body-fixed reference frame  $S_b$ , and we introduce the angles  $\psi$  and  $\phi$ , as shown in the figure. The body point  $P$  on the disk, with position  $\mathbf{r}_b^P = R\mathbf{i}_b$ , feels a repulsive force from point  $A$ , located at  $\mathbf{r}_0^A = -6R\mathbf{j}_0$ , whose magnitude is equal to a known constant  $k$  times the distance between  $A$  and  $P$ . Initially,  $\psi_0 = \phi_0 = 0$  and the disk is at rest.

**Data:** the moment of inertia of the disk about a line perpendicular to the drawing passing through  $G$  is  $I = mR^2/2$ .



Assuming that the disk and the cylinder are always in contact:

- How many effective degrees of freedom does the system have? Define  $\mathbf{i}_b$ ,  $\mathbf{j}_b$ , and any other unit vectors that you want to use to solve the problem, in terms of  $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ .
- Give the general expressions for the position, velocity, and acceleration of  $G$  and the angular velocity and acceleration of  $b$  with respect to  $S_0$ , as a function of  $\psi$ ,  $\phi$  and their derivatives.
- Find the mathematical relation that exists between  $\phi$  and  $\psi$ . Rewrite the expressions of the last question in terms of  $\psi$  and its derivatives only.
- Identify and give the mathematical expression of all forces on the disk, stating clearly their points of application, and introducing new unknowns as necessary. Then, reduce the system of forces to a resultant force  $\mathbf{F}$  and a resultant torque  $\mathbf{M}_G$  about  $G$ .
- Write down a first-order differential equation for  $\psi$ , free of any other unknowns in the problem, and give the initial conditions for its integration.

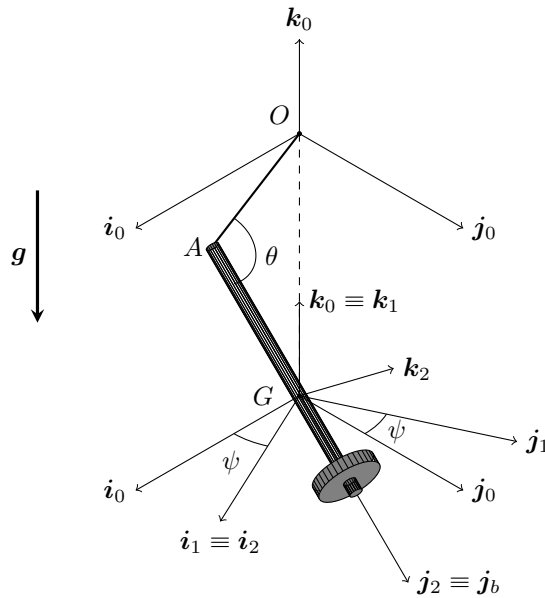
Assuming now that the disk may separate from the cylinder:

- Find the condition for which the disk detaches from the cylinder in terms of  $\psi$  only (i.e., no  $\dot{\psi}$  can appear in the solution).

38. We consider a heavy, homogeneous, axisymmetric rigid body  $b$  of mass  $m$ . The body point  $A$  hangs from a fixed point  $O$  on the ceiling with a massless, inextensible string of length  $\ell$ , as shown in the figure. We know that the string always remains tense. The center of mass  $G$  of  $b$  is located a distance  $\ell$  from  $A$ , and is constrained to move along the vertical that passes through  $O$ . We define the inertial reference frame  $S_0$  with center on  $O$ . We introduce the angle  $\psi$  between the  $OAG$  plane and the  $Oy_0z_0$  plane. We define the auxiliary reference frame  $S_1$  as shown in the figure, so that the segment  $AG$  is contained in the  $Gy_1z_1$  plane. We introduce the angle  $\theta$  between the string  $OA$  and the axis of the body  $AG$ , and we define the auxiliary reference frame  $S_2$  so that  $j_2$  points in the  $AG$  direction. Finally, we define the body-fixed reference frame  $S_b$  and the angle  $\varphi$ , which represents the spin of the rigid body about the  $j_2$  vector, such that when  $\varphi = 0$ ,  $S_b$  coincides with  $S_2$ . The tensor of inertia of the rigid body about  $G$  expressed in the basis of  $S_b$  is:

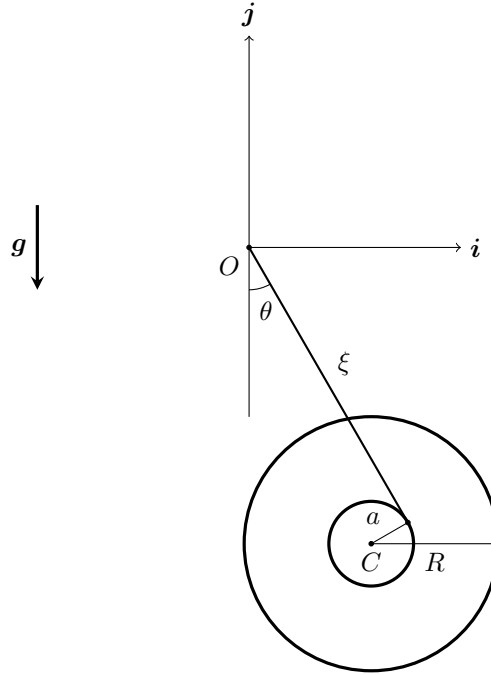
$$\bar{\bar{I}}_G = \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & A \end{bmatrix}_b,$$

where  $A = 3B = 6m\ell^2$ .



- How many configuration degrees of freedom does this system have? Justify your answer.
- Give the expressions of the unit vectors of  $S_1$  in terms of those of  $S_0$ , the unit vectors of  $S_2$  in terms of those of  $S_1$ , and the unit vectors of  $S_b$  in terms of those of  $S_2$ .
- Find the general expression for the position, velocity, and acceleration of  $G$ , and the angular velocity and acceleration of  $b$ , with respect to  $S_0$ , in terms of  $\psi$ ,  $\theta$  and their derivatives.
- Identify all the forces/torques acting on  $b$  and reduce them to a single resultant force and a single resultant torque at  $G$ .
- Write down a system of (differential) equations that allows solving for the time evolution of all unknowns in the problem.
- What is the condition that  $\dot{\psi}$  and  $\dot{\varphi}$  must satisfy for the nutation angle  $\theta$  to be constant in time?
- Find three first integrals of motion (i.e., equations that do not depend on second derivatives). Justify your procedure.

39. The rigid body yo-yo in the figure consists of two heavy, homogeneous disks of radius  $R$  and an inner cylinder of radius  $a$ , about which an ideally thin, infinitely long, massless, inextensible string winds up and down. The mass of the yo-yo is  $m$ , and its moment of inertia about its spin axis is  $mR^2/2$ . We denote the yo-yo with symbol  $b$ , and its center point as  $C$ . The other end of the string is attached to a fixed point on the ceiling, and we know that the string remains taut at all times. We introduce the generalized coordinates  $\xi$ ,  $\theta$  to determine the configuration of the system.



- Determine the position, velocity, and acceleration of the center point  $C$  of the yo-yo, and its angular velocity and angular acceleration, as a function of the generalized coordinates and their time derivatives.
- Identify all the forces/torques acting on the yo-yo  $b$  and reduce them to a single resultant force and a single resultant torque at  $C$ .
- Write down a system of (differential) equations that allows solving for the time evolution of all unknowns in the problem.
- Find a first integral of motion (i.e., an equation that do not depend on second derivatives). Justify your procedure.

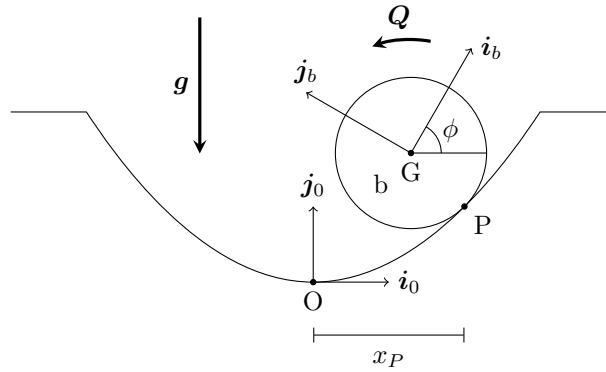
In the following, assume that the yo-yo is further constrained to stay with  $\theta = 0$ :

- If the yo-yo is initially at rest, determine the initial value of  $\ddot{\xi}_0$ . Sketch the graph of  $\ddot{\xi}_0$  as a function of the ratio  $R/a \in [1, \infty[$ .
- If initially  $\xi_0 = 2R$ , find the minimum value of  $\Omega_0$ , the magnitude of the initial angular velocity of the yo-yo, for it to be able to reach the origin (i.e., for  $\xi$  to become equal to 0 at a later instant of time). Assuming  $a \ll R$ , sketch the graph of  $\Omega$  as a function of  $R$ .

40. A homogeneous, heavy disk  $b$  of mass  $m$ , radius  $R$ , and center  $G$  rolls without slip on the surface of a cavity with parabolic shape, whose geometry is given by

$$y = \frac{x^2}{4R},$$

and has a maximum depth of  $3R$ . The moment of inertia of the disk about  $G$  is  $I = mR^2/2$ . To study the motion of the disk, we define an inertial reference frame  $S_0$  and a body-fixed reference frame  $S_b$  as shown below. We call  $P$  the instantaneous point of contact between the disk and the parabola. We introduce the variables  $x_P$  and  $\phi$ . An external, constant torque  $Q\mathbf{k}_0$  is applied on the disk through an axle passing through  $G$ . Assume that the friction coefficient between the disk and the parabola  $\mu$  is infinite, and that the disk and the parabola are always in contact. Initially, the disk is at rest with  $x_P = 0$  and  $\phi = 0$ .



- How many degrees of freedom does the system have after taking into account all constraints? Write down the general expression of a tangent unit vector  $\mathbf{e}_t$  and a normal unit vector  $\mathbf{e}_n$  to the parabola at  $P$  as a function of  $x_P$ , in terms of  $\{\mathbf{i}_0, \mathbf{j}_0\}$ .
- Give the general expressions for the position and velocity of  $G$ , and the angular velocity of the disk  $b$  with respect to  $S_0$ , as a function of  $x_P$ ,  $\phi$  and their derivatives.
- Identify and give the mathematical expression of all forces and torques on the disk, introducing new unknowns as necessary. For each force, state clearly the point of application. Then, reduce the system to a resultant force  $\mathbf{F}$  and a resultant torque  $\mathbf{M}_G$  about  $G$ .
- Express  $\phi$  as a function of  $x_P$ .

**Hint:** one of the resulting integrals is immediate; the other can be solved as

$$\int \frac{\sqrt{u^2 + 4a^2}}{2a^2} du = \frac{u\sqrt{u^2 + 4a^2}}{4a^2} + \ln(\sqrt{u^2 + 4a^2} + u) + \text{const}$$

From this point on, assume the function  $\phi(x_P)$  is known, even if you did not reach the correct result above.

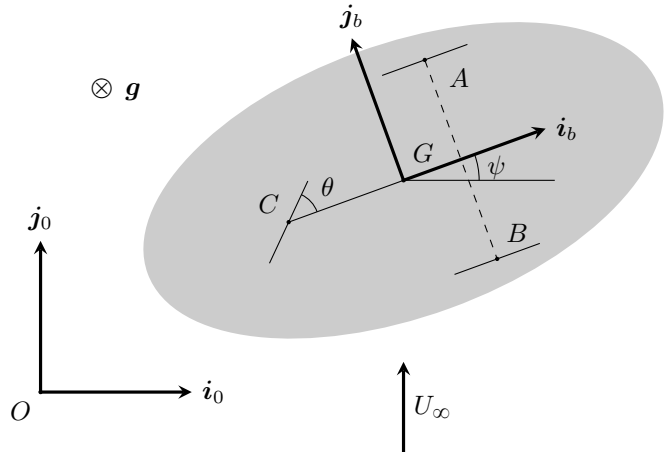
- Noting that the power of the constant torque is  $Q\dot{\phi}$ , write down a first-order differential equation for  $x_P$ , free of any other unknowns in the problem.
- Determine the condition on the value of  $Q$  to ensure that the disk gets out of the cavity.

41. A sail sledge is a special vehicle designed to slide over ice on blades, equipped with a vertical sail to catch the wind and propel itself. The sledge is moving on a horizontal plane of ice. We model it as a heavy rigid body  $b$  of mass  $m$  and moment of inertia  $I$  about its center of mass  $G$ , which is located at a negligible height with respect to the ice. The contact points of the blades with the ice are points  $A$ ,  $B$  and  $C$ , which form an equilateral triangle of side  $\sqrt{3}\ell$  and center below  $G$  as shown. The configuration of the sledge is defined by the  $x$  and  $y$  coordinates of its center of mass in  $S_0$ , and the attitude angle  $\psi$ . The blades at  $A$  and  $B$  are oriented parallel to  $\mathbf{i}_b$ , while the blade at  $C$  forms an angle  $\theta$  with this direction. Each blade may slide along the direction parallel to it without friction, but it cannot slip in the direction perpendicular to it.

The (horizontal) aerodynamic forces on the sail are a lift force  $\mathbf{L}$  and a drag force  $\mathbf{D}$ , respectively perpendicular and parallel to the upstream wind velocity  $\mathbf{u}_b^w$  relative to  $S_b$ . The point of application of these forces is the center of pressure  $P$  of the sail, which is found at a distance  $\ell$  above  $G$ , and their magnitudes are:

$$L = b_L(u_b^w)^2; \quad D = b_D(u_b^w)^2;$$

where  $b_L$  and  $b_D$  are two known constants (we can assume the sail is automatically readjusted to keep them constant under all conditions). Wind blows in the direction of the arrow with a groundspeed  $u_0^w = U_\infty$  relative to  $S_0$ .



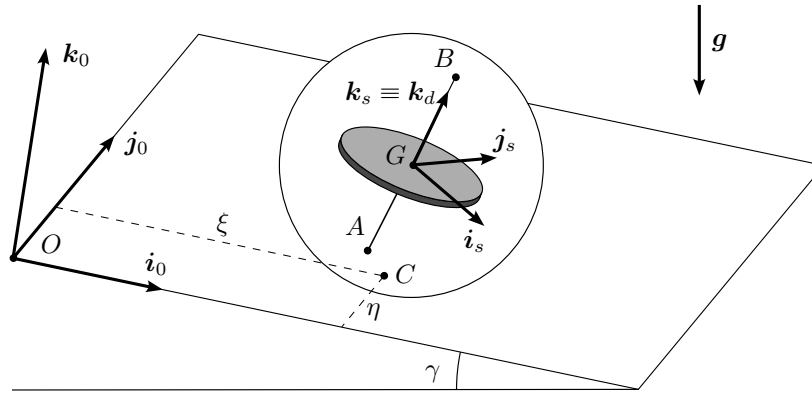
- Find the general expression for the position, velocity, and acceleration of  $G$ , and the angular velocity and acceleration of the body  $b$ , with respect to  $S_0$ , in terms of  $x$ ,  $y$ ,  $\psi$ , and their derivatives.
- Find the position  $\mathbf{r}_b^I$  of the instantaneous center of rotation  $I$  of  $b$  as a function of  $\theta$ .
- Express  $\dot{x}$  and  $\dot{y}$  as a function of  $\dot{\psi}$  and  $\theta$ , assuming  $\theta \neq 0$ .
- Write down all the forces acting on  $b$  and specify their point of application. Introduce new unknowns in the problem as needed. Assume that the reaction forces on  $A$  and  $B$  in the  $\mathbf{j}_b$  direction are equal in magnitude.

In the rest of the problem, consider that  $\psi = \theta = 0$ .

- If the sledge is at rest and  $b_D = 0$ ,  $\mathbf{L} \cdot \mathbf{i}_b \geq 0$ , what is the maximum value of  $U_\infty$  that the vehicle can withstand without tipping over at that instant?
- Assuming that the sledge does not tip over, what is the maximum speed it can continuously travel at, as a function of  $b_L$ ,  $b_D$ , and  $U_\infty$ ?

42. A heavy, thin, homogeneous, hollow sphere  $s$  of radius  $R$  and mass  $m$  contains a heavy, homogeneous disk  $d$  of mass  $2m$ . The disk is supported by a massless axle that is rigidly connected to the sphere at two polar opposite points  $A$  and  $B$ , so that the centers of mass of the sphere and disk coincide,  $G_s \equiv G_d \equiv G$ . The disk can spin freely about the axle. The sphere rolls without slipping over an inclined plane that forms a known, fixed angle  $\gamma$  with the horizontal. We call  $C$  the instantaneous contact point of the sphere with the plane, and we assume that the sphere cannot separate from the plane. We define the inertial reference frame  $S_0$ , the sphere-fixed reference frame  $S_s$ , and the disk-fixed reference frame  $S_d$  (see figure). The attitude of  $S_s$  with respect to  $S_0$  is given by the 3-1-3 Euler angles  $\psi, \theta, \phi$ . The attitude of  $S_d$  with respect to  $S_s$  is given by angle  $\beta$ , knowing that when  $\beta = 0$ ,  $S_d$  coincides with  $S_s$ . The position of the contact point on the plane is described with coordinates  $\xi, \eta$ . The tensors of inertia of the sphere and the disk about  $G$ , expressed in the basis  $B_s$  of  $S_s$ , are:

$$\bar{I}_G^s = 2 \frac{mR^2}{3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_s; \quad \bar{I}_G^d = \frac{mR^2}{6} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}_s.$$



- Starting from the vector basis  $B_0$  of  $S_0$ , write down the expressions for the change-of-basis associated to the Euler angles, which define the intermediate vector bases ( $B_1, B_2$ ) and the vector basis  $B_s$  of  $S_s$ .
- Give the general expressions for  $\mathbf{r}_0^G$ ,  $\boldsymbol{\omega}_{s0}$ , and  $\boldsymbol{\omega}_{ds}$  as a function of the coordinates of the problem and their derivatives.
- Write down the equations of the rolling without slipping constraint between the sphere and the plane, in terms of the coordinates of the problem and their derivatives. Classify this constraint according to [integrable / non-integrable], [stationary / non-stationary].
- Draw an exploded view diagram of the disk, the sphere (including the axle as part of it), and the plane, and correctly represent all reactions on each body.
- Express all forces and torques on the sphere mathematically, using the unit vectors you have defined before and introducing new unknowns as needed. Reduce this system of forces and torques to a resultant force and a resultant torque at  $G$ .
- Write down the total angular momentum about  $G$  and the total kinetic energy of the system, as seen from  $S_0$ .
- Find two first integrals of motion.

### 3 Solution keys

Partial answers to some of the problems are provided next to facilitate students to check their work. Full solutions are not provided to encourage exploration and critical thinking; in case of doubt about any of the problems, please ask for office hours with your problem session professor. Please only check these keys after having attempted the problems on your own.

#### 3.1 Particle exercises and problems

1.

2.

$$3. \phi = \arcsin \sqrt{\frac{k}{3a}} t, \quad s = \frac{kt^2}{2}.$$

4.

$$5. v_0^B = \sqrt{(v_A + r\dot{\theta} \sin \theta)^2 + (r\dot{\theta} \cos \theta)^2}, \quad a_0^B = \sqrt{r^2 \ddot{\theta}^2 + r^2 \dot{\theta}^4}$$

6.

7. The Coriolis acceleration is  $\mathbf{a}_{cor} = -2a\omega\dot{\psi}(\cos \psi \mathbf{i}_1 + \sin \psi \mathbf{j}_1)$ .The centrifugal acceleration is  $\mathbf{a}_{cf} = -a\omega^2[(1 + \cos \psi)\mathbf{i}_1 + \sin \psi \mathbf{j}_1]$ The Equation of Motion (EOM) is  $\dot{\psi} = -2\omega \cos(\psi/2)$  for  $-\pi < \psi \leq 0$ .

The tension on the bar is

$$T = 2ma\omega^2 \cos \frac{\psi}{2} \left( 3 \cos \frac{\psi}{2} - 2 \right).$$

The string becomes loose at  $\psi = -2 \arccos(2/3)$ .

8.

9.

10.

11. a) The particle can *move towards* positive or negative infinity.b) The particle can *move towards* positive/negative infinity directly or can be *reflected* to positive/negative infinity.c) The particle can *move towards* positive/negative infinity directly, can be *reflected* to positive/negative infinity, or can *oscillate* back and forth.12. Hint: use separation of variables to integrate  $dx/\sqrt{E - V(x)} = \sqrt{2/m} dt$ .13. The terminal velocity is  $\sqrt{8ga}$ .

The position of the particle is given by

$$\begin{aligned} \operatorname{arctanh} \left( \sqrt{\frac{2-y}{3}} \right) &= -\sqrt{\frac{3g}{2a}} t + \operatorname{arctanh} \left( \frac{1}{\sqrt{3}} \right), \quad \text{for } 0 < t < \sqrt{\frac{2a}{3g}} \operatorname{arctanh} \left( \frac{1}{\sqrt{3}} \right). \\ \operatorname{arctanh} \left( \sqrt{\frac{2-y}{3}} \right) &= \sqrt{\frac{3g}{2a}} t - \operatorname{arctanh} \left( \frac{1}{\sqrt{3}} \right), \quad \text{for } t > \sqrt{\frac{2a}{3g}} \operatorname{arctanh} \left( \frac{1}{\sqrt{3}} \right). \end{aligned}$$

14. If  $v_0 > a\sqrt{2k}$ , the particle moves towards minus infinity. If  $v_0 < a\sqrt{2k}$ , the particle moves towards infinity after decelerating before arriving to  $x = 0$ . If  $v_0 = a\sqrt{2k}$ , the particle take infinite time to (i.e. cannot) reach  $x = 0$ .

15. If  $x$  is defined positive downwards, we have the Equation Of Motion (**EOM**) as  $4r^2\ddot{x} + 9\nu\dot{x} = 2r^2g$ . After we define the Initial Condition (**IC**) as  $x(t = 0) = 0$ ,  $\dot{x}(t = 0) = 0$ , the trajectory is found as

$$x = \frac{2r^2g}{81\nu^2} \left[ -4r^2 + 4r^2 \exp\left(-\frac{9\nu}{4r^2}t\right) + 9\nu t \right] .$$

16. The terminal velocity is

$$v_t = \sqrt{\frac{2mg}{\rho SC_D}} .$$

After we define the IC as  $v(t = 0) = 0$  and  $x(t = 0) = 0$  (positive downwards), the velocity and the position of the football is

$$v(t) = v_t \tanh\left(\frac{g}{v_t}t\right) , \quad s(t) = \frac{v_t^2}{g} \ln \left[ \cosh\left(\frac{g}{v_t}t\right) \right] .$$

17. The motion of the particle is described by

$$\mathbf{r}_0^M = \frac{\sqrt{3}g}{k^2} \cos kt \mathbf{i} + \frac{2g}{k^2} \sin kt \mathbf{j} + \frac{g}{k^2} (\cos kt - 1) \mathbf{k} .$$

It describes a circular trajectory as the intersection of two surfaces:

$$\begin{aligned} \text{(Sphere)} \quad & x^2 + y^2 + (z + g/k^2)^2 = 4g^2/k^4 , \\ \text{(Plane)} \quad & x - \sqrt{3}z = \sqrt{3}g/k^2 , \end{aligned}$$

Note that other pairs of surfaces can also be chosen to represent the same curve. The velocity of the particle constant,  $v_0^M = 2g/k$ .

- 18.

19. The quadrature of the EOM is given by

$$\begin{aligned} \dot{r}^2(1 + 4r^2/a^2) + r^2(\dot{\theta}^2 + 2g/a) &= v_0^2 + 2ga , \\ r^2\dot{\theta} &= av_0 , \end{aligned}$$

where  $\theta$  is the azimuthal coordinate of the cylindrical coordinates. If  $v_0 < \sqrt{2ga}$ ,  $r_{max} = a$  and  $r_{min} = v_0\sqrt{a/2g}$ . If  $v_0 = \sqrt{2ga}$ ,  $r = a$  is constant. If  $v_0 > \sqrt{2ga}$ ,  $r_{min} = a$  and  $r_{max} = v_0\sqrt{a/2g}$ .

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25. The acceleration of the spacecraft with respect to  $Fxy$  can be written as

$$\mathbf{a}_0^S = (\ddot{r} - r\dot{\nu}^2) \mathbf{i}_1 + (2\dot{r}\dot{\nu} + r\ddot{\nu}) \mathbf{j}_1 ,$$

with

$$\begin{aligned} \dot{r} &= \frac{pe\dot{\nu} \sin \nu}{(1 + e \cos \nu)^2} , \\ \ddot{r} &= \frac{pe(\ddot{\nu} \sin \nu + \dot{\nu}^2 \cos \nu)(1 + e \cos \nu)^2 + 2pe^2\dot{\nu}^2 \sin^2 \nu (1 + e \cos \nu)}{(1 + e \cos \nu)^4} . \end{aligned}$$

Since the true anomaly is a function of time (i.e.  $\nu(t)$ ), in principle  $\ddot{\nu} \neq 0$ , the Euler force also appears among other inertia forces as

$$\mathbf{F}_{Eul} = mL\ddot{\nu}(\sin\theta\mathbf{i}_1 - \cos\theta\mathbf{j}_1).$$

The tension as a function of time can be calculated once solved for  $\theta(t)$  as

$$\begin{aligned} \mathbf{T} &= -T(\cos\theta\mathbf{i}_1 + \sin\theta\mathbf{j}_1), \\ T(t) &= mL\dot{\theta}^2 - \frac{\mu_F m \cos\theta}{r^2} + mL\dot{\nu}^2 + 2mL\dot{\theta}\dot{\nu} + m(r\dot{\nu}^2 - \ddot{r})\cos\theta - m(2\dot{r}\dot{\nu} + r\ddot{\nu})\sin\theta, \end{aligned}$$

where  $\mu_F = GM_F$  is the gravitational parameter of the celestial body  $F$ .

26.

27. The quadrature of the EOM is

$$m\dot{x}_1^2(1 + \cos^2 x_1)/2 + \sin x_1(mg - 2k) + k\sin^2 x_1/2 - m\omega^2 x_1^2/2 = C,$$

with the constant  $C$  to be evaluated at the initial condition. The reaction forces from the wire are given by

$$\begin{aligned} \mathbf{N}_1 &= \frac{m(\omega^2 x_1 \cos x_1 - \dot{x}_1^2 \sin x_1) - k(2 - \sin x_1) + mg}{1 + \cos^2 x_1}(-\cos x_1\mathbf{i}_1 + \mathbf{j}_1), \\ \mathbf{N}_2 &= -2m\omega\dot{x}_1\mathbf{k}_1. \end{aligned}$$

For  $\omega = 0$  and  $x_1 \in [0, 2\pi]$ , the equilibrium points and their stability are

- $x_1 = \pi/2$ , this equilibrium always exists and it is stable if  $\alpha > 1$ ;
- $x_1 = 3\pi/2$ , this equilibrium always exists and it is stable if  $\alpha < 1/3$ ;
- $\sin x_1 = 2 - 1/\alpha$ , this equilibrium exists if  $1/3 < \alpha < 1$  and it is also stable if  $1/3 < \alpha < 1$ .

For  $\omega \neq 0$ ,  $\alpha = 0$ ,  $\beta = \omega^2/g$ , and the bead dropped with no initial velocity at  $x_{1,0} = \pi$ , two different behaviours are identified:

- $\beta > \beta_c$ , the bead oscillates;
- $\beta < \beta_c$ , the bead moves towards infinity.

For  $\beta = \beta_c$ , the potential curve has a local maximum at  $x_c$ , with  $V(x_c) = V(x_{1,0})$ . The values of  $x_c$  and  $\beta_c$  are given by

$$\begin{aligned} \tilde{V}(x_c) &= 2\sin x_c - \beta x_c^2 = -\beta_c \pi^2 \\ \dot{\tilde{V}}(x_c) &= 2\cos x_c - 2x_c\beta_c = 0. \end{aligned}$$

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31. There are two D.O.F for the system of  $P_1$  and  $P_2$ . The quadrature form of the EOM is

$$\begin{aligned} \xi^2\dot{\theta} &= C_1, \\ 2\dot{\xi}^2 + \xi^2\dot{\theta}^2\sin^2\alpha + 2g\xi(1 - \cos\alpha) &= C_2. \end{aligned}$$

For the string to be tense, it requires  $\ddot{\xi} > g$ .

For  $P_1$  to have a horizontal circular trajectory over the cone,  $\theta$  and  $\xi$  should satisfy

$$\xi\dot{\theta}^2\sin^2\alpha = g(1 - \cos\alpha),$$

with the normal force from the cone and the tension (on  $P_1$ ) given by

$$\mathbf{T} = mg(-\sin\alpha\mathbf{e}_R + \cos\alpha\mathbf{k}), \quad \mathbf{N} = \frac{mg(1 - \cos\alpha)}{\sin\alpha}(\cos\alpha\mathbf{e}_R + \sin\alpha\mathbf{e}_\theta).$$

32. The two equations to solve for  $\phi$  and  $T$  (tension force) are

$$\begin{aligned} \mathbf{i}_1 : \quad & a[(3\pi - \omega t + \phi)\ddot{\phi} - 2\omega\dot{\phi} + \dot{\phi}^2] = -g \sin \phi , \\ \mathbf{j}_1 : \quad & ma(3\pi - \omega t + \phi)\dot{\phi}^2 = -mg \cos \phi + T . \end{aligned}$$

There is conservation of mechanical energy for  $\omega = 0$  and the quadrature form of the EOM reads

$$\frac{1}{2}m[a(3\pi + \phi)]^2\dot{\phi}^2 + mg[a \sin \phi - a(3\pi + \phi) \cos \phi] = E .$$

33. The required torque in the  $\mathbf{k}$  direction is  $2m\omega(\dot{x}_1x_1 + \dot{x}_2x_2)$ .

The equilibrium configurations are

- For  $\gamma < 1$ : no equilibrium exists;
- For  $\gamma = 1$ : equilibrium exists only if  $\kappa = 0$ , for which all positions are equilibrium configurations;
- For  $\gamma > 1$ : symmetric equilibrium ( $x_1x_2 < 0$ ) exists under the condition  $x_1 = -x_2$  and  $x_1^2 = [\kappa L^3/4(\gamma - 1)]^{2/3}$ ; asymmetric equilibrium configurations also exist.

The minimum and maximum values of  $x_1$  occur at  $x_1^3/L^3 - 3x_1/L + 1 = 0$  for  $x_1 > 0$  and  $x_1^3/L^3 - 3x_1/L - 1 = 0$  for  $x_1 < 0$ .

34. The EOM that needs to be solved for  $\phi$  is

$$\ddot{\varphi} + \frac{g \cos \varphi}{R} + A^2\omega^2(1 + \cos \varphi) \sin^2(\omega t) \sin \varphi = 0 .$$

The reaction force from the wire to  $P$  is

$$\begin{aligned} \mathbf{N} &= N_1 \cos \phi \mathbf{i} + N_1 \sin \phi \mathbf{j} + N_2 \mathbf{k} , \\ N_1 &= mg \sin \varphi - mA^2R\omega^2(1 + \cos \varphi) \sin^2(\omega t) \cos \varphi - mR\dot{\varphi}^2 , \\ N_2 &= 2mAR\omega\dot{\varphi} \sin \varphi \sin(\omega t) - mAR\omega^2(1 + \cos \varphi) \cos(\omega t) . \end{aligned}$$

Under the condition  $A^2 \ll g/(R\omega^2)$  and  $\varphi = \delta - \pi/2$  with  $\delta \ll 1$ , the EOM becomes

$$\ddot{\delta} + \frac{g}{R}\delta = \frac{A^2\omega^2}{2} [1 - \cos(2\omega t)] ,$$

which oscillates about  $\delta = A^2R\omega^2/(2g)$  with the natural frequency  $\omega_n = \sqrt{g/R}$ . The approximation  $\delta \ll 1$  will fail if  $2\omega \simeq \omega_n$ , since then the oscillator resonates and its amplitude will be large.

35. The expression of the spring forces are

$$\mathbf{F}_{S,P} = -\mathbf{F}_{S,Q} = k\mathbf{PQ} = k(\mathbf{r}_0^Q - \mathbf{r}_0^P) = kR(\mathbf{e}_B - \mathbf{e}_A) + k[L - \xi - R(\psi - \theta)]\mathbf{e}_\psi + k\xi\mathbf{e}_\theta$$

36. Some of the forces are the centrifugal force and the Coriolis force:

$$\begin{aligned} \mathbf{F}_{\text{cent}} &= m\omega^2(\xi + \ell \cos \phi)\mathbf{i}_1 - m\omega^2\ell \sin \phi \sin \theta \mathbf{j}_1 \\ \mathbf{F}_{\text{Cori}} &= -2m\omega(\dot{\ell} \sin \phi \sin \theta + \ell \dot{\phi} \cos \phi \sin \theta + \ell \dot{\theta} \sin \phi \cos \theta)\mathbf{i}_1 - 2m\omega(\dot{\xi} + \dot{\ell} \cos \phi - \ell \dot{\phi} \sin \phi)\mathbf{j}_1 \end{aligned}$$

37. The two equations sought are:

$$\begin{aligned} \mathbf{e}_\zeta : \quad & \ddot{\zeta} + (\ddot{\zeta} - \zeta\dot{\beta}^2)(1 + 3\sin^2 \beta) + 3(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}) \cos \beta \sin \beta = -g + \Omega^2\zeta(1 + 3\sin^2 \beta) + 4\Omega\zeta\dot{\beta} \\ \mathbf{e}_\beta : \quad & 3(\ddot{\zeta} - \zeta\dot{\beta}^2) \cos \beta \sin \beta + (2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta})(1 + 3\cos^2 \beta) = 3\Omega^2\zeta \cos \beta \sin \beta - 4\Omega\dot{\zeta} \end{aligned}$$

38. The force on  $P$  due to  $Q$  is:

$$\mathbf{F}_P = 2[\xi - (2 + \cos \phi) \sin \theta]\mathbf{i}_0 + 2[\eta - \sin \phi]\mathbf{j}_0 - 2[4 + (2 + \cos \phi) \cos \theta]\mathbf{k}_0$$

39. No, the tension does not exert work; it is easy to prove it with  $\mathbf{e}_r \cdot \mathbf{v}_0^P = 0$ .
40. One of the forces acting on  $Q$  is the reaction of the ring,  $\mathbf{N}_Q = N_3(-\sin \phi \mathbf{i}_1 + \cos \phi \mathbf{j}_1) + N_4 \mathbf{k}_1$ .
41.  $\mathbf{v}_0^B = -W\omega \cos \theta \mathbf{i}_1 + (H\omega - W\dot{\theta} \sin \theta) \mathbf{j}_1 + W\dot{\theta} \cos \theta \mathbf{k}_1$   
 $\mathbf{a}_0^B = (2W\omega \dot{\theta} \sin \theta - H\omega^2) \mathbf{i}_1 - (W\ddot{\theta} \sin \theta + W\dot{\theta}^2 \cos \theta + W\omega^2 \cos \theta) \mathbf{j}_1 + (W\ddot{\theta} \cos \theta - W\dot{\theta}^2 \sin \theta) \mathbf{k}_1$
- 42.
- 43.
- 44.
45. Calling  $\omega_0 = \sqrt{k/m}$ , as long as  $\omega \neq \omega_0$ , the particular solution is

$$x_0^G(t) = \frac{\omega_0^2 \ell}{\omega_0^2 - \omega^2} \cos(\omega t); \quad x_0^C(t) = \frac{\omega^2 \ell}{\omega_0^2 - \omega^2} \cos(\omega t) \quad (6)$$

- 46.
47. The final quantities are:

$$\alpha = 0.744,$$
$$\beta = 1.4365.$$

- 48.
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- 50.
- 51.

### 3.2 Rigid body exercises and problems

1.  $v_{min} = v_0 \cdot \boldsymbol{\omega}_{B/0} = (v_0/a)\mathbf{j}$ . The IAR exists and is the line with direction  $\mathbf{e}_\omega \equiv \mathbf{j}$  that passes through point A.
2.  $\boldsymbol{\omega}_{B/0} = \dot{\theta}(\mathbf{i}_B - \sin\theta\mathbf{j}_B - \cos\theta\mathbf{k}_B) = \dot{\theta}(\cos\theta\mathbf{i} - \sin\theta\mathbf{j} - \mathbf{k})$ .  
 $\mathbf{v}_0^P = \frac{a}{\cos^2\theta}\dot{\theta}\mathbf{j}$ .
3. Some partial results are:

$$\begin{aligned}\boldsymbol{\omega}_{B0} &= \mathbf{i} \left( \dot{\theta} \sin \gamma \sin \beta + \dot{\gamma} \cos \beta \right) + \mathbf{j} \left( \dot{\theta} \cos \gamma + \dot{\beta} \right) + \mathbf{k} \left( \dot{\theta} \sin \gamma \cos \beta - \dot{\gamma} \sin \beta \right), \\ &= \mathbf{i}_B \left( \dot{\gamma} \cos \theta + \dot{\beta} \sin \theta \sin \gamma \right) + \mathbf{j}_B \left( \dot{\theta} + \dot{\beta} \cos \gamma \right) + \mathbf{k}_B \left( \dot{\gamma} \sin \theta - \dot{\beta} \sin \gamma \cos \theta \right), \\ \boldsymbol{\alpha}_{B0} &= \mathbf{i} \left[ \dot{\theta} \left( \dot{\gamma} \cos \gamma \sin \beta + \dot{\beta} \cos \beta \sin \gamma \right) + \ddot{\gamma} \cos \beta - \dot{\gamma} \dot{\beta} \sin \beta \right] \\ &\quad + \mathbf{j} \left[ \ddot{\beta} - \dot{\theta} \dot{\gamma} \sin \gamma \right] + \mathbf{k} \left[ \dot{\theta} \left( \dot{\gamma} \cos \gamma \cos \beta - \dot{\beta} \sin \beta \sin \gamma \right) - \ddot{\gamma} \sin \beta - \dot{\gamma} \dot{\beta} \cos \beta \right], \\ &= \mathbf{i}_B \left[ \dot{\beta} \left( \dot{\theta} \cos \theta \sin \gamma + \dot{\gamma} \cos \gamma \sin \theta \right) + \ddot{\gamma} \cos \theta - \dot{\gamma} \dot{\theta} \sin \theta + \ddot{\beta} \sin \theta \sin \gamma \right] \\ &\quad + \mathbf{j}_B \left[ \ddot{\beta} \cos \gamma - \dot{\gamma} \dot{\beta} \sin \gamma \right] \\ &\quad + \mathbf{k}_B \left[ \ddot{\gamma} \sin \theta + \dot{\theta} \dot{\gamma} \cos \theta - \ddot{\beta} \sin \gamma \cos \theta - \dot{\beta} \left( \dot{\gamma} \cos \gamma \cos \theta - \dot{\theta} \sin \theta \sin \gamma \right) \right].\end{aligned}$$

4. For the letter T, we first consider a frame  $S_0$ , whose  $y$  axis is the vertical symmetric line and the  $x$  axis coincides with the bottom line of the upper rectangle. The center of mass is located in  $S_0$  as

$$\mathbf{r}_0^G = \frac{wt - (h - t)^2}{2(w + h - t)} \mathbf{j}.$$

In the frame with origin on  $G$ , we have

$$\begin{aligned}I_{Gx} &= \frac{\sigma}{3} [wt^3 + (h - t)^3 t] - mr^2, \quad r = -\frac{wt - (h - t)^2}{2(w + h - t)}, \\ I_{Gy} &= \frac{\sigma t}{12} [w^3 + (h - t)t^2].\end{aligned}$$

The product of inertia  $P_{xz,yz}$  is zero.

5. (a) The radius of the “slender” bar is negligible.

$$\bar{\bar{I}}_G = \frac{mL^2}{12} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)

$$\bar{\bar{I}}_G = \frac{m}{12} \begin{bmatrix} 3R^2 + L^2 & 0 & 0 \\ 0 & 3R^2 + L^2 & 0 \\ 0 & 0 & 6R^2 \end{bmatrix}$$

(c)

$$\bar{\bar{I}}_G = \frac{m}{12} \begin{bmatrix} b^2 + c^2 & 0 & 0 \\ 0 & a^2 + c^2 & 0 \\ 0 & 0 & a^2 + b^2 \end{bmatrix}$$

(d)

$$\bar{\bar{I}}_G = \frac{2mR^2}{5} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(e)

$$\bar{I}_G = mR^2 \begin{bmatrix} 2/5 - 9/64 & 0 & 0 \\ 0 & 2/5 - 9/64 & 0 \\ 0 & 0 & 2/5 \end{bmatrix}$$

(f)

$$\bar{I}_G = \frac{3m}{80} \begin{bmatrix} 8R^2 & 0 & 0 \\ 0 & 4R^2 + L^2 & 0 \\ 0 & 0 & 4R^2 + L^2 \end{bmatrix}$$

(g)

$$\bar{I}_G = \frac{mR^2}{4} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

6. The IAR passes point I that can be located from G by  $\mathbf{GI} = -L(\mathbf{i}_B + \sqrt{2}\mathbf{k}_B)/2$  and it has the same direction as the angular velocity  $\boldsymbol{\omega}_{B0} = \Omega(\sqrt{2}\mathbf{i}_B - \mathbf{k}_B)$ .

$$T_0 = \frac{M\Omega^2 L^2}{8} (8B + 4D - 8\sqrt{2}E + 9) .$$

$$\mathbf{H}_{G,0} = M\Omega L^2 [(\sqrt{2}B - E)\mathbf{i}_B - F\mathbf{j}_B + (\sqrt{2}E - D)\mathbf{k}_B] .$$

$$\mathbf{H}_{A,0} = M\Omega L^2 [(\sqrt{2}B - E)\mathbf{i}_B - F\mathbf{j}_B + (\sqrt{2}E - D + 3/4)\mathbf{k}_B] .$$

7. Mordor ( $\mathcal{M}$ ) exerts a force on The Ring ( $\mathcal{R}$ ) that can be written as:

$$\mathbf{F}_{\mathcal{M}} = \int_{\mathcal{R}} \beta \mathbf{r}_0^P dm = \beta \mathbf{r}_0^G M .$$

The moment exerted by Mordor on The Ring about G can be written as:

$$\mathbf{M}_{G,\mathcal{M}} = \int_{\mathcal{R}} \mathbf{r}_B^P \times \beta \mathbf{r}_0^P dm = \beta \int_{\mathcal{R}} \mathbf{r}_B^P \times (\mathbf{r}_0^G + \mathbf{r}_B^P) dm = \beta \int_{\mathcal{R}} \mathbf{r}_B^P \times \mathbf{r}_0^G dm = \beta \int_{\mathcal{R}} \mathbf{r}_B^P dm \times \mathbf{r}_0^G = 0 .$$

8. The EOM for 2 DOF are

$$\begin{aligned} m\ddot{y}_G &= -mg - k(y_G + \sqrt{3}a \cos \theta/3) , \\ ma\ddot{\theta} &= 4\sqrt{3}k(y_G \sin \theta + 2a \cos \theta) . \end{aligned}$$

The equation to solve the reaction force from the vertical line  $R$  (defined positive in  $\mathbf{i}$  direction) is

$$R = ak(2 - \sqrt{3} \sin \theta/3) .$$

The 4 initial conditions required to solve the EOM are

$$t = 0 \rightarrow y_G = \sqrt{3}/6 , \quad \dot{y}_G = v_0 , \quad \theta = 0 , \quad \dot{\theta} = -\sqrt{3}u_0/a .$$

At equilibrium for  $\theta = \pi/4$ , we have

$$y_G = -2a , \quad k = \frac{mg}{a(2 - \sqrt{6}/6)} , \quad R = ka \left( 2 - \frac{\sqrt{6}}{6} \right) .$$

After fixing the position of  $G$ , the linearized EOM for the perturbation  $\phi$

$$\ddot{\phi} + \frac{8\sqrt{6}k}{m}\phi = 0 ,$$

which is a harmonic oscillator with the natural frequency being  $\Omega = \sqrt{8\sqrt{6}k/m}$ .

- 9.

10. The EOM is  $L\dot{\theta}^2/2 - g \cos \theta = C$ , with  $C$  a constant to be determined by IC.
11. The vector basis  $B_1\{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$  is defined as:  $\mathbf{k}_1//\mathbf{k}$ ;  $\mathbf{i}_1$  is parallel to the perpendicular axis of the disk, from the axis  $Oz$  to the disc;  $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ . The angular velocity and acceleration of the disk is  $\boldsymbol{\omega}_{B0} = \omega(-\mathbf{i}_1 + \mathbf{k}_1)$  and  $\boldsymbol{\alpha}_{B0} = -\omega^2 \mathbf{j}_1$ . The acceleration of the contact point  $C$  is  $\mathbf{a}_0^{C(B)} = \omega^2 R(\mathbf{i}_1 + \mathbf{k}_1)$ .

12.

13. For  $\mu = 0$ ,

$$6\dot{x} + \dot{\theta}R \sin \theta = 6\omega R, \quad 24\dot{x}^2 + 13\dot{\theta}^2 R^2 + 8\dot{x}\dot{\theta}R \sin \theta - 8gR \sin \theta = 37\omega^2 R^2$$

For  $\mu \rightarrow \infty$ ,

$$\dot{\theta}^2 R(37 - 8 \sin \theta) - 8g \sin \theta = 37\omega^2 R.$$

14.

15.

16.

17. The only forces that appear on the fork-disc system arise from the collar joint and the weight of both disk and fork. From the equations of motion in basis  $B_2$  it can be noticed that for the disk alone the angular momentum with respect to  $\mathbf{i}_2$  is conserved. Therefore, one of the quadratures of the problem is:

$$\dot{\phi} = C_1. \quad (7)$$

Furthermore, mechanical energy is conserved due to the fact that (i) the resultant reaction force at the collar does not work since  $\mathbf{F} \cdot \mathbf{v}_0^{O_1} = 0$ , and (ii) because the resultant torque has no component along the direction of  $\boldsymbol{\omega}_{21}$ , the resultant reaction torque does not work either. (iii) Last but not least the other forces present are the disk and fork respective weights, which are conservative forces.

This provide with the following conservation law:

$$\dot{\theta}^2 \left( \frac{I_2}{2} + \frac{mR^2}{8} + \frac{mb^2}{2} \right) + \dot{\phi}^2 \left( \frac{mR^2}{4} \right) - (mb + Ma)g \cos \theta = C_2. \quad (8)$$

The resultant reaction force and torque that the axis  $AB$  applies on the fork-disc system at  $O_1$  is  $\mathbf{R}_{O_1} = \left[ \ddot{\theta}(Ma + mb) + g \sin \theta (M + m) \right] \mathbf{i}_2 - \left[ \dot{\theta}^2 (Ma + mb) + g \cos \theta (M + m) \right] \mathbf{k}_2$  and  $\mathbf{Q}_{O_1} = -\frac{mR^2}{2} \dot{\phi} \dot{\theta} \mathbf{k}_2$ , respectively. The resultant (internal) reaction force and torque that the axis  $CD$  applies on the disk at  $O$  is  $\mathbf{R}_O = (mb\ddot{\theta} + mg \sin \theta) \mathbf{i}_2 - (mb\dot{\theta}^2 + mg \cos \theta) \mathbf{k}_2$  and  $\mathbf{Q}_O = \frac{mR^2}{4} \ddot{\theta} \mathbf{j}_2 - \frac{mR^2}{2} \dot{\phi} \dot{\theta} \mathbf{k}_2$ , respectively.

18. We define  $\xi = \dot{x}_G$ ,  $\eta = \dot{y}_G$ ,  $\boldsymbol{\omega}_{B0} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$ .

For rolling with slip, there are 5 DOF. The EOM are

$$\begin{aligned} \dot{\xi} &= \frac{-\mu g(\xi - a\omega_y)}{\sqrt{(\xi - a\omega_y)^2 + (\eta + a\omega_x)^2}}, \quad \dot{\eta} = \frac{-\mu g(\eta + a\omega_x)}{\sqrt{(\xi - a\omega_y)^2 + (\eta + a\omega_x)^2}}, \\ 2a\dot{\omega}_x &= 5\dot{\eta}, \quad 2a\dot{\omega}_y = -5\dot{\xi}, \quad \dot{\omega}_z = 0. \end{aligned}$$

For rolling without slip, there are 3 DOF. The problem can be reduced to quadratures as

$$\xi = a\omega_y = C_1, \quad \eta = -a\omega_x = C_2, \quad \omega_z = C_3,$$

with the constants to be determined by IC.

The ball hits the pocket with velocity and angular velocity being

$$\mathbf{v}_0^G = \frac{2\xi_0}{\sqrt{5}} \mathbf{i}, \quad \boldsymbol{\omega}_{B0} = \frac{\xi_0}{a} \left( \frac{11}{4} - \sqrt{5} \right) \mathbf{j}.$$

19.

20. The length  $L$  required to achieve the final angular velocity  $\omega_f$  is  $L^2 = \frac{(I + mR^2)(\omega_0 - \omega_f)}{m(\omega_0 + \omega_f)}$ .

The tension is  $T = \frac{2KI\omega_0^3 t}{R(1 + K\omega_0^2 t^2)^2}$ , with  $K = \frac{mR^2}{I + mR^2}$ .

21.

22.

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24.

25. This problem has 4 DOF. To solve the motion of the cylinder, it requires 8 equations to solve for the 8 *unknowns*  $x_G, y_G, z_G, \beta, \theta, \varphi, T_C$ , and  $T_D$ . In this case, the *unknown coordinates* have to be solved together with the *unknown reaction forces*. The constraint equations can not be decoupled from the Newton ordinary differential equations.

$$\begin{aligned}
 &(\cos \theta \cos \beta - x_G/l)^2 + (\cos \theta \sin \beta - y_G/l)^2 + (\sin \theta + z_G/l)^2 = 1 \\
 &(\cos \theta \cos \beta + x_G/l)^2 + (3 - \cos \theta \sin \beta - y_G/l)^2 + (\sin \theta - z_G/l)^2 = 1 \\
 &T\mathbf{i} : m\ddot{x}_G = (T_C - T_D) \cos \theta \cos \beta - (T_C + T_D) x_G/l \\
 &T\mathbf{j} : m\ddot{y}_G = (T_C - T_D) \cos \theta \sin \beta - (T_C + T_D) y_G/l + 3T_D \\
 &T\mathbf{k} : m\ddot{z}_G = -mg + (T_D - T_C) \sin \theta - (T_C + T_D) z_G/l \\
 &R\mathbf{i}_2 : \ddot{\varphi} - \ddot{\beta} \sin \theta - \dot{\beta} \dot{\theta} \cos \theta = 0 \rightarrow \dot{\varphi} - \dot{\beta} \sin \theta = \text{const.} \\
 &R\mathbf{j}_2 : ml^2 \left[ \ddot{\theta} \mu - \dot{\beta}^2 (\lambda - \mu) \cos \theta \sin \theta + \dot{\beta} \dot{\varphi} \lambda \cos \theta \right] \\
 &= (T_D - T_C) (x_G \sin \theta \cos \beta + y_G \sin \theta \sin \beta + z_G \cos \theta) - 3T_D l \sin \theta \sin \beta \\
 &R\mathbf{k}_2 : ml^2 \left[ \ddot{\beta} \mu \cos \theta - \dot{\beta} \dot{\theta} (2\mu - \lambda) \sin \theta - \dot{\theta} \dot{\varphi} \lambda \right] \\
 &= (T_D - T_C) (x_G \sin \beta - y_G \cos \beta) + 3T_D l \cos \beta
 \end{aligned}$$

If the body-fixed vector basis is used to decompose the equations for rotation, it leads to:

$$\begin{aligned}
 ml^2 \lambda \alpha_{x_B} &= 0 \Rightarrow \dot{\beta} \sin \theta - \dot{\varphi} = \text{const} \\
 ml^2 \mu \alpha_{y_B} - ml^2 (\mu - \lambda) \omega_{x_B} \omega_{z_B} &= -(T_1 - T_2) [x_G (\cos \beta \sin \theta \cos \varphi + \sin \beta \sin \varphi) \\
 &\quad + y_G (\sin \beta \sin \theta \cos \varphi - \cos \beta \sin \varphi) + z_G \cos \theta \cos \varphi] \\
 &\quad - 3T_2 l (\sin \beta \sin \theta \cos \varphi - \cos \beta \sin \varphi) \\
 ml^2 \mu \alpha_{z_B} + ml^2 (\mu - \lambda) \omega_{x_B} \omega_{y_B} &= (T_1 - T_2) [x_G (\cos \beta \sin \theta \sin \varphi - \sin \beta \cos \varphi) \\
 &\quad + y_G (\sin \beta \sin \theta \sin \varphi + \cos \beta \cos \varphi) + z_G \cos \theta \sin \varphi] \\
 &\quad + 3T_2 l (\sin \beta \sin \theta \sin \varphi + \cos \beta \cos \varphi) ,
 \end{aligned}$$

with  $T_1 = T_C$  and  $T_2 = T_D$ . Conservation of mechanical energy gives

$$\frac{m}{2} (\dot{x}_G^2 + \dot{y}_G^2 + \dot{z}_G^2) + \frac{ml^2}{2} \left[ \lambda (\dot{\varphi} - \dot{\beta} \sin \theta)^2 + \mu \dot{\theta}^2 + \mu (\dot{\beta} \cos \theta)^2 \right] + mgz_G = \text{const.}$$

26. The EOM of the system is

$$\frac{425}{64} md^2 \dot{\theta}^2 + 5mg \frac{\sqrt{2}-1}{4} d(1 - \sin \theta) = 5mg \frac{\sqrt{2}-1}{4} d .$$

There are two equilibrium positions, given by  $\theta = \pi/2$  and  $\theta = -\pi/2$ . After expanding about the stable equilibrium  $\theta = \pi/2$  as  $\theta = \pi/2 + \delta$  with  $\delta \ll 1$ , the EOM reads:

$$\frac{425}{64} md^2 \dot{\delta}^2 + 5mg \frac{\sqrt{2}-1}{4} \frac{d\delta^2}{2} = E ,$$

whose angular frequency is

$$\sqrt{\frac{8g}{85d}(\sqrt{2}-1)}$$

27. Vector  $\mathbf{i}_1$  is  $\mathbf{OG}$  and  $\mathbf{j}_1$  is calculated following the right-hand rule so that  $\mathbf{k} = \mathbf{k}_1$ . Two equations come from projecting Newton's second law along  $\mathbf{i}_1$  and  $\mathbf{j}_1$  so that:

$$\begin{aligned} -2m[\ddot{\xi} + \dot{\theta}^2(4L - \xi)] &= 2mg \cos \theta - k[4L - \xi - L \sin(\phi - \theta) - 2L \sin \theta] - T, \\ 2m[\ddot{\theta}(4L - \xi) - 2\dot{\xi}\dot{\theta}] &= -2mg \sin \theta - k[L \cos(\phi - \theta) - 2L \cos \theta]. \end{aligned}$$

Tension can be eliminated using Newton's second law for particle P in the  $\mathbf{j}$  direction:

$$-m\ddot{\xi} = -mg + T,$$

The third equation is the equation of angular momentum of the block at  $G$  about  $\mathbf{k}$ . The moment of inertia of the block at  $G$  about  $\mathbf{k}$  is  $I_z^G = 5mL^2/6$ . Only the spring causes a torque about  $G$ , so:

$$5mL^2\ddot{\phi}/6 = kL[(4L - \xi) \cos(\phi - \theta) - 2L \sin \phi] \quad (9)$$

28.

29.

30. The ellipsoid and the plane are tangent if their normals at  $P$  coincide.

31.

32.

33. The smaller wheel will leave the ground when the normal at  $P$  is  $N_P = 0$ . Imposing this condition on the first rotational equation, using the rest of equations to reduce the problem to an equation on  $N_P$  and  $\psi$ ,

$$\frac{10(A + 3B)}{ma^2} \sqrt{3} \dot{\psi}^2/4 + 9\sqrt{3}(27\dot{\psi}^2/10) = 13\frac{g}{a} \quad \Rightarrow \quad \dot{\psi}_{\max}^2 = \frac{104g}{323\sqrt{3}a}$$

34.

35.

36. To obtain the force  $\mathbf{F}$  we use the equation of motion of the particle:

$$m\mathbf{a}_0^P = -mg[\sin \gamma \mathbf{i}_0 + \cos \gamma \mathbf{j}_0] - \mathbf{F}.$$

Hence,

$$\mathbf{F} = -mg[\sin \gamma \mathbf{i}_0 + \cos \gamma \mathbf{j}_0] - m\mathbf{a}_0^P.$$

37.

38.

39.

40.

41.

42.